MIL-M-28787/193A(EC)
3 August 1979
SUPERSEDING
MIL-M-28787/193(EC)
2 June 1977

MILITARY SPECIFICATION

* MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, DIGITAL, KEY CODE SYT

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital random access memory module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Memory, random access
Family	Digital
Key code	SYT
Size	1A
Weight	35 grams maximum
Failure rate	12.527 failures/10^6 hours maximum
* Environment	Class 1 as specified in MIL-STD-1389
Maximum power dissipation	3.6 W (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current	
ADI0-ADI3, BDI0-BDI3	
Low level (V_{IL} = 0.0 V)	-0.88 mA maximum
(V_{IL} = 0.4 V)	-0.8 mA maximum
High level (V_{IH} = 2.4 V)	20 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Input current -Continued		
AADD0-AADD7, BADD0-BADD7, AE1, BE1, AWE, BWE		
Low level	(V_{IL} = 0.0 V)	-3.5 mA maximum
	(V_{IL} = 0.4 V)	-3.2 mA maximum
High level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	4 mA maximum
AE0, BE0		
Low level	(V_{IL} = 0.0 V)	-7.0 mA maximum
	(V_{IL} = 0.4 V)	-6.4 mA maximum
High level	(V_{IH} = 2.4 V)	240 µA maximum
	(V_{IH} = 5.5 V)	8 mA maximum
Input clamp diode voltage		
V_{CC} = 4.5 V, I_{clamp} = 12.0 mA		-0.2 V minimum, -1.5 V maximum
Output voltage		
Low level	(I_{OL} = 16.0 mA)	0.5 V maximum
High level	(I_{OH} = -2.0 mA)	2.4 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.4 V)	16 0 mA
High level	(V_{OH} = 2.4 V)	-2.0 mA
Output leakage current		
High-impedance state		
V_{OL} = 0.4 V		-30 µA minimum, 30 µA maximum
V_{OH} = 2.4 V		0 µA minimum, 30 µA maximum
Input capacitance		
ADI0-ADI3, BDI0-BDI3		15 pF maximum
AADD0-AADD7, BADD0-BADD7, AE1, BE1, AWE, BWE		60 pF maximum
AE0, BE0		120 pF maximum
* Output capacitance		20 pF maximum
Dynamic characteristics (see figure 1)		
Setup time		
Address to write enable (t_{AS})		15 ns minimum
Enable to write enable (t_{ES})		10 ns minimum
Data to write enable (t_{DS})		40 ns minimum
Hold time		
Address to write enable (t_{AH})		10 ns minimum
Enable to write enable (t_{EH})		10 ns minimum
Data to write enable (t_{DH})		5 ns minimum

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Dynamic characteristics (see figure 1) -Continued	
Write pulse width (t_{WP})	40 ns minimum
Address access time (t_{AA})	
Address to output	
C_L = 50 pF	60 ns maximum
C_L = 150 pF	70 ns maximum
Enable access time (t_{EA})	
Enable to output enabled	
C_L = 50 pF	30 ns maximum
C_L = 150 pF	40 ns maximum
Enable recovery time	
Low-state to high-impedance state (t_{ERL})	
C_L < 10 pF	30 ns maximum
High-state to high-impedance state (t_{ERH})	
C_L < 10 pF	30 ns maximum
Write recovery propagation delay time (t_{WR})	
C_L = 50 pF	50 ns maximum
C_L = 150 pF	60 ns maximum

* 1.3 Module function. This module contains 2 random access memories of 256 words by 4 bits with three-state outputs.

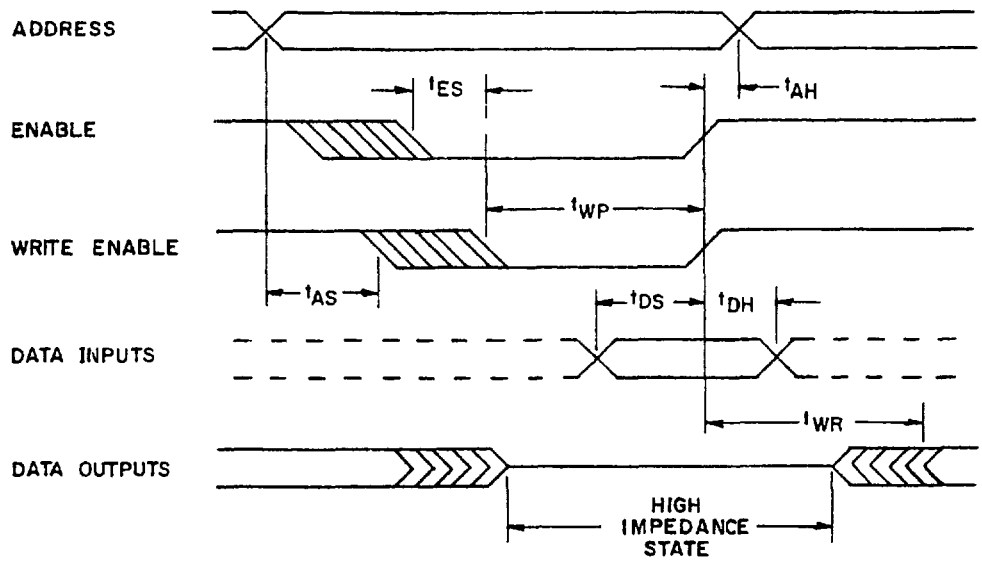
1.3.1 Input to output relationship. Each random access memory (RAM) has the input to output relationship of table I. The following describes the essential module operation.

1.3.1.1 Read. Memory A is addressed with AADD0-AADD7 inputs which select one of 256 words. The memory is enabled by making all of memory A's enables (AE0, AE1) low. If the write enable ($\overline{AWE}$) is high and the memory is enabled, the stored data is read out on the four data out pins. The data read out is the complement of the data written in during the write cycle. Memory B is addressed with BADD0-BADD7 inputs which select one of 256 words. The memory is enabled by making all of memory B's enables (BE0, BE1) low. If the write enable ($\overline{BWE}$) is high and the memory is enabled, the stored data is read out on the four data out pins. The data read out is the complement of the data written in during the write cycle.

1.3.1.2 Write. Both memories are addressed and enabled as in the read cycle. If the memory A write enable ($\overline{AWE}$) is low, the data on the four data input pins ($\overline{ADIO}$-ADI3) is written into the addressed word of memory A. If the memory B write enable ($\overline{BWE}$) is low, the data on the four data input pins (BDI0-BDI3) is written into the addressed word of memory B.

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WRITE CYCLE WAVEFORMS



READ CYCLE WAVEFORMS

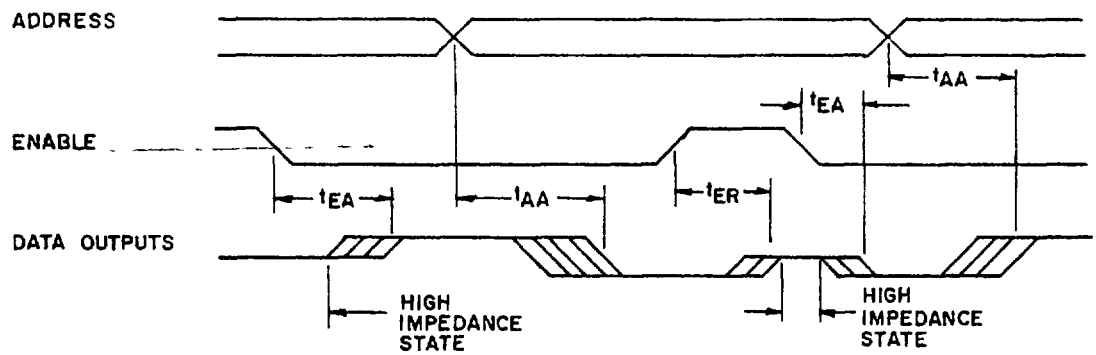


FIGURE 1. Dynamic characteristics waveforms.

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TABLE I. Input to output relationship for each RAM.

Inputs			Outputs	Mode
$\overline{E}$	$\overline{WE}$	DI	DO	
H	X	X	OFF	Not selected
L	L	L	OFF	Write "L"
L	L	H	OFF	Write "H"
L	H	X	$\overline{DI}(t_n)$	Read

NOTES:
1. H = High-voltage level.
2. L = Low-voltage level.
3. X = Don't care.
4. OFF = Floating output level is determined by external circuitry connected to the output.

1.3.1.3 Three-state outputs. The three-state output mode allows each memory to be connected to a common bus with several like memories. The memory A outputs (AD00-AD03) can be placed in the high-impedance state by setting any of memory A's enables at a high level or by setting memory A's write enable to a low level. The memory B outputs (BD00-BD03) can be placed in the high-impedance state by setting any of memory B's enables at a high level or by setting memory B's write enable to a low level.

1.3.2 Parallel outputs. Bussing is allowed between like outputs of a number of memories. When outputs of several SYT memories are bussed together, all address lines should be addressed using synchronous clocking.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

* MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 12.527 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be SYT.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Hold-down holes. The module shall have no additional requirements for location or quantity of hold-down holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/193-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 2.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 $\pm$0.5 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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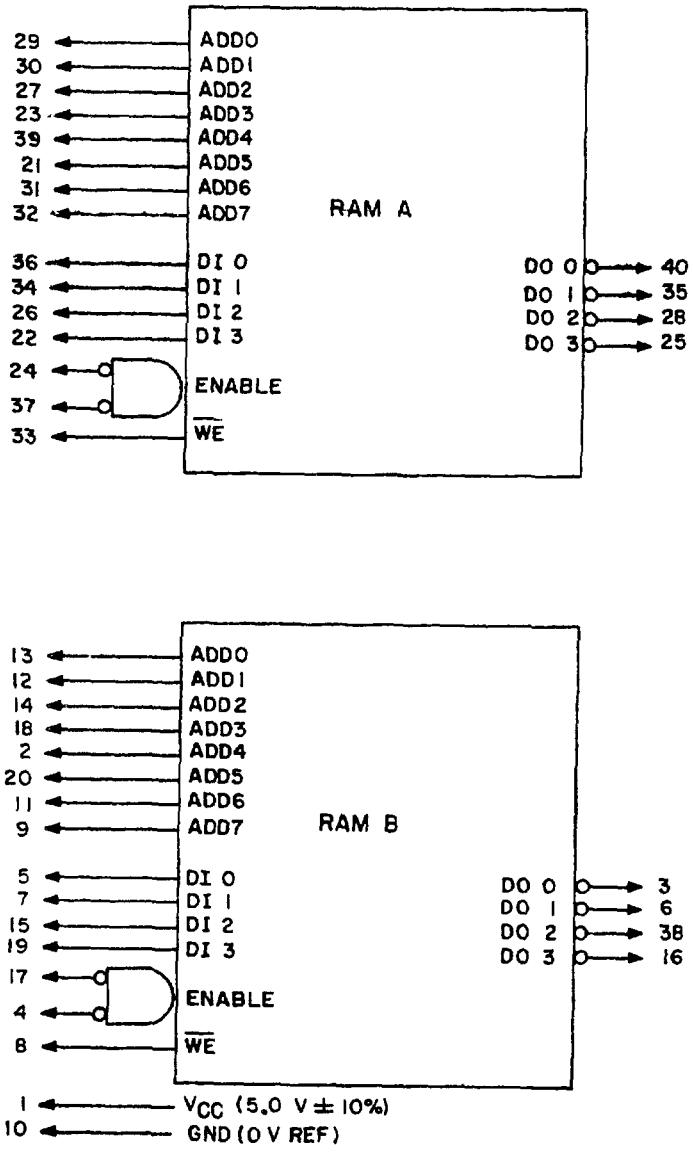


FIGURE 2. SYT module function diagram, random access memory (2x256x4).

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1 0	∞	10	∞	1.0	∞	MΩ
B1-B8	V_{out} (low-loaded)	IV	S	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
C1, C3, C8-C16, C18, C23-C30	Input breakdown current	V	S	0.0	4.0	0 0	4.0	0.0	4.0	0 0	4.0	mA
C4-C7, C19-C22	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C2, C17	Input breakdown current	V	S	0.0	8 0	0.0	8.0	0.0	8.0	0.0	8.0	mA
D1-D8	Output leakage current (V_{OL} = 0.4 V)	VI	X	-30	+30	-30	+30	-30	+30	-30	+30	µA
D9-D16	Output leakage current (V_{OH} = 2.4 V)	VI	X	0	30	0	30	0	30	0	30	µA
E1-E8	Output short circuit current	VII	X	-12	-70	-12	-70	-12	-70	-12	-70	mA
F1-F30	Input clamp diode voltage	VIII	X	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	V
G1-G8	Static output voltage, low level	IX	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
G9-G16	Static output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1, H2, H7-H16, H21-H28	Input current, low level	X	X	-0.01	-3.5	-0.01	-3.5	-0.01	-3.5	-0.01	-3.5	mA
H3-H6, H17-H20	Input current, low level	X	X	-0.01	-0.88	-0.01	-0.88	-0.01	-0.88	-0.01	-0.88	mA

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
H29,H30	Input current, low level	X	X	-0.01	-7.0	-0 01	-7.0	-0.01	-7.0	-0.01	-7.0	mA
J1, J2, J7-J16, J21-J28	Input current, high level	XI	X	-1	80	-1	80	-1	80	-1	80	µA
J3-J6, J17-J20	Input current, high level	XI	X	-1	20	-1	20	-1	20	-1	20	µA
J29,J30	Input current, high level	XI	X	-1	240	-1	240	-1	240	-1	240	µA
K1-K3	Power supply current, low-level outputs, high-level outputs, and high-impedance outputs (V_{CC} = 5.5 V, t_{AMB} = 25° ±5°C)	XII	X	58	582	58	582	← See table XII →				mA
K4-K6	Power supply current, low-level outputs, high-level outputs, and high-impedance outputs (V_{CC} = 5.0 V, t_{AMB} = 25° ±5°C)	XII	X	54	540	54	540	← See table XII →				mA

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements at temperature specified in test				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K7-K9	Power supply current, low-level outputs, high-level outputs, and high-impedance outputs (V_{CC} = 5.0 V, t_{FIN} = 0°C)	XII	Q	58	580	58	580	← See table XII →				mA
K10-K12	Power supply current, low-level outputs, high-level outputs, and high-impedance outputs (V_{CC} = 5.0 V, t_{FIN} = 80°C)	XII	Q	44	440	44	440	← See table XII →				mA
K13-K15	Power supply current, low-level outputs, high-level outputs, and high-impedance outputs (V_{CC} = 5.0 V, t_{FIN} = 60°C)	XII	Q	50	500	50	500	← See table XII →				mA

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1-L8	Address threshold, low level	XIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
M1-M8	Address threshold, high level	XIV	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
N1-N8	Transition time (t_{THL}), C_L = 50 pF	XV	S	3	10	3	10	3	10	3	10	ns
N9-N16	Transition time (t_{THL}), C_L = 150 pF	XV	Q	5	20	5	20	5	20	5	20	ns
N17-N24	Transition time (t_{TLH}), C_L = 50 pF	XV	S	4	38	4	38	4	38	4	38	ns
N25-N32	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	5	50	5	50	5	50	5	50	ns
P1-P16	Enable recovery time (t_{ERL} and t_{ERH})	XVI	S	3	30	3	30	3	30	3	30	ns
Q1-Q8	Propagation delay time, write enable low to high-impedance state, t_{PLHLH}	XVII	S	3	30	3	30	3	30	3	30	ns
Q9-Q16	Propagation delay time, write enable low to high-impedance state, t_{PLHHL}	XVII	S	3	30	3	30	3	30	3	30	ns
R1,R2	Walking-one-walking-zero test pattern (C_L = 50 pF)	XVIII	X	← See table XVIII →								
R3,R4	Walking-one-walking-zero test pattern (C_L = 150 pF)	XVIII	Q	← See table XVIII →								
S1,S2	Galloping-one-galloping-zero test pattern (C_L = 50 pF)	XIX	X	← See table XIX →								

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S3, S4	Galloping-one-galloping-zero test pattern (C_L = 150 pF)	XIX	Q	See table XIX								
T1, T2	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 50 pF)	XX	X	See table XX								
T3, T4	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 150 pF)	XX	Q	See table XX								
U1, U2	Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 50 pF)	XXI	X	See table XXI								
U3, U4	Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 150 pF)	XXI	Q	See table XXI								
V	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	mV
W1, W3, W8-W15, W16, W18, W23-W30	Input capacitance	XXIII	Q	3	60	3	60	3	60	3	60	pF
W2, W17	Input capacitance	XXIII	Q	3	120	3	120	3	120	3	120	pF
W4-W7, W19-W22	Input capacitance	XXIII	Q	3	15	3	15	3	15	3	15	pF
* X1-X8	Output capacitance	XXIV	Q	3	20	3	20	3	20	3	20	pF
Y	Power supply filter capacitance	XXV	S	1.8	6.0	1.8	6.0	1.8	6.0	1.8	6.0	µF

NOTES:

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* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below.

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	4
Pulse generator	Item J01	2
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
LSI tester	Item U03	1
Capacitance bridge	Item C01	1
Capacitance meter	Item C05	1
Resistors		
51Ω ±5%, 1 W		3
324Ω ±1%, 1/8 W		9
2.2 kΩ ±1%, 1/8 W		3
1 kΩ ±5%, 1/4 W		1
1.15 kΩ ±1%, 1/4 W		1
Value as required for figure 14, note 2, 1/8 W		8
1.21 kΩ ±1%, 1/8 W		8
Capacitors		
Value as required for figure 14, note 4, test type S		8
Value as required for figure 14, note 4, test type Q		1
Diode		
1N3600, 1N4150, or equivalent		8
Components		
5476, or equivalent		1
5400, or equivalent		1
9316, or equivalent		2
Push button		1
Switch, SPDT		1
Item U03 LSI tester		
Load boards as required to tables XVIII, XIX, XX, and XXI.		
All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.		

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V, and Y, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test, electrical tests D through V, and Y, shall be performed in the order specified in 4.4.2 using the 0° to 60°C end-of-life requirements of table II. After the module has been returned to 25°C, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the 25° ±5°C end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/193-1 supersedes part number M28787/193. The two parts are interchangeable.

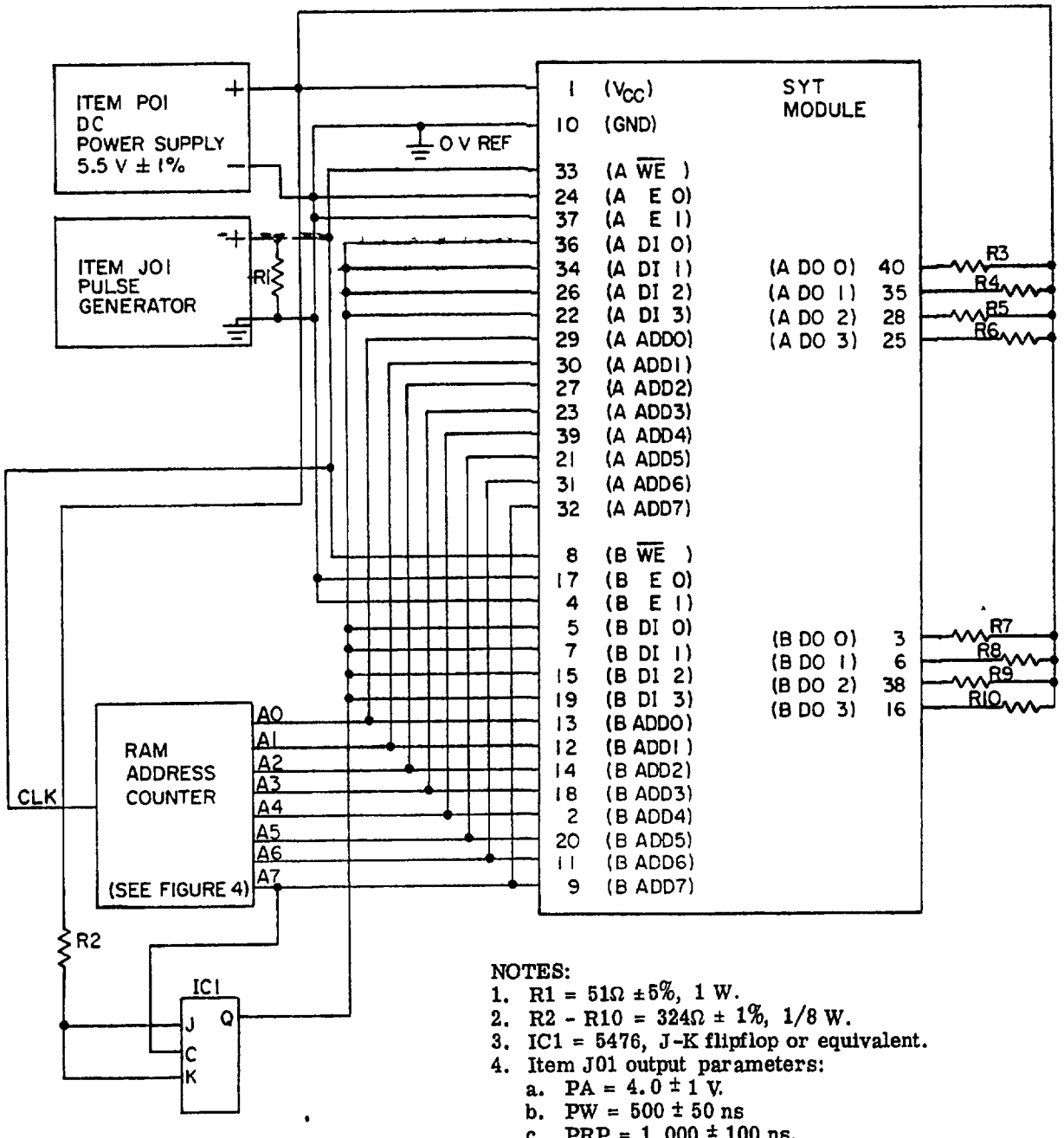
6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5963-N044)

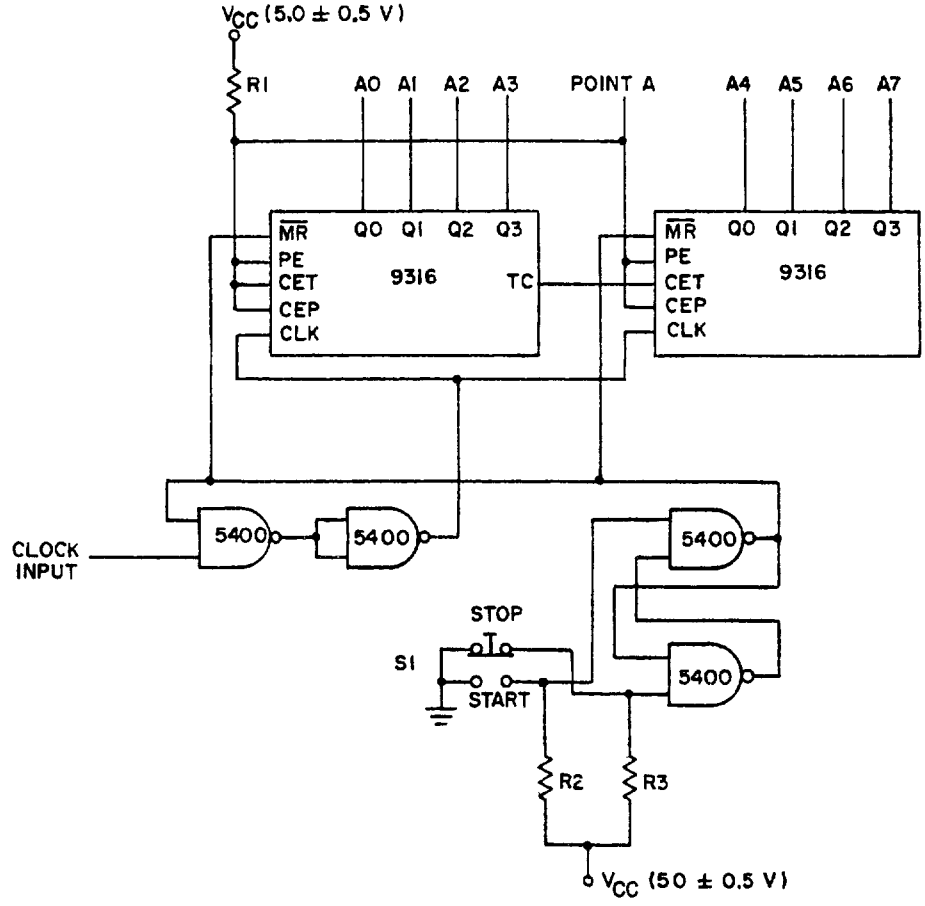
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 - R10 = 324Ω ± 1%, 1/8 W.
3. IC1 = 5476, J-K flipflop or equivalent.
4. Item J01 output parameters:
   a. PA = 4.0 ± 1 V.
   b. PW = 500 ± 50 ns
   c. PRP = 1,000 ± 100 ns.
   d. $t_{TLH} = t_{THL}$ = 25 ± 10 ns.

FIGURE 3. Operating temperature and life test circuit.

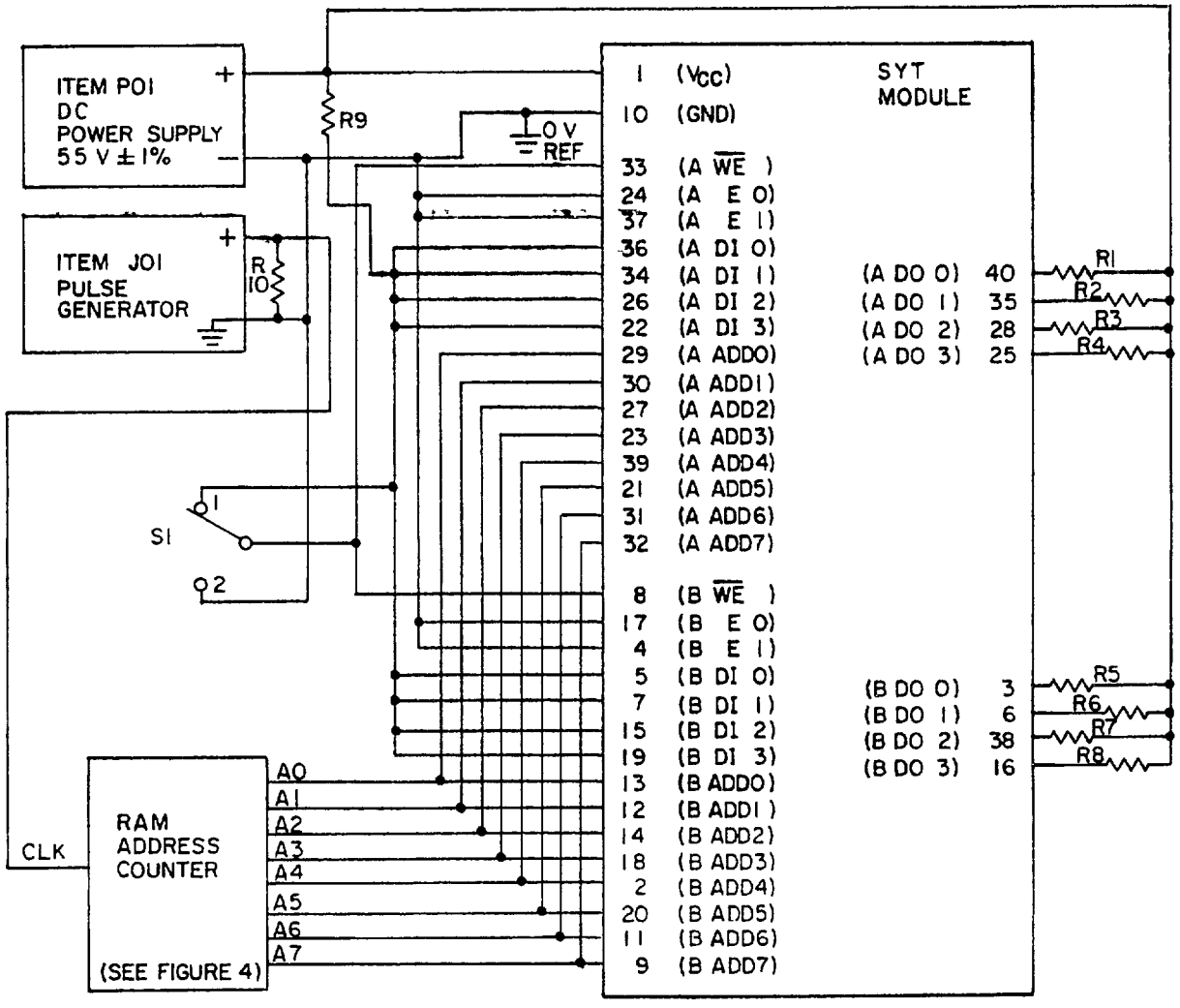
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NOTES:
1. R1 - R3 = 2.2 kΩ ± 1%, 1/8 W.
2. S1 = push button.
3. 2 each 9316 high-speed synchronous 4-bit binary counters.
4. 1 each 5400 quad 2-input NAND gate.

FIGURE 4. RAM address counter.

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NOTES:
1. R1 - R9 = 324Ω ± 1%, 1/8 W.
2. R10 = 51Ω ± 5%, 1 W.
3. S1 = SPDT switch.
4. Load all memory addresses with low levels by connecting the RAM address counter to address inputs and cycling through all addresses with switch S1 in position 2. Then place switch S1 in position 1 and connect pins 2, 9, 11, 12, 13, 14, 18, 20, 21, 23, 27, 29, 30, 31, 32 and 39 to pin 10.

FIGURE 5. Component temperature test circuit.

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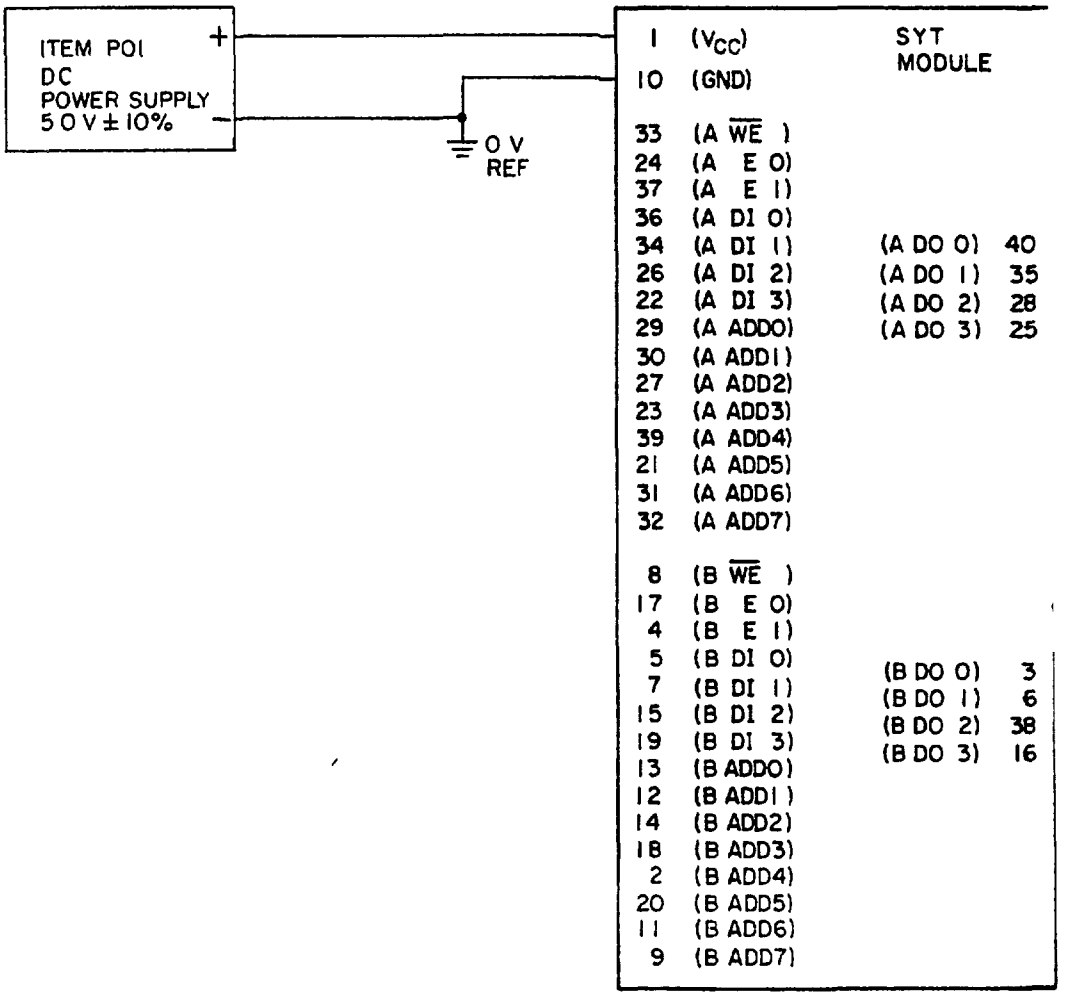


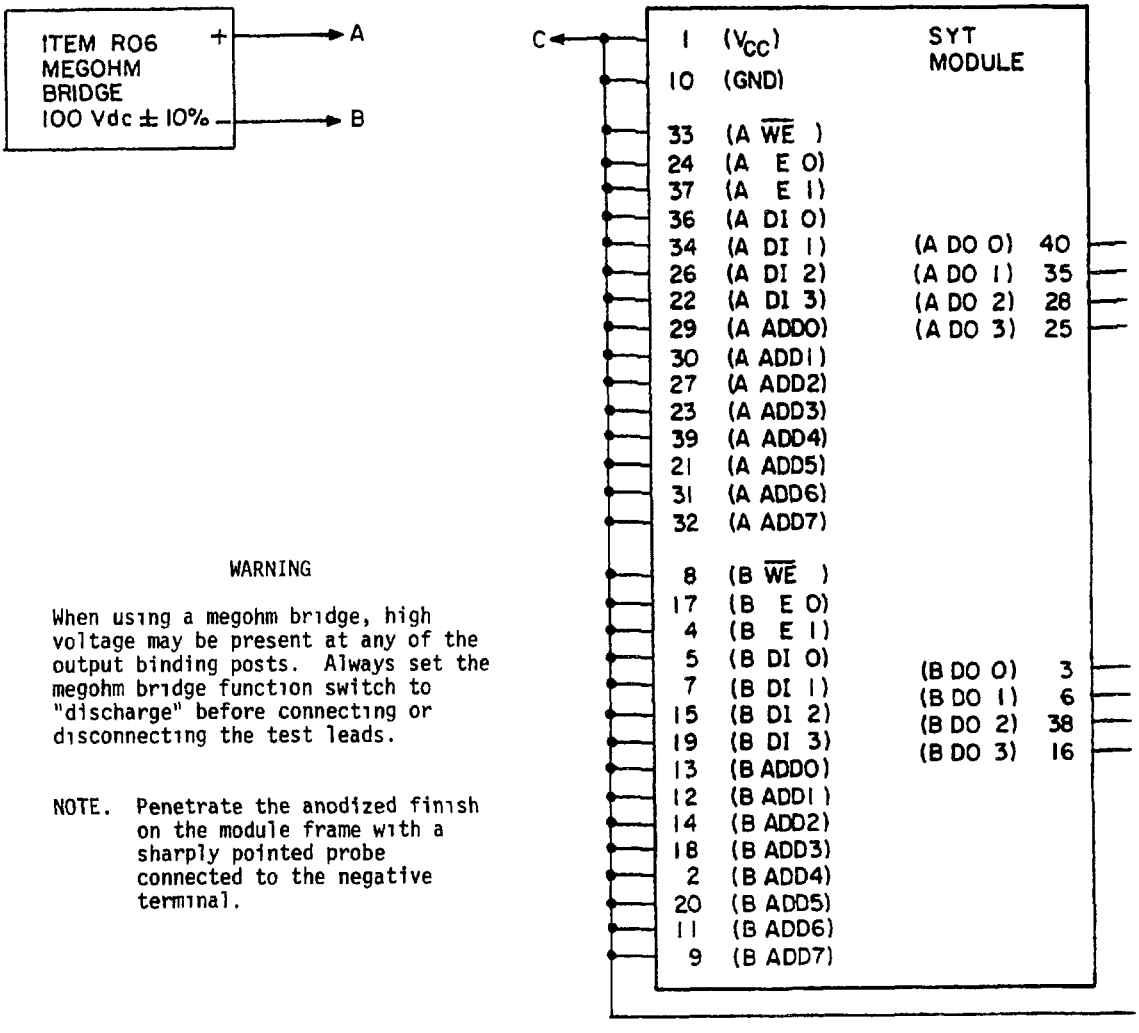
FIGURE 6. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

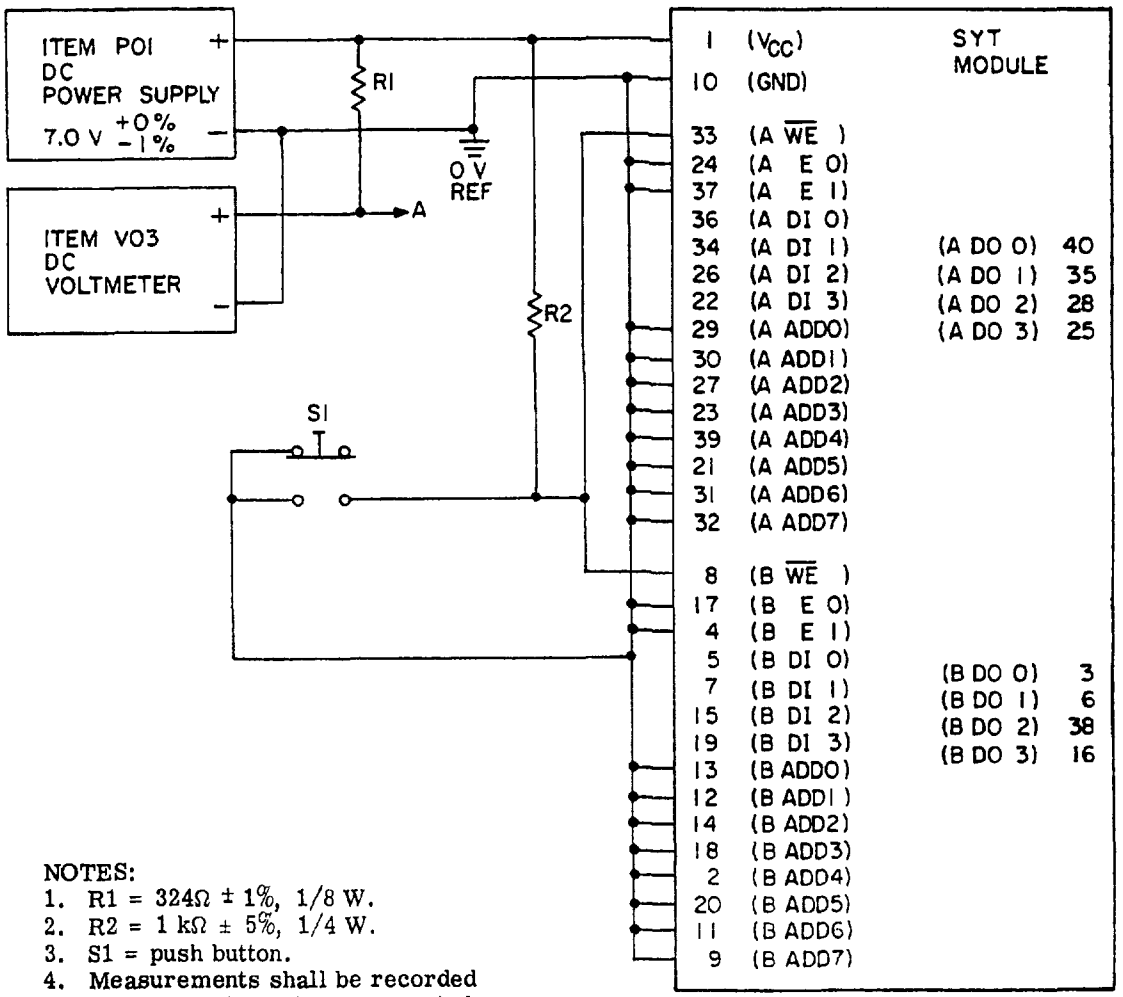
FIGURE 7. Isolation test circuit for table III.

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TABLE IV. V_{out} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	ADO0 V_{out} (low-loaded)	40	0.0	0.8	V	8
B2	ADO1	35				
B3	ADO2	28				
B4	ADO3	25				
B5	BDO0	3				
B6	BDO1	6				
B7	BDO2	38				
B8	BDO3 V_{out} (low-loaded)	16	0.0	0.8	V	8

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NOTES:
1. R1 = 324Ω ± 1%, 1/8 W.
2. R2 = 1 kΩ ± 5%, 1/4 W.
3. S1 = push button.
4. Measurements shall be recorded after point A has been connected to the appropriate pin for a minimum of 5 seconds.
5. S1 shall be pushed and released once before starting the test.

FIGURE 8. V_{out} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	$\overline{AWE}$	Input breakdown current	33	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,34,36,37,39	0.0	4.0	mA	9
C2	AE0		24	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,26,27,29,30, 31,32,33,34,36,37,39		8.0		
C3	AE1		37	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,39		4.0		
C4	ADI0		36	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,37,39		1.0		
C5	ADI1		34	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,36,37,39				
C6	ADI2		26	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,27,29,30, 31,32,33,34,36,37,39				
C7	ADI3		22	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,23,24,26,27,29,30, 31,32,33,34,36,37,39		1.0		
C8	AADD0		29	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,30, 31,32,33,34,36,37,39		4.0		
C9	AADD1	Input breakdown current	30	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 31,32,33,34,36,37,39	0.0	4.0	mA	9

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C10	AADD2	Input breakdown current	27	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,29,30, 31,32,33,34,36,37,39	0.0	4.0	mA	9
C11	AADD3		23	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,24,26,27,29,30, 31,32,33,34,36,37,39				
C12	AADD4		39	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,37				
C13	AADD5		21	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,22,23,24,26,27,29,30, 31,32,33,34,36,37,39				
C14	AADD6		31	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,32,33,34,36,37,39				
C15	AADD7		32	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,33,34,36,37,39				
C16	$\overline{BWE}$		8	2,4,5,7,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,37,39		4.0		
C17	BE0		17	2,4,5,7,8,9,11,12,13,14,15,18, 19,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39		8.0		
C18	BE1	Input breakdown current	4	2,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,37,39	0.0	4.0	mA	9

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TABLE V. Input breakdown current test - Continued.

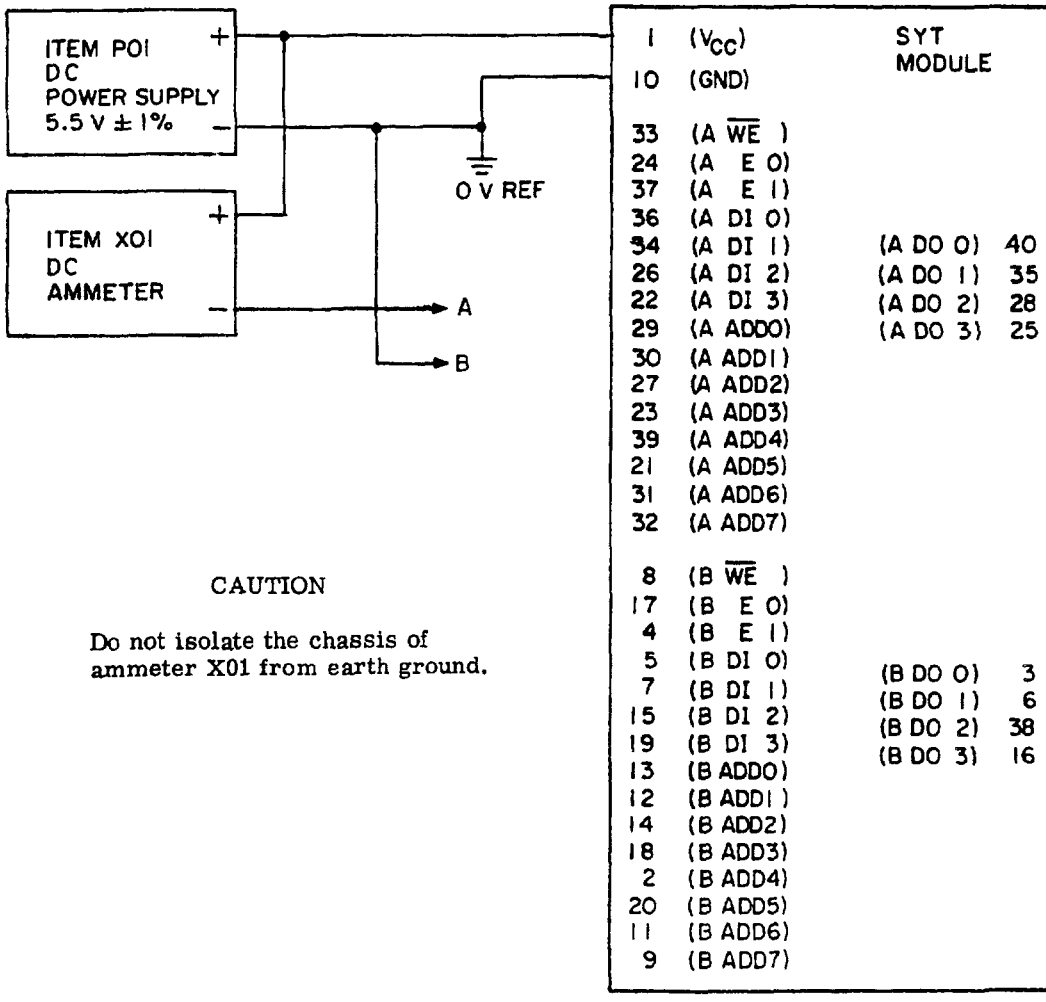
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C19	BDIO	Input breakdown current	5	2,4,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,37,39	0.0	1.0	mA	9
C20	BDI1		7	2,4,5,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,37,39				
C21	BDI2		15	2,4,5,7,8,9,11,12,13,14,17,18, 19,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39				
C22	BDI3		19	2,4,5,7,8,9,11,12,13,14,15,17, 18,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39		1.0		
C23	BADDO		13	2,4,5,7,8,9,11,12,14,15,17,18, 19,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39		4.0		
C24	BADD1		12	2,4,5,7,8,9,11,13,14,15,17,18, 19,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39				
C25	BADD2		14	2,4,5,7,8,9,11,12,13,15,17,18, 19,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39				
C26	BADD3		18	2,4,5,7,8,9,11,12,13,14,15,17, 19,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39				
C27	BADD4	Input breakdown current	2	4,5,7,8,9,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,37,39	0.0	4.0	mA	9

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C28	BADD5	Input breakdown current	20	2,4,5,7,8,9,11,12,13,14,15,17, 18,19,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39	0.0	4.0	mA	9
C29	BADD6	Input breakdown current	11	2,4,5,7,8,9,12,13,14,15,17,18, 19,20,21,22,23,24,26,27,29,30, 31,32,33,34,36,37,39	0 0	4.0	mA	9
C30	BADD7	Input breakdown current	9	2,4,5,7,8,11,12,13,14,15,17, 18,19,20,21,22,23,24,26,27,29, 30,31,32,33,34,36,37,39	0.0	4.0	mA	9

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

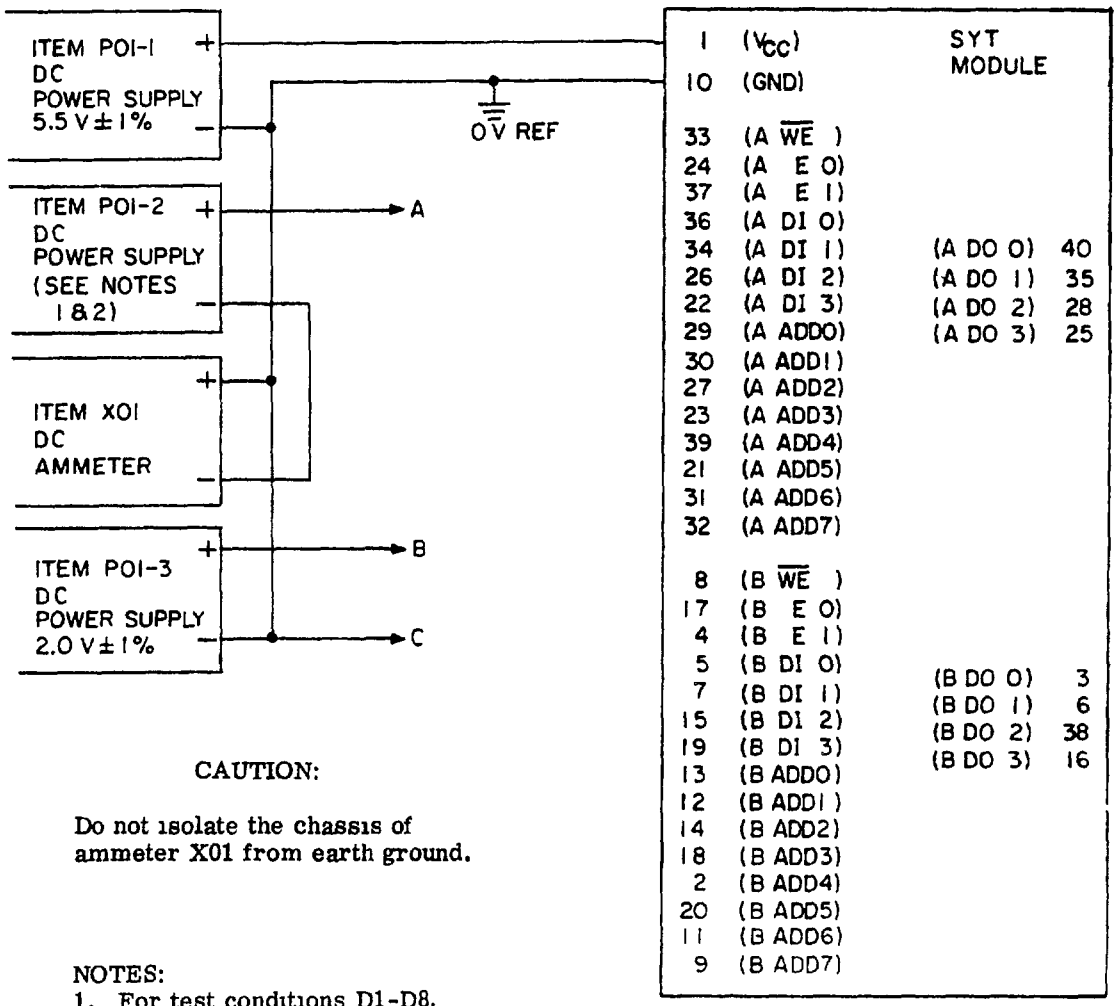
FIGURE 9. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	ADO0	Output leakage current (V_{OL} = 0.4 V)	40	4,17, 24,37	5,7,15, 19,22,26, 34,36	-30	30	µA	10
D2	ADO1		35						
D3	ADO2		28						
D4	ADO3		25						
D5	BDO0		3						
D6	BDO1		6						
D7	BDO2		38						
D8	BDO3	Output leakage current (V_{OL} = 0.4 V)	16		5,7,15, 19,22,26, 34,36	-30			
D9	ADO0	Output leakage current (V_{OH} = 2.4 V)	40		Open	0			
D10	ADO1		35						
D11	ADO2		28						
D12	ADO3		25						
D13	BDO0		3						
D14	BDO1		6						
D15	BDO2		38						
D16	BDO3	Output leakage current (V_{OH} = 2.4 V)	16	4,17, 24,37	Open	0	30	µA	10

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. For test conditions D1-D8, set item P01-2 to 0.4 V ±1%.
2. For test conditions D9-D16, set item P01-2 to 2.4 V ±1%.

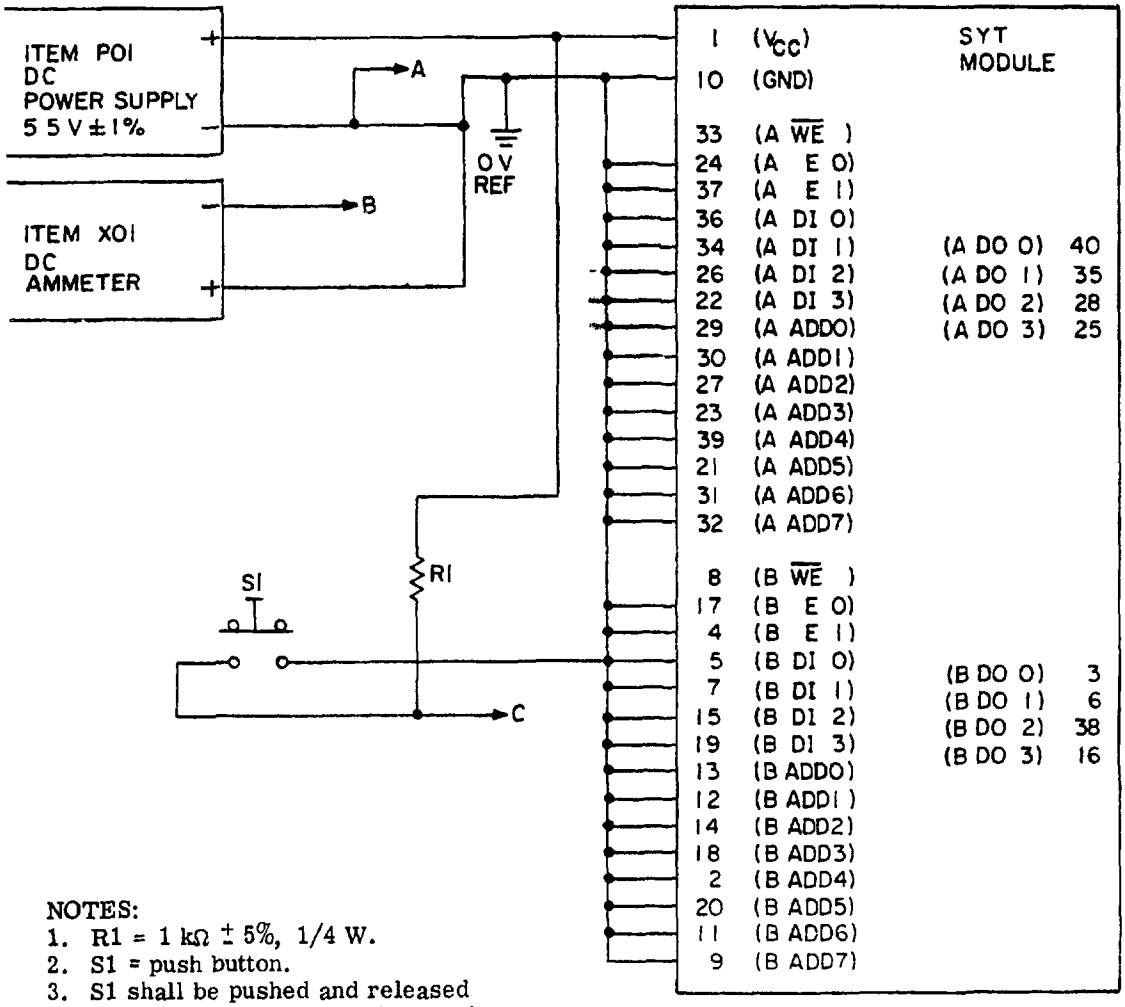
FIGURE 10. Output leakage current test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	AD00	Output short circuit current	8	40	33	-12	-70	mA	11
E2	AD01			35					
E3	AD02			28					
E4	AD03		8	25	33				
E5	BD00		33	3	8				
E6	BD01			6					
E7	BD02			38					
E8	BD03	Output short circuit current	33	16	8	-12	-70	mA	11

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NOTES:
1. R1 = 1 kΩ ± 5%, 1/4 W.
2. S1 = push button.
3. S1 shall be pushed and released before starting test condition E1 and before starting test condition E5.

CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 11. Output short circuit current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

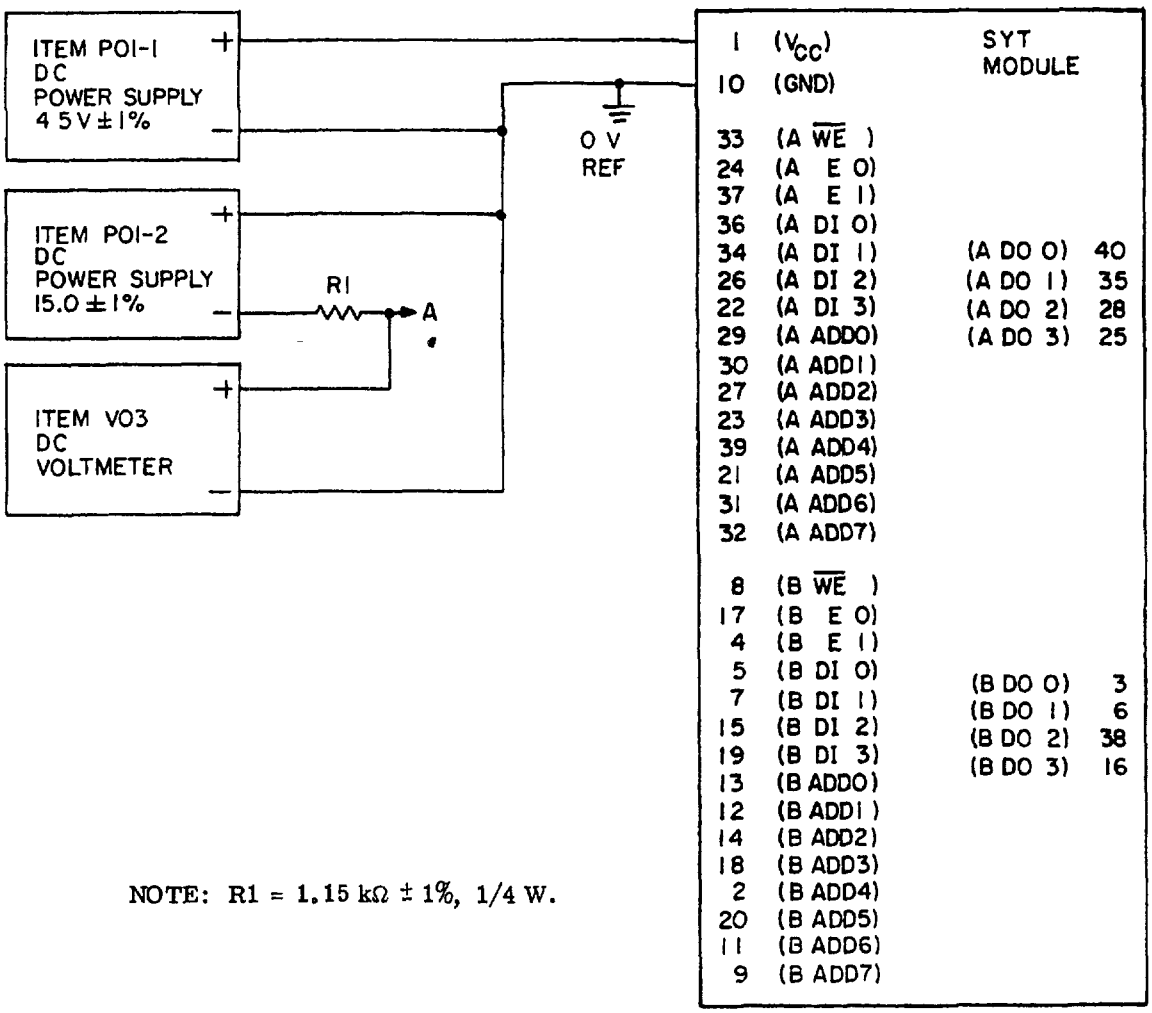
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
F1	$\overline{AWE}$	Input clamp diode voltage	33	-0.2	-1.5	V	12
F2	AE0		24				
F3	AE1		37				
F4	ADI0		36				
F5	ADI1		34				
F6	ADI2		26				
F7	ADI3		22				
F8	AADD0		29				
F9	AADD1		30				
F10	AADD2		27				
F11	AADD3		23				
F12	AADD4		39				
F13	AADD5		21				
F14	AADD6		31				
F15	AADD7		32				
F16	$\overline{BWE}$		8				
F17	BE0		17				
F18	BE1		4				
F19	BDI0	Input clamp diode voltage	5	-0.2	-1.5	V	12

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TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
F20	BDI1	Input clamp diode voltage	7	-0.2	-1.5	V	12
F21	BDI2		15				
F22	BDI3		19				
F23	BADD0		13				
F24	BADD1		12				
F25	BADD2		14				
F26	BADD3		18				
F27	BADD4		2				
F28	BADD5		20				
F29	BADD6		11				
F30	BADD7	Input clamp diode voltage	9	-0.2	-1.5	V	12

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NOTE: R1 = 1.15 kΩ ± 1%, 1/4 W.

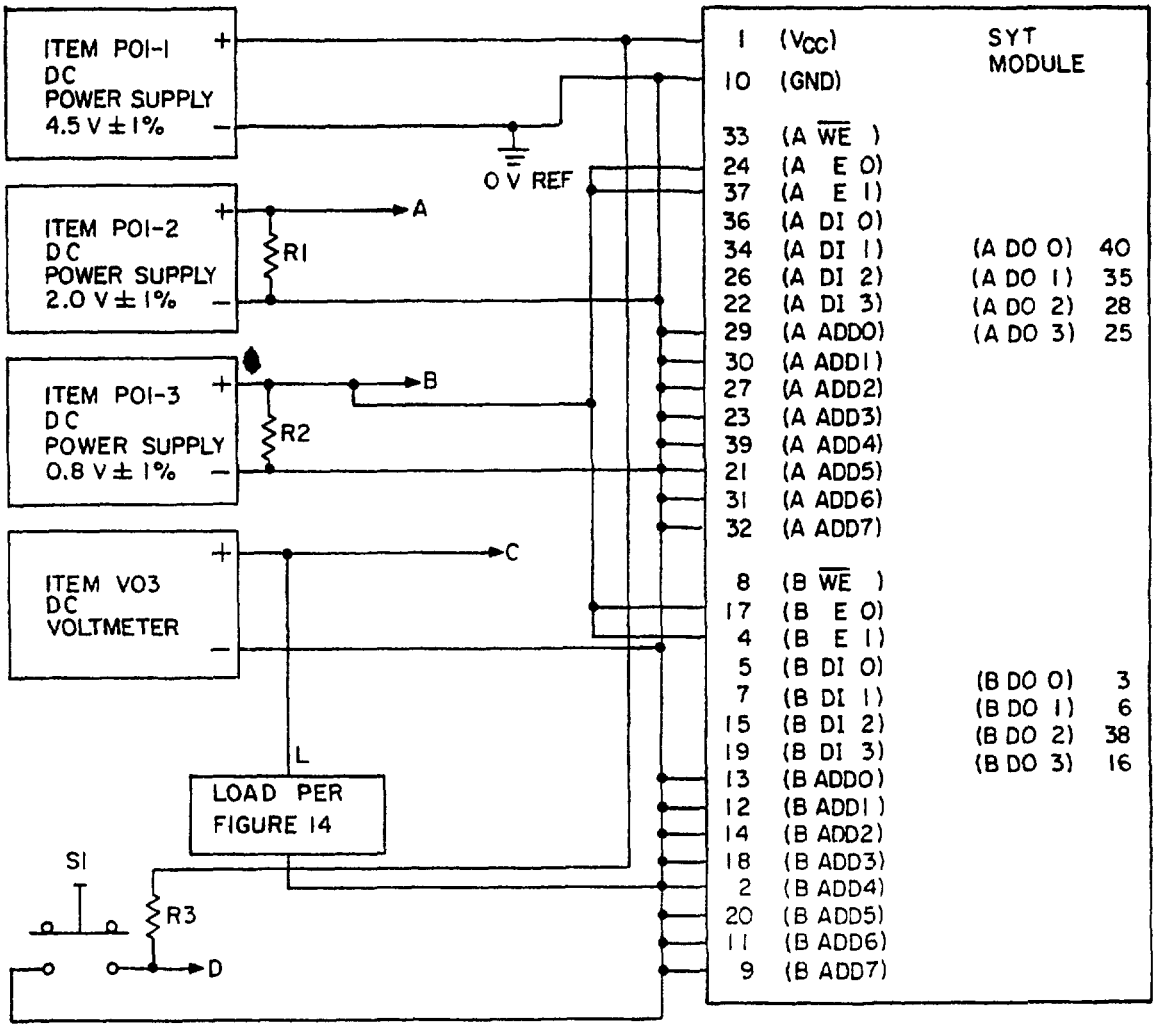
FIGURE 12. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Static output voltage (low-level and high-level) test

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	AD00	Static output voltage, low level	36	Open	40	33	0.0	0.5	V	13
G2	AD01		34		35					
G3	AD02		26		28					
G4	AD03		22		25	33				
G5	BD00		5		3	8				
G6	BD01		7		6					
G7	BD02		15		38					
G8	BD03	Static output voltage, low level	19	Open	16	8	0.0	0.5		
G9	AD00	Static output voltage, high level	Open	36	40	33	2.4	4.5		
G10	AD01			34	35					
G11	AD02			26	28					
G12	AD03			22	25	33				
G13	BD00			5	3	8				
G14	BD01			7	6					
G15	BD02			15	38					
G16	BD03	Static output voltage, high level	Open	19	16	8	2.4	4.5	V	13

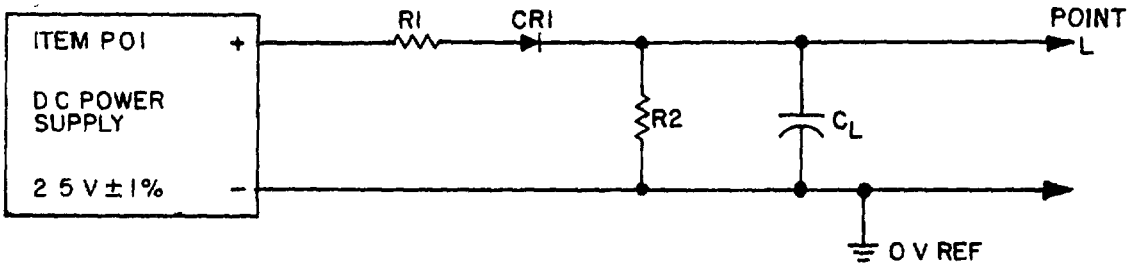
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NOTES:
1. R1 = R2 = 51Ω ± 5%, 1 W.
2. R3 = 1 kΩ ± 5%, 1/4 watt.
3. S1 = push button.
4. S1 shall be pushed and released before each test condition.

FIGURE 13. Static output voltage (low-level and high-level) test circuit for tables IX.

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NOTES.
1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5%, out of point L with point L held at 0.4 V.
3 R2 = 1 21 kΩ ±1%, 1/8 W.
4 Test type S
C_L = 50 ±5 pF (including probe and parasitic capacitance).
Test type Q·
C_L = 150 ±5 pF (including probe and parasitic capacitance)
5. Capacitor C_L is not required for electrical test conditions G1-G16.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 14. SYT logic load circuit.

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TABLE X. Input current (low-level) test.

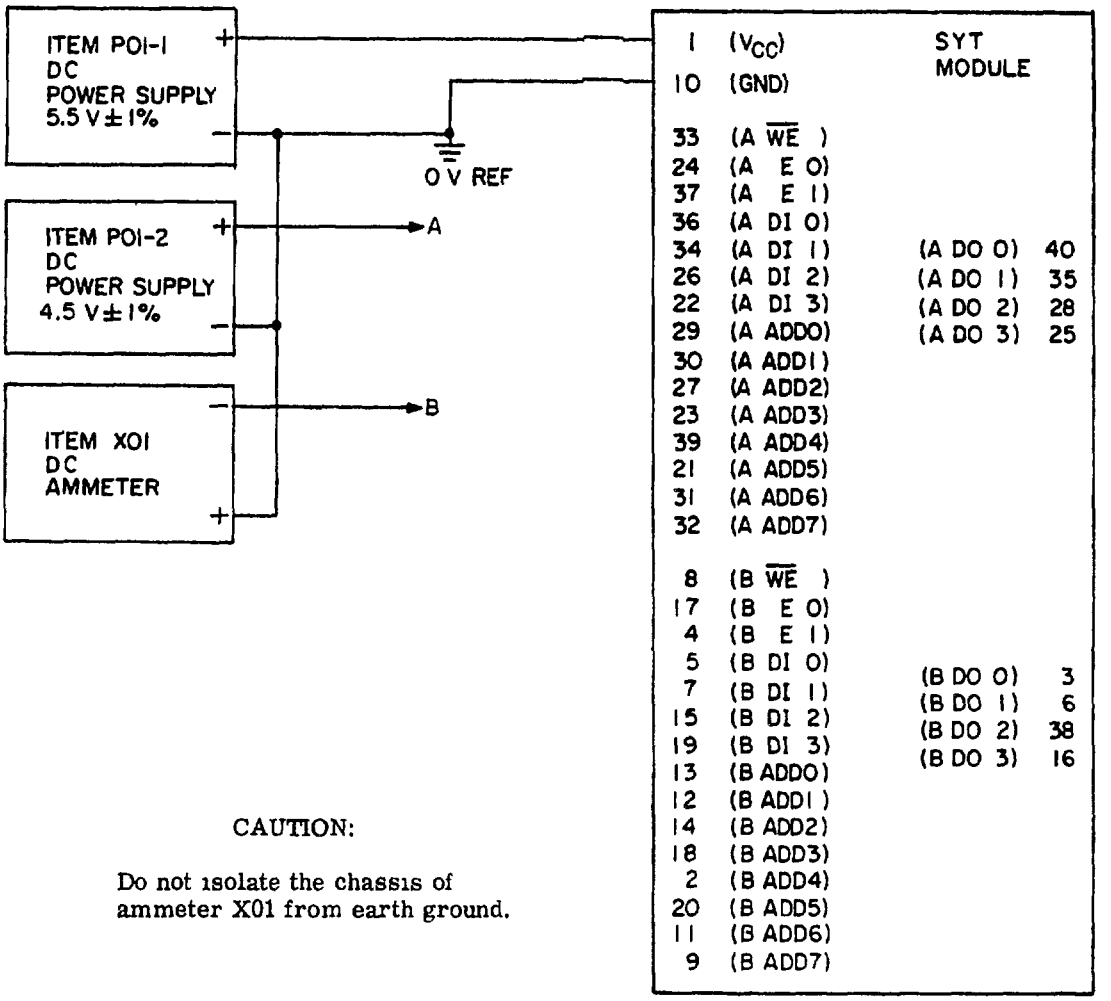
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H1	$\overline{AWE}$	Input current, low level	Open	33	-0.01	-3.5	mA	15
H2	AE1		24	37		-3.5		
H3	ADI0		Open	36		-0.88		
H4	ADI1			34				
H5	ADI2			26				
H6	ADI3		Open	22		-0.88		
H7	AADD0		21,23,27,30, 31,32,39	29		-3.5		
H8	AADD1		21,23,27,29, 31,32,39	30				
H9	AADD2		21,23,29,30, 31,32,39	27				
H10	AADD3		21,27,29,30, 31,32,39	23				
H11	AADD4		21,23,27,29, 30,31,32	39				
H12	AADD5		23,27,29,30, 31,32,39	21				
H13	AADD6		21,23,27,29, 30,32,39	31				
H14	AADD7		21,23,27,29, 31,30,39	32				
H15	$\overline{BWE}$	Input current, low level	Open	8	-0.01	-3.5	mA	15

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TABLE X. Input current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H16	BE1	Input current, low level	17	4	-0.01	-3.5	mA	15
H17	BDI0		Open	5		-0.88		
H18	BDI1			7				
H19	BDI2			15				
H20	BDI3		Open	19		-0.88		
H21	BADD0		2,9,11,12,14, 18,20	13		-3.5		
H22	BADD1		2,9,11,13,14, 18,20	12				
H23	BADD2		2,9,11,12,13, 18,20	14				
H24	BADD3		2,9,11,12,13, 14,20	18				
H25	BADD4		9,11,12,13, 14,18,20	2				
H26	BADD5		2,9,11,12,13, 14,18	20				
H27	BADD6		2,9,12,13,14, 18,20	11				
H28	BADD7		2,11,12,13, 14,18,20	9		-3.5		
H29	AE0		37	24		-7.0		
H30	BE0	Input current, low level	4	17	-0.01	-7.0	mA	15

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15 Input current (low-level) test circuit for table X.

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TABLE XI. Input current (high-level) test.

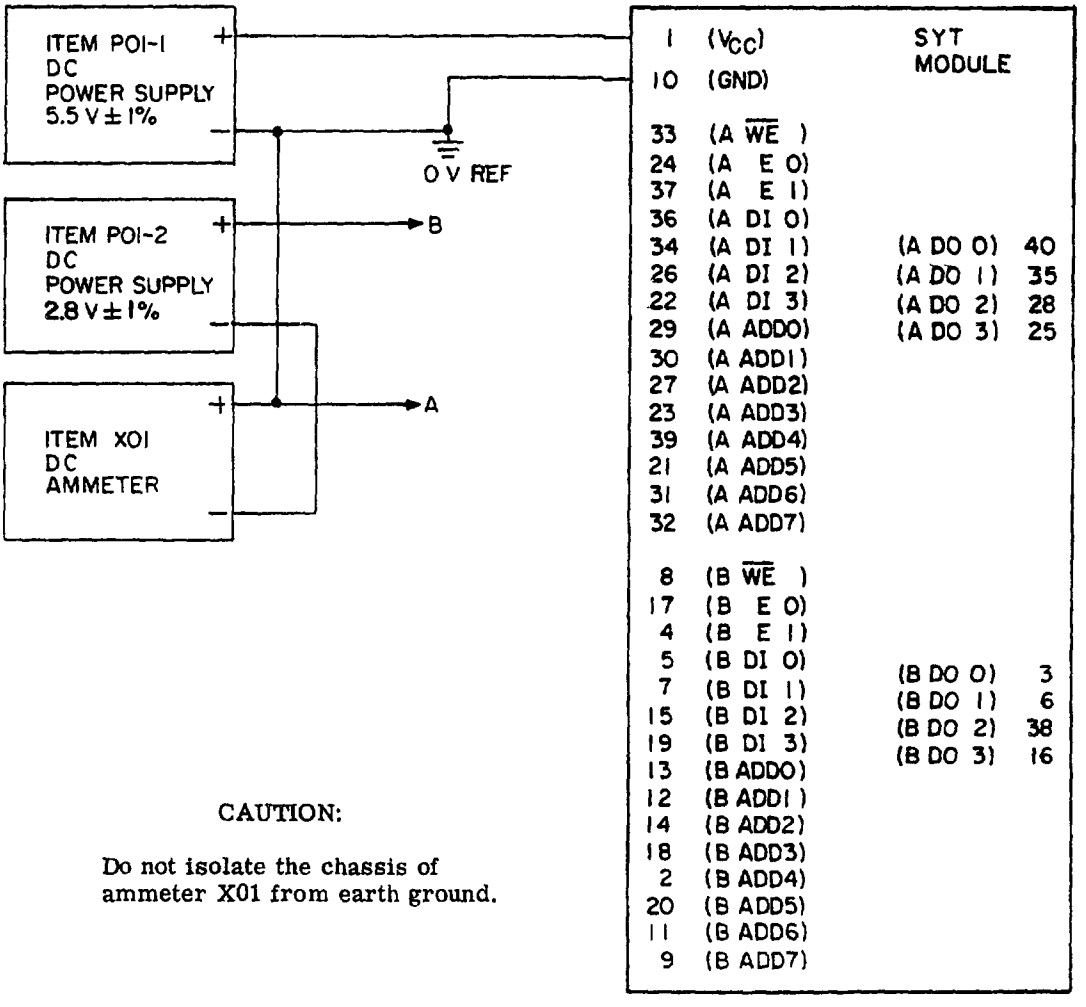
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	$\overline{AWE}$	Input current, high level	Open	33	-1	80	μA	16
J2	AE1		24	37		80		
J3	ADI0		Open	36		20		
J4	ADI1			34				
J5	ADI2			26				
J6	ADI3		Open	22		20		
J7	AADD0		21,23,27,30, 31,32,39	29		80		
J8	AADD1		21,23,27,29, 31,32,39	30				
J9	AADD2		21,23,29,30, 31,32,39	27				
J10	AADD3		21,27,29,30, 31,32,39	23				
J11	AADD4		21,23,27,29, 30,31,32	39				
J12	AADD5		23,27,29,30, 31,32,39	21				
J13	AADD6		21,23,27,29, 30,32,39	31				
J14	AADD7		21,23,27,29, 31,30,39	32				
J15	$\overline{BWE}$	Input current, high level	Open	8	-1	80	μA	16

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J16	BE1	Input current, high level	17	4	-1	80	µA	16
J17	BDI0		Open	5		20		
J18	BDI1			7				
J19	BDI2			15				
J20	BDI3		Open	19		20		
J21	BADD0		2,9,11,12,14, 18,20	13		80		
J22	BADD1		2,9,11,13,14, 18,20	12				
J23	BADD2		2,9,11,12,13, 18,20	14				
J24	BADD3		2,9,11,12,13, 14,20	18				
J25	BADD4		9,11,12,13, 14,18,20	2				
J26	BADD5		2,9,11,12,13, 14,18	20				
J27	BADD6		2,9,12,13,14, 18,20	11				
J28	BADD7		2,11,12,13, 14,18,20	9		80		
J29	AE0		37	24		240		
J30	BE0	Input current, high level	4	17	-1	240	µA	16

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 16. Input current (high-level) test circuit for table XI.

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TABLE XII. Power supply current (low-level outputs, high-level outputs, and high-impedance outputs) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply current, low-level outputs (V_{CC} = 5.5 V, t_{AMB} = 25° ±5°C)	D	C	58	582	mA	17
K2	Power supply current, high-level outputs (V_{CC} = 5.5 V, t_{AMB} = 25° ±5°C)	Open	C, D	58	582		
K3	Power supply current, high-impedance outputs (V_{CC} = 5.5 V, t_{AMB} = 25° ±5°C)	C	Open	58	582		
K4	Power supply current, low-level outputs (V_{CC} = 5.0 V, t_{AMB} = 25° ±5°C)	D	C	54	540		
K5	Power supply current, high-level outputs (V_{CC} = 5.0 V, t_{AMB} = 25° ±5°C)	Open	C, D	54	540		
K6	Power supply current, high-impedance outputs (V_{CC} = 5.0 V, t_{AMB} = 25° ±5°C)	C	Open	54	540		
K7	Power supply current, low-level outputs (V_{CC} = 5.0 V, t_{FIN} = 0°C)	D	C	58	580		
K8	Power supply current, high-level outputs (V_{CC} = 5.0 V, t_{FIN} = 0°C)	Open	C, D	58	580		
K9	Power supply current, high-impedance outputs (V_{CC} = 5.0 V, t_{FIN} = 0°C)	C	Open	58	580		
K10	Power supply current, low level outputs (V_{CC} = 5.0 V, t_{FIN} = 80°C)	D	C	44	440	mA	17

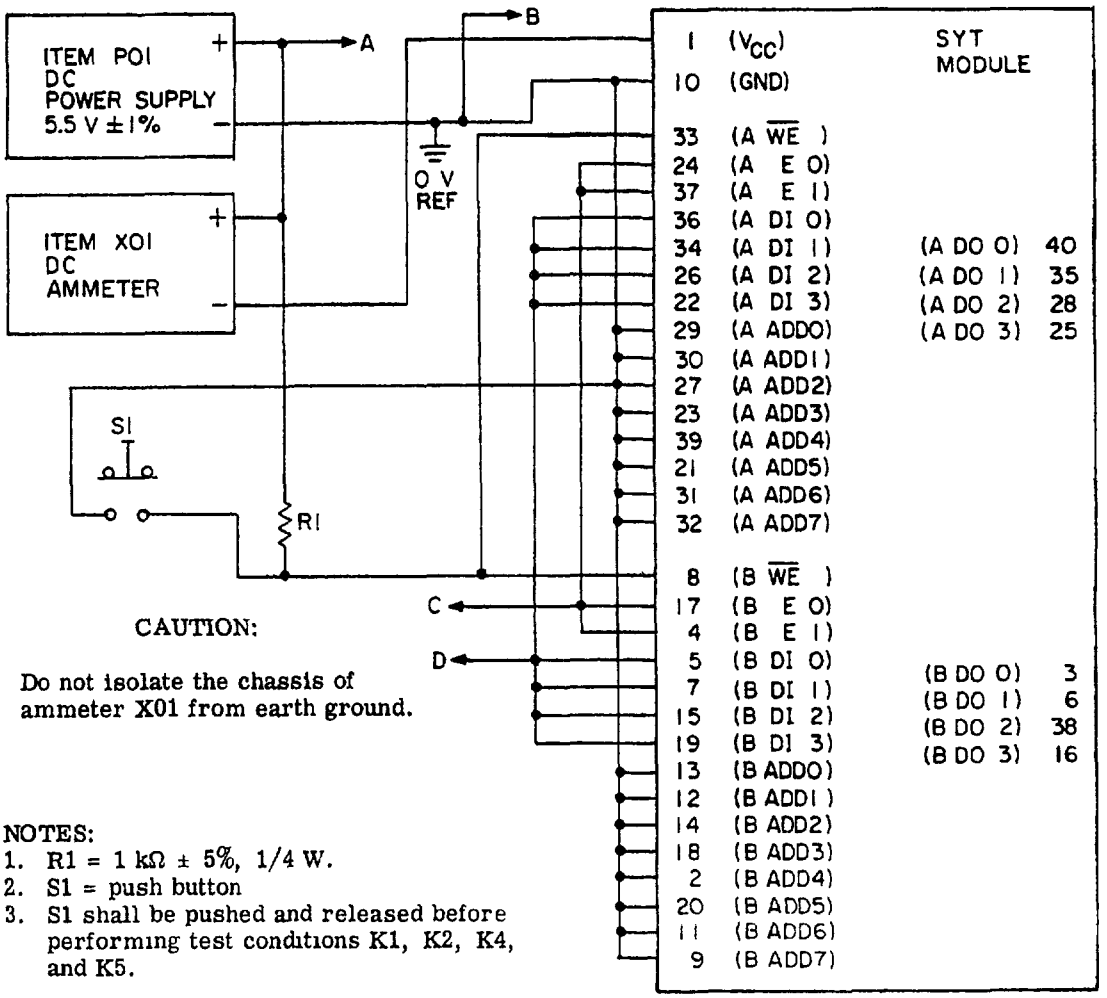
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TABLE XII. Power supply current (low-level outputs, high-level outputs, and high-impedance outputs) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K11	Power supply current, high-level outputs (V_{CC} = 5.0 V, t_{FIN} = 80°C)	Open	C, D	44	440	mA	17
K12	Power supply current, high-impedance outputs (V_{CC} = 5.0 V, t_{FIN} = 80°C)	C	Open	44	440		
K13	Power supply current, low-level outputs (V_{CC} = 5.0 V, t_{FIN} = 60°C)	D	C	50	500		
K14	Power supply current, high-level outputs (V_{CC} = 5.0 V, t_{FIN} = 60°C)	Open	C, D	50	500		
K15	Power supply current, high-impedance outputs (V_{CC} = 5.0 V, t_{FIN} = 60°C)	C	Open	50	500	mA	17

NOTE: Test conditions K1-K15 shall be performed only at the specified temperature.

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NOTES:
1. R1 = 1 kΩ ± 5%, 1/4 W.
2. S1 = push button
3. S1 shall be pushed and released before performing test conditions K1, K2, K4, and K5.

FIGURE 17. Power supply current (low-level outputs, high-level outputs, and high-impedance outputs) test circuit for table XII.

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TABLE XIII. Address threshold (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Address threshold, low level	8, 33	24,37,36, 34,26,22, 29,30,27, 23,39,21, 31,32,17, 4,5,7,15, 19,13,12, 14,18,2, 20,11,9	40	0.0	0.4	V	18
L2				35				
L3				28				
L4				25				
L5				3				
L6				6				
L7				38				
L8	Address threshold, low level	8, 33	24,37,36, 34,26,22, 29,30,27, 23,39,21, 31,32,17, 4,5,7,15, 19,13,12, 14,18,2, 20,11,9	16	0.0	0.4	V	18

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TABLE XIV. Address threshold (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Address threshold, high level	33,36,34, 26,22,29, 30,27,23, 39,21,31, 32,5,7, 15,19,13, 12,14,18, 2,20,11, 9,8	24,37,17,4	40	2.4	4.5	V	18
M2				35				
M3				28				
M4				25				
M5				3				
M6				6				
M7				38				
M8	Address threshold, high level	33,36,34, 26,22,29, 30,27,23, 39,21,31, 32,5,7, 15,19,13, 12,14,18, 2,20,11, 9,8	24,37,17,4	16	2.4	4.5	V	18

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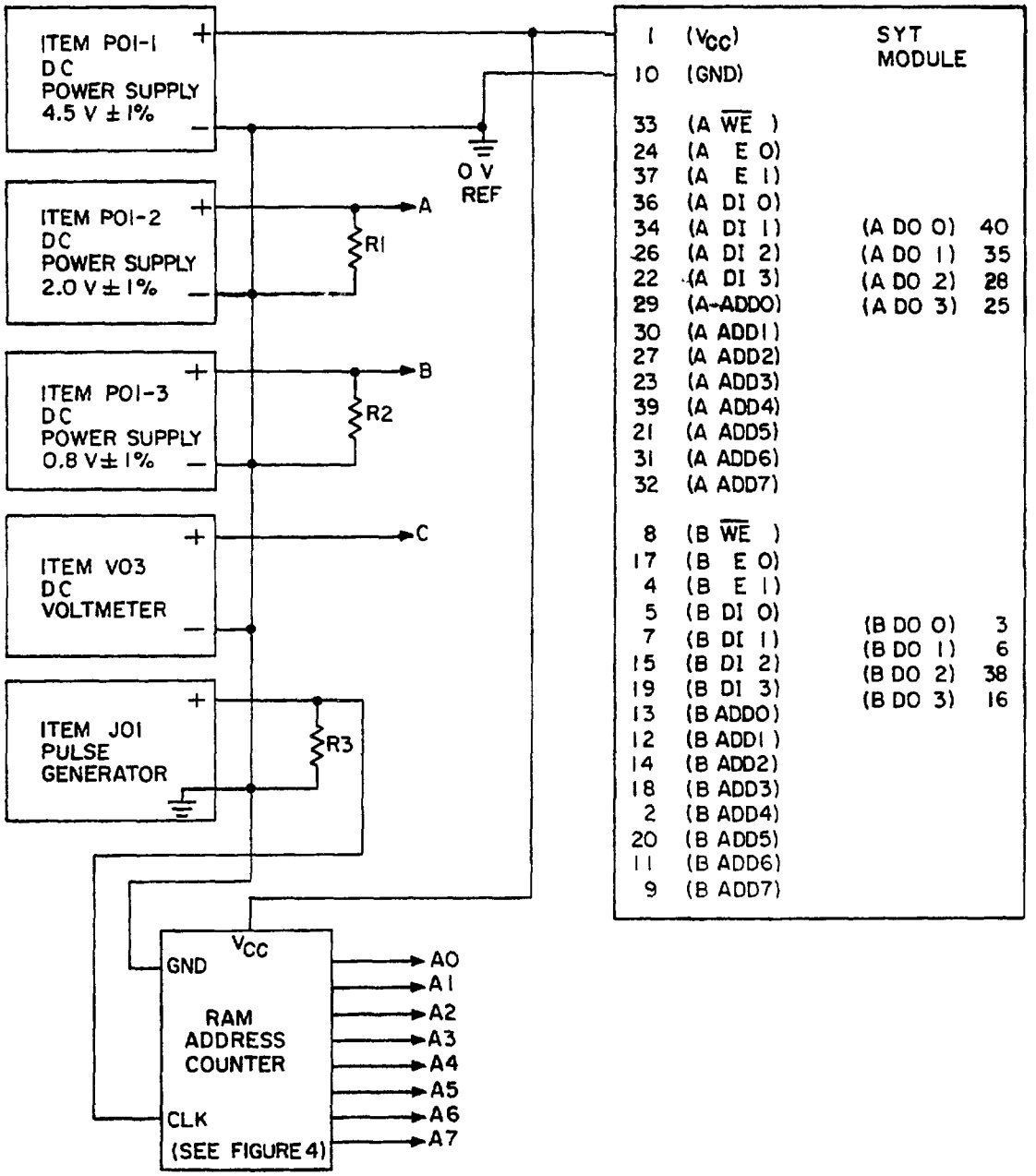


FIGURE 18. Address threshold (low-level and high-level) test circuit for tables XIII and XIV.

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NOTES:
1. R1-R3 = 51Ω ± 5%, 1 W.
2. Item J01 output parameters:

   a. PRP = 1 MHz ± 10%.
   b. PW = 500 ± 50 ns.
   c. PA = 4.0 ± 0.1 V.
   d. $t_{TLH} = t_{THL}$ = 25 ± 20 ns.

3. The following sequence must be completed before performing test conditions L1-L8:

   a. Connect the RAM address counter to address inputs.
   b. Connect pins 33, 24, 37, 36, 34, 26, 22, 8, 17, 4, 5, 7, 15, and 19 to module ground.
   c. Cycle RAM address counter for 1 second. This will load all locations with a high level.
   d. Disconnect pins 8 and 33 from ground and reconnect to V_{CC}.
   e. Disconnect RAM address counter. Load location zero with a low level by connecting pins 29, 30, 27, 23, 39, 21, 31, 32, 13, 12, 14, 18, 2, 20, 11, and 9 to ground and pins 36, 34, 26, 22, 5, 7, 15, and 19 to V_{CC}
   f. Switch pins 8 and 33 to ground for 100 µs minimum, then connect to point A.
   g. Address location zero should be at a low level and all other address locations should be at a high level.
   h. Without disconnecting pins 8 and 33 or interrupting module power, perform test conditions L1-L8 as specified in table XIII.

4. The following sequence must be completed before performing test conditions M1-M8.

   a. Connect the RAM address counter to address inputs.
   b. Connect pins 4, 8, 17, 24, 33, and 37 to ground, and pins 5, 7, 15, 19, 22, 26, 34, and 36 to V_{CC}.
   c. Cycle RAM address counter for 1 second. This will load all locations with a low level.
   d. Disconnect pins 8 and 33 from ground and reconnect to V_{CC}.
   e. Disconnect RAM address counter and address location 256 by connecting pins 36, 34 26, 22, 5, 7, 15, and 19 to ground and pins 29, 30, 27, 23, 39, 21, 31, 32, 13, 12, 14, 18, 2, 20, 11, and 9 to V_{CC}.
   f. Switch pins 8 and 33 to ground for 100 µs minimum, then connect to point A.
   g. Address location 256 should be at a high level and all other locations should be at a low level.
   h. Without disconnecting pins 8 and 33 or interrupting module power, perform test conditions M1-M8 as specified in table XIV.

FIGURE 18. Address threshold (low-level and high-level) test circuit for tables XIII XIV - Continued.

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TABLE XV. Transition time (t_{THL} and t_{TLH}) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N1	ADO0	Transition time (t_{THL}), C_L = 50 pF	Open	40	Open	3	10	ns	19,20
N2	ADO1			35					
N3	ADO2			28					
N4	ADO3			25					
N5	BDO0			3					
N6	BDO1			6					
N7	BDO2			38					
N8	BDO3	Transition time (t_{THL}), C_L = 50 pF		16		3	10		
N9	ADO0	Transition time (t_{THL}), C_L = 150 pF		40		5	20		
N10	ADO1			35					
N11	ADO2			28					
N12	ADO3			25					
N13	BDO0			3					
N14	BDO1			6					
N15	BDO2			38					
N16	BDO3	Transition time (t_{THL}), C_L = 150 pF	Open	16	Open	5	20	ns	19,20

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TABLE XV. Transition time (t_{THL} and t_{TLH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N17	AD00	Transition time (t_{TLH}), C_L = 50 pF	Open	40	Open	4	38	ns	19,20
N18	AD01			35					
N19	AD02			28					
N20	AD03			25					
N21	BD00			3					
N22	BD01			6					
N23	BD02			38					
N24	BD03	Transition time (t_{TLH}), C_L = 50 pF		16		4	38		
N25	AD00	Transition time (t_{TLH}), C_L = 150 pF		40		5	50		
N26	AD01			35					
N27	AD02			28					
N28	AD03			25					
N29	BD00			3					
N30	BD01			6					
N31	BD02			38					
N32	BD03	Transition time (t_{TLH}), C_L = 150 pF	Open	16	Open	5	50	ns	19,20

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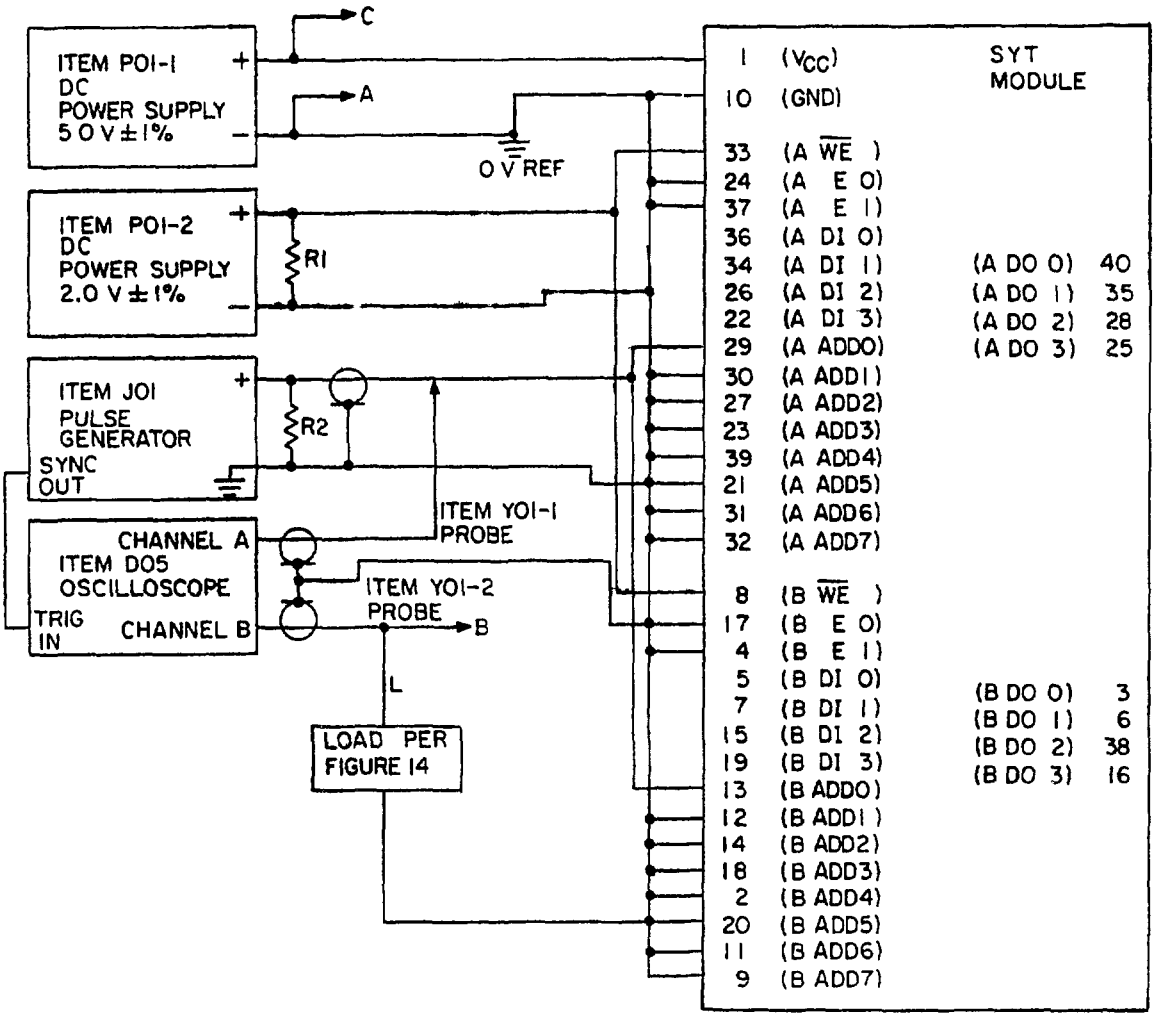


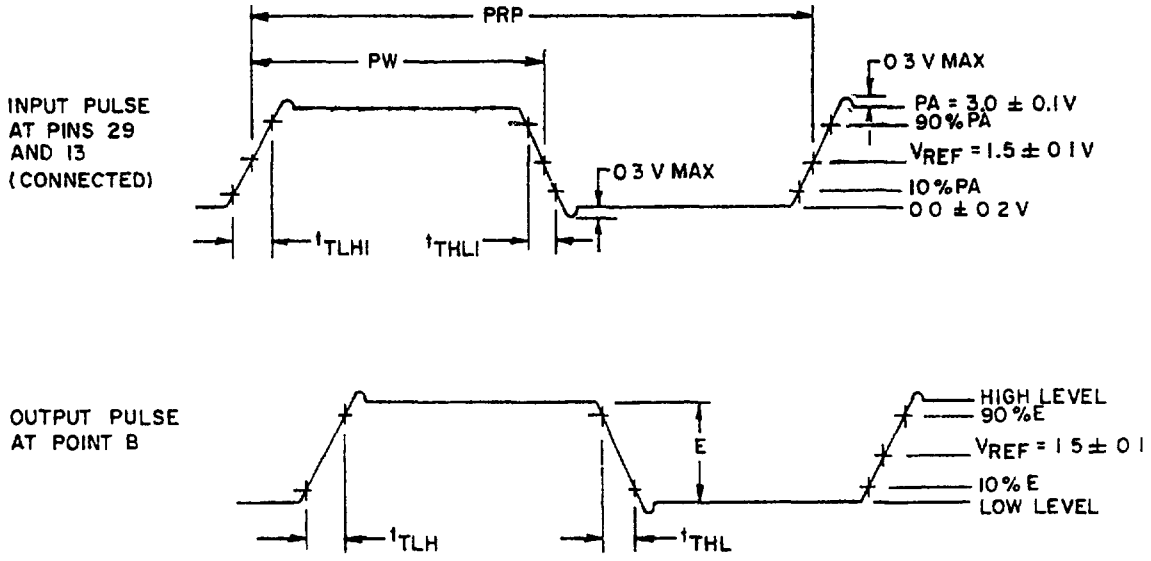
FIGURE 19. Transition time (t_{THL} and t_{TLH}) test circuit for table XV.

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. The following sequence must be completed before performing test conditions N1-N32:
   a. Tie pins 8 and 33 to C (V_{CC}).
   b. Load a low level into all bits of address zero by connecting pins 2, 4, 9, 11, 12, 13, 14, 17, 18, 20, 21, 23, 24, 27, 29, 30, 31, 32, 37, and 39 to A (GND) while pins 5, 7, 15, 19, 22, 26, 34, and 36 are connected to C (V_{CC}).
   c. Switch pins 8 and 33 from C (V_{CC}) to A (GND) for 100 ns minimum, then back to C (V_{CC}).
   d. Switch pins 5, 7, 15, 19, 22, 26, 34, and 36 from C (V_{CC}) to A (GND) and switch pins 13 and 29 from A (GND) to C (V_{CC}).
   e. Switch pins 8 and 33 from C (V_{CC}) to A (GND) for 100 ns minimum, then connect to item P01-2 as shown on figure 19.
   f. Disconnect pins 5, 7, 15, 19, 22, 26, 34, and 36 and switch pins 13 and 29 from V_{CC} to item J01 without disturbing pins 8 and 33 or module power. Perform test conditions N1-N32.

FIGURE 19. Transition time (t_{THL} and t_{TLH}) test circuit for table XV - Continued.

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NOTES.
1. Input pulse parameters.
   t_{TLH1} = THL1 = 6 ± 1 ns.
   PW = 500 ± 50 ns.
   PRP = 1,000 ± 100 ns.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$0.4 V.
3. The lead length of electrical conductors between resistor R2 and input pin; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 20. Transition time (t_{THL} and t_{TLH}) test circuit waveforms for table XV.

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TABLE XVI. Enable recovery time (low-state to high-impedance state t_{ERL} and high-state to high-impedance state t_{ERH}) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	AD00	Enable recovery time (t_{ERL})	4,17,24,37	13,29	40	3	30	ns	21,22
P2	AD01				35				
P3	AD02				28				
P4	AD03				25				
P5	BD00				3				
P6	BD01				6				
P7	BD02				38				
P8	BD03	Enable recovery time (t_{ERL})			16				
P9	AD00	Enable recovery time (t_{ERH})			40				
P10	AD01				35				
P11	AD02				28				
P12	AD03				25				
P13	BD00				3				
P14	BD01				6				
P15	BD02				38				
P16	BD03	Enable recovery time (t_{ERH})	4,17,24,37	13,29	16	3	30	ns	21,22

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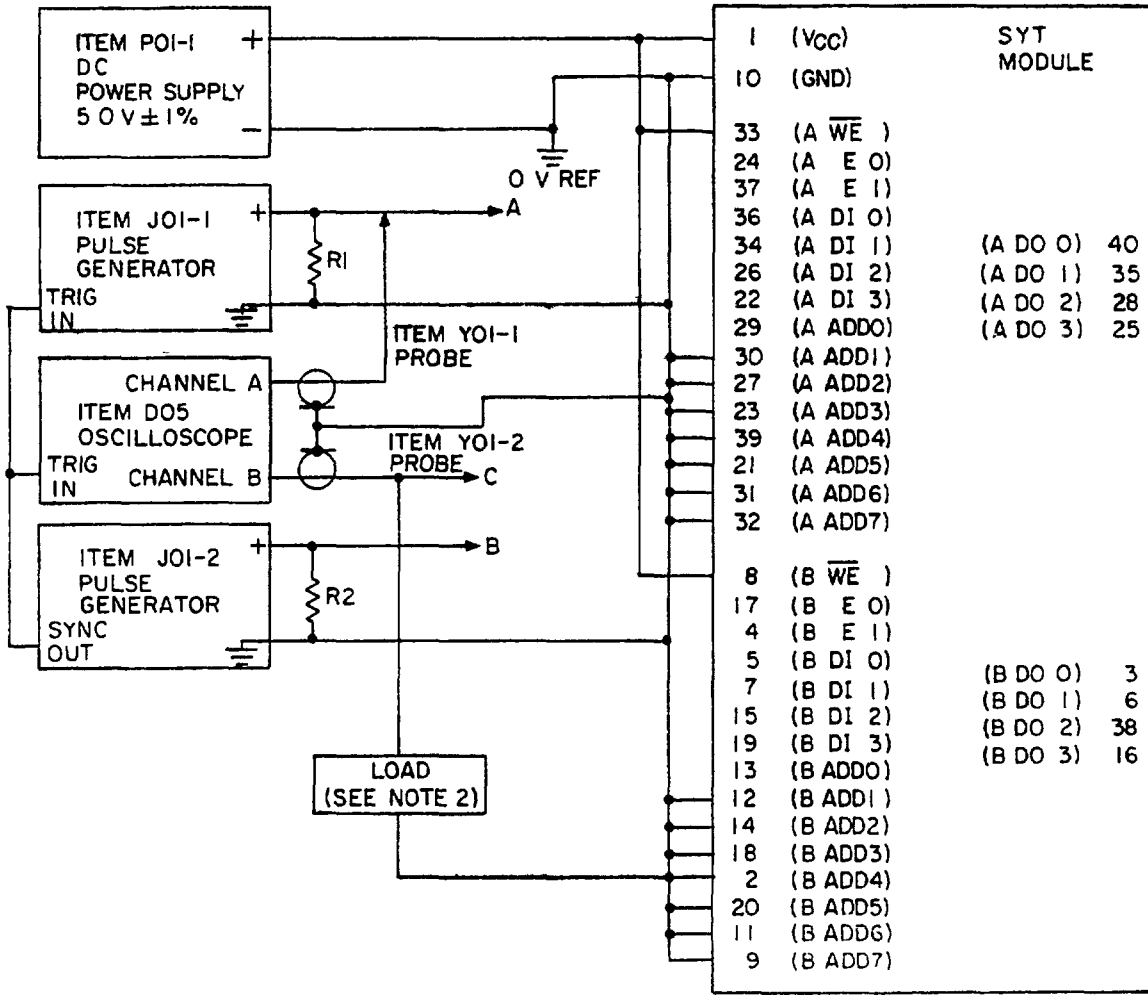


FIGURE 21. Enable recovery time (low-state to high-impedance state t_{ERL} and high-state to high-impedance state t_{ERH}) test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:35Z
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NOTES:
1. R1 = R2 = 51Ω ± 5%, 1 W.
2. Load per figure 14 except C_L < 10 pF.
3 The following sequence must be completed before performing test conditions P1-P16.

   a. Store a high level in address location zero.

      1. Connect pins 29, 30, 27, 23, 39, 21, 31, 32, 13, 12, 14, 18, 2, 20, 11, 9, 36, 34, 26, 22, 5, 7, 15, 19, 24, 37, 17, and 4 to pin 10.
      2. After step 1 connect pins 8 and 33 to pin 10 for 100 ns minimum, then connect pins 8 and 33 to pin 1. Proceed to step 3.b. without disconnecting.

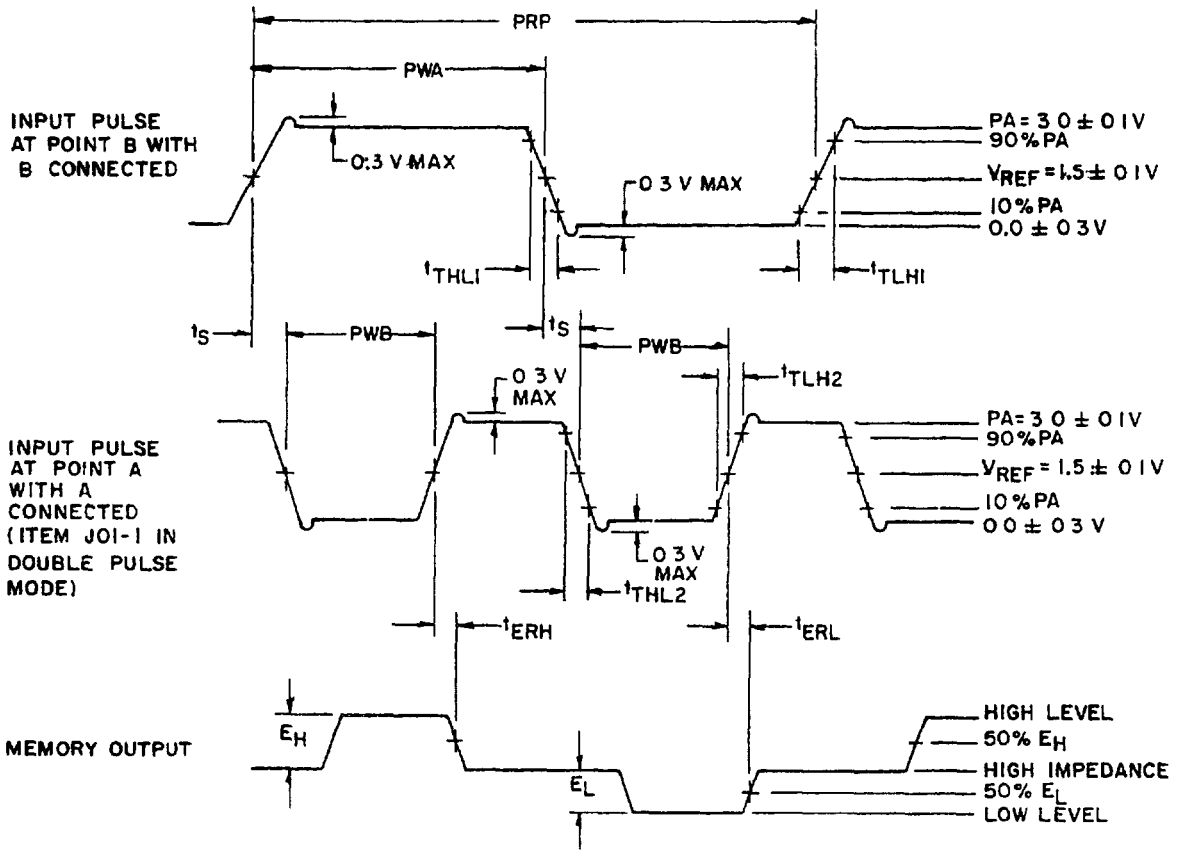
   b. Store a low level in address location one.

      1. Switch pins 5, 7, 13, 15, 19, 22, 26, 29, 34, and 36 from pin 10 to pin 1.
      2. After step 1 switch pins 8 and 33 from pin 1 to pin 10 for 100 ns minimum, then reconnect to pin 1.

   c. Disconnect pins 4, 17, 24, and 37 and finish connecting per figure 21. Verify that address location zero is a high level and address location one is a low level, then perform test conditions P1-P16 without disturbing pins 8 and 33 or module power.

FIGURE 21. Enable recovery time (low-state to high-impedance state t_{ERL} and high-state to high-impedance state t_{ERH}) test circuit for table XVI - Continued.

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NOTE: Input pulse parameters
PWA = 500 ±50 ns
PWB = 100 ±10 ns
PRP = 1,000 ±100 ns.
t_S = 200 ±20 ns
$t_{TLH1} = t_{THL1}$ = 10 ±2 ns
$t_{TLH2} = t_{THL2}$ = 10 ±2 ns

FIGURE 22. Enable recovery time (low-state to high-impedance state t_{ERL} and high-state to high-impedance state t_{ERH}) test circuit waveforms for table XVI.

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TABLE XVII. Propagation delay time (write enable low to high-impedance state, t_{PLH}) test.

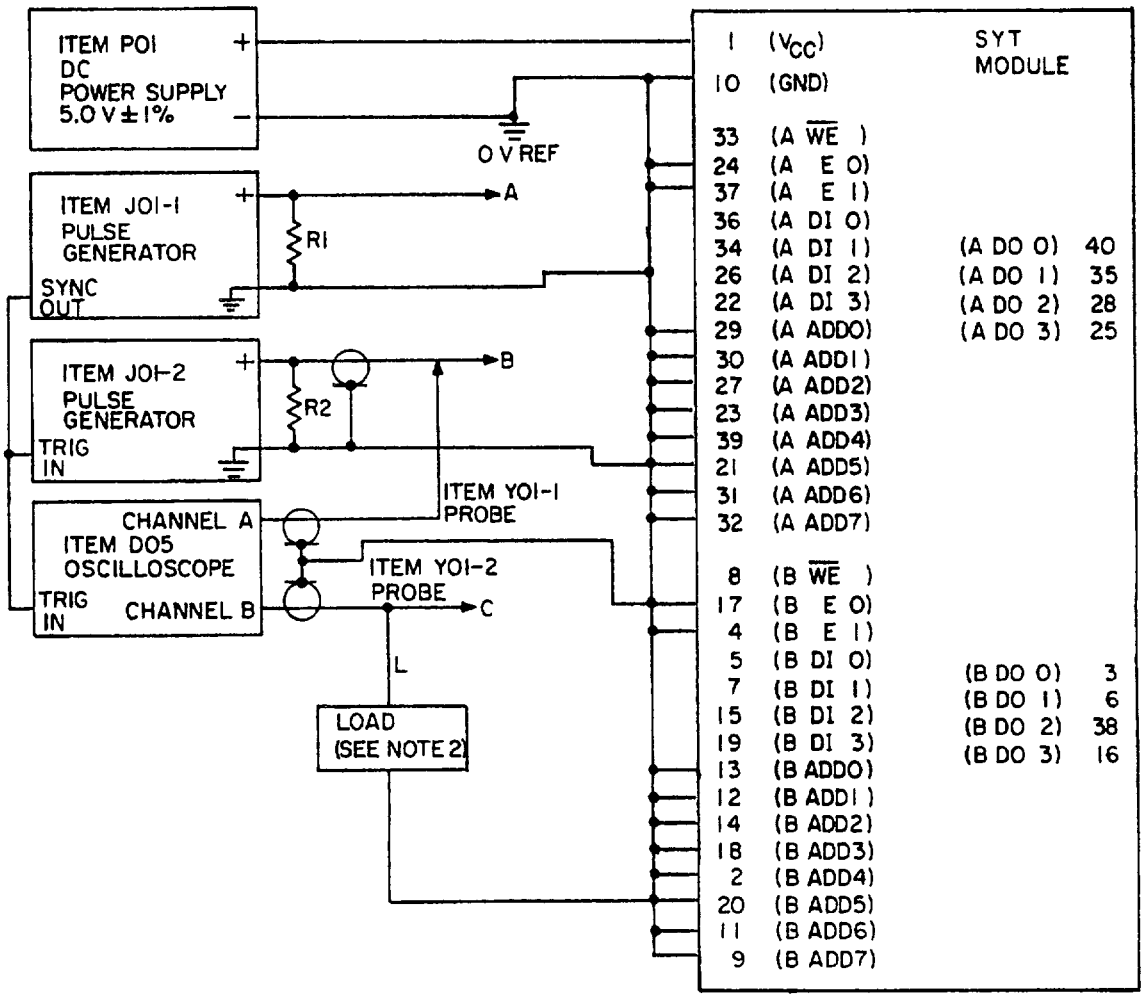
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	ADO0	Propagation delay time, write enable low to high-impedance state, t_{PLHLH}	36	33	40	3	30	ns	23,24
Q2	ADO1		34		35				
Q3	ADO2		26		28				
Q4	ADO3		22	33	25				
Q5	BDO0		5	8	3				
Q6	BDO1		7		6				
Q7	BDO2		15		38				
Q8	BDO3	Propagation delay time, write enable low to high-impedance state, t_{PLHLH}	19	8	16				
Q9	ADO0	Propagation delay time, write enable low to high-impedance state, t_{PLHHL}	36	33	40				
Q10	ADO1		34		35				
Q11	ADO2		26		28				
Q12	ADO3		22	33	25				
Q13	BDO0		5	8	3				
Q14	BDO1	Propagation delay time, write enable low to high-impedance state, t_{PLHHL}	7	8	6	3	30	ns	23,24

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TABLE XVII. Propagation delay time (write enable low to high-impedance state, t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q15	BDO2 Propagation delay time, write enable low to high-impedance state, t_{PLHHL}	15	8	38	3	30	ns	23,24
Q16	BDO3 Propagation delay time, write enable low to high-impedance state, t_{PLHHL}	19	8	16	3	30	ns	23,24

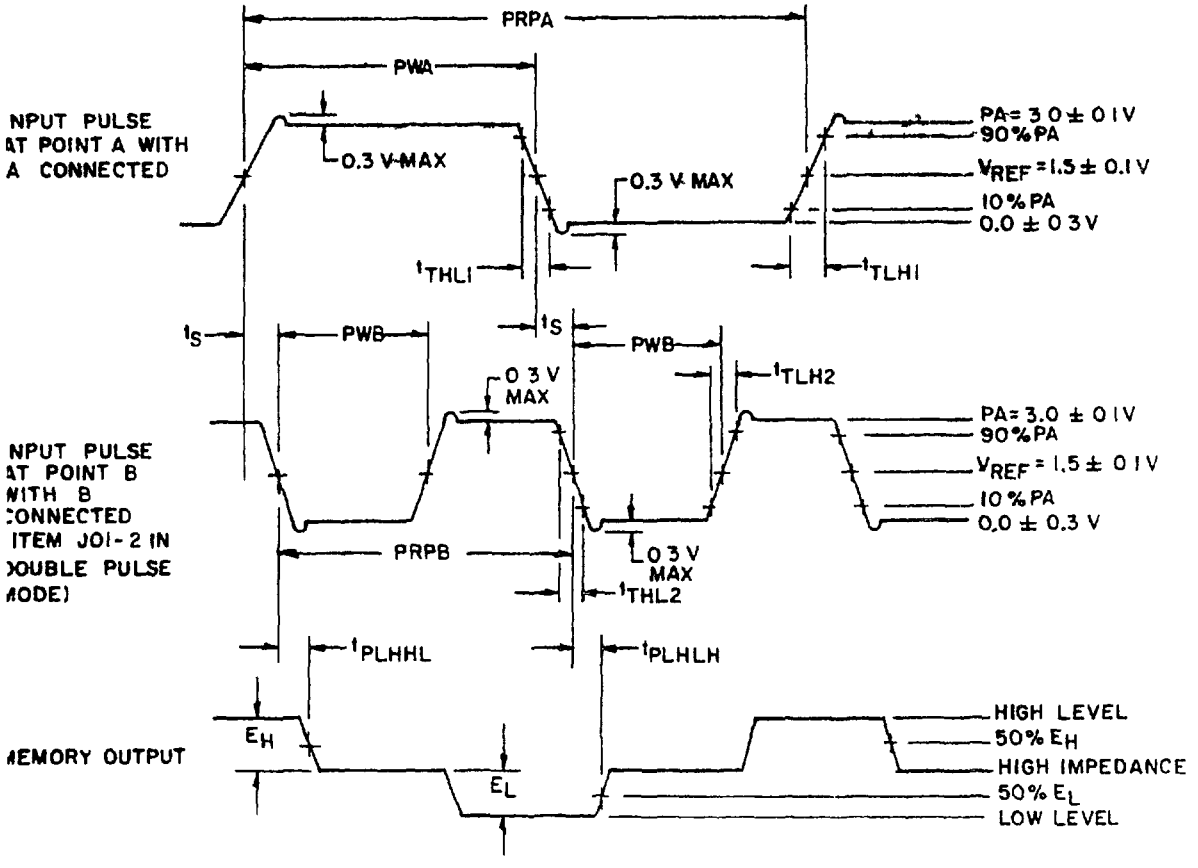
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:35Z
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NOTES.
1. R1 = R2 = 51Ω ± 5%, 1 W.
2. Load per figure 14 except C_L <10 pF.

FIGURE 23. Propagation delay time (write enable low to high-impedance state, t_{PLH}) test circuit for table XVII.

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NOTE: Input pulse parameters:
PWA = 500 ± 50 ns.
PWB = 100 ± 10 ns.
PRPA = 1,000 ± 100 ns.
t_S = 200 ± 20 ns.
$t_{TLH} = t_{THL}$ = 10 ± 2 ns.
PRPB = 500 ± 50 ns.

FIGURE 24. Propagation delay time (write enable low to high-impedance state, t_{PLH}) test circuit waveforms for table XVII.

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TABLE XVIII. Walking-one-walking-zero test pattern.

Test condition	Parameter		Initial limits 25° ±5°C Ambient		Test figure
			Min	Max	
R1	Walking-one-walking-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	See notes 1, 2, 3, 4, and 6	See notes		25,26, 27,28
R2	Walking-one-walking-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	See notes 1, 2, 3, 4, and 7			
R3	Walking-one-walking-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	See notes 1, 2, 3, 5, and 6			
R4	Walking-one-walking-zero test pattern (C_L = 150 pF, V_{cc} = 5.5 V)	See notes 1, 2, 3, 5, and 7	See notes		25,26, 27,28

NOTES

1. Walking-one-walking-zero pattern testing. The objective of walking-one-walking-zero pattern testing (writing a one in a field of zeros and a zero in a field of ones) is to show the independence of each bit in both the zero and one state with respect to all other bits which are in the complement state. This pattern also verifies minimum write pulse width (t_{WP}), address setup time (t_{AS}), data setup time (t_{DS}), enable setup time (t_{ES}), data hold time (t_{DH}), enable hold time (t_{EH}), address access time (t_{AA}), and enable access time (t_{EA}).

2. A program that fulfills the walking-one test pattern would be as follows:

   a. Write low levels at all memory B and A address locations.
   b. Read low levels at all memory A and B address locations.
   c. Write a high level in memory output AD00 at address location 0.
   d. Read low levels in all other memory B and A bit locations.

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TABLE XVIII. Walking-one-walking-zero test pattern - Continued.

NOTES: -Continued

2. -Continued
   e. Read a high level in memory output AD00 at address location 0.
   f. Write a low level in memory output AD00 at address location 0.
   g. Write a high level in memory output AD01 at address location 0.
   h. Read low levels in all other memory B and A bit locations.
   i. Read a high level in memory output AD01 at address location 0.
   j. Write a low level in memory output AD01 at address location 0.
   k. Write a high level in memory output AD02 at address location 0.
   l. Read low levels in all other memory B and A bit locations.
   m. Read a high level in memory output AD02 at address location 0.
   n. Write a low level in memory output AD02 at address location 0.
   o. Write a high level in memory output AD03 at address location 0.
   p. Read low levels in all other memory B and A bit locations
   q. Read a high level in memory output AD03 at address location 0.
   r. Write a low level in memory output AD03 at address location 0.
   s. Write a high level in memory output BD00 at address location 0.
   t. Read low levels in all other memory A and B bit locations.
   u. Read a high level in memory output BD00 at address location 0.
   v. Write a low level in memory output BD00 at address location 0.
   w. Write a high level in memory output BD01 at address location 0.
   x. Read low levels in all other memory A and B bit locations
   y. Read a high level in memory output BD01 at address location 0.
   z. Write a low level in memory output BD01 at address location 0.
   aa. Write a high level in memory output of BD02 at address location 0.
   bb. Read low levels in all other memory A and B bit locations.
   cc. Read a high level in memory output BD02 at address location 0.
   dd. Write a low level in memory output of BD02 at address location 0.
   ee. Write a high level in memory output BD03 at address location 0.
   ff. Read low levels in all other memory A and B bit locations.
   gg. Read a high level in memory output BD03 at address location 0.
   hh. Write a low level in memory output BD03 at address location 0.
   ii. Continue the pattern until a high level has been walked through every memory B and A bit location.

3 A program that fulfills the walking-zero test pattern would be as follows:

   a Write high levels at all memory B and A address locations.
   b. Read high levels in all memory B and A address locations.
   c. Write a low level in memory output AD00 at address location 0.
   d. Read high levels in all other memory B and A bit locations.
   e. Read a low level in memory output AD00 at address location 0.
   f. Write a high level in memory output AD00 at address location 0.
   g. Write a low level in memory output AD01 at address location 0.
   h. Read high levels in all other memory B and A bit locations.
   i. Read a low level in memory AD01 at address location 0.
   j. Write a high level in memory output AD01 at address location 0.
   k. Write a low level in memory output AD02 at address location 0.
   l. Read high levels in all other memory B and A bit locations.
   m. Read a low level in memory output AD02 at address location 0.
   n. Write a high level in memory output AD02 at address location 0.
   o. Write a low level in memory output AD03 at address location 0.
   p. Read high levels in all other memory B and A bit locations.
   q. Read a low level in memory output AD03 at address location 0.
   r. Write a high level in memory output BD00 at address location 0.

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TABLE XVIII. Walking-one-walking-zero test pattern - Continued.

NOTES -Continued

3. -Continued
   - s. Write a low level in memory output BDO0 at address location 0.
   - t. Read high levels in all other memory A and B bit locations.
   - u. Read a low level in memory output BDO0 at address location 0.
   - v. Write a high level in memory output BDO0 at address location 0.
   - w. Write a low level in memory output BDO1 at address location 0.
   - x. Read high levels in all other memory A and B bit locations.
   - y. Read a low level in memory output BDO1 at address location 0.
   - z. Write a high level in memory output BDO1 at address location 0.
   - aa. Write a low level in memory output BDO2 at address location 0.
   - bb. Read high levels in all other memory A and B bit locations.
   - cc. Read a low level memory output BDO2 at address location 0.
   - dd. Write a high level in memory output BDO2 at address location 0.
   - ee. Write a low level in memory output BDO3 at address location 0.
   - ff. Read high levels in all other memory A and B bit locations.
   - gg. Read a low level in memory BDO3 at address location 0.
   - hh. Write a high level in memory output BDO3 at address location 0.
   - ii. Continue the pattern until a low level has been walked through every memory B and A bit location.

4. For test conditions R1 and R2, load output pins with the load specified on figure 14, note 4, test type S.

5. For test conditions R3 and R4, load output pins with the load specified on figure 14, note 4, test type Q.

6. For test conditions R1 and R3, set V_{CC} = 4.5 V $\pm$1%.

7. For test conditions R2 and R4, set V_{CC} = 5.5 V $\pm$1%.

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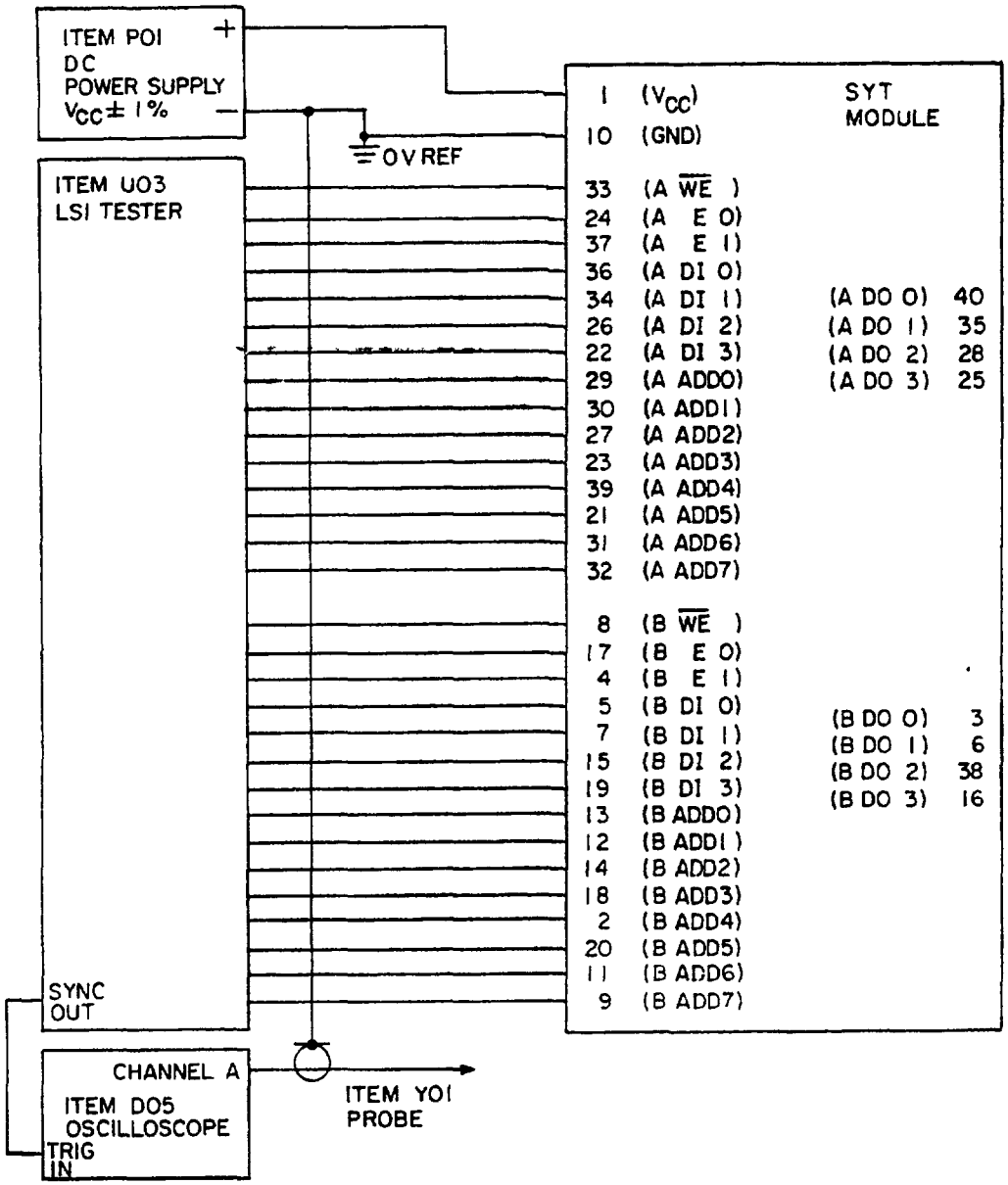


FIGURE 25 Walking-one-walking-zero test pattern circuit for table XVIII.

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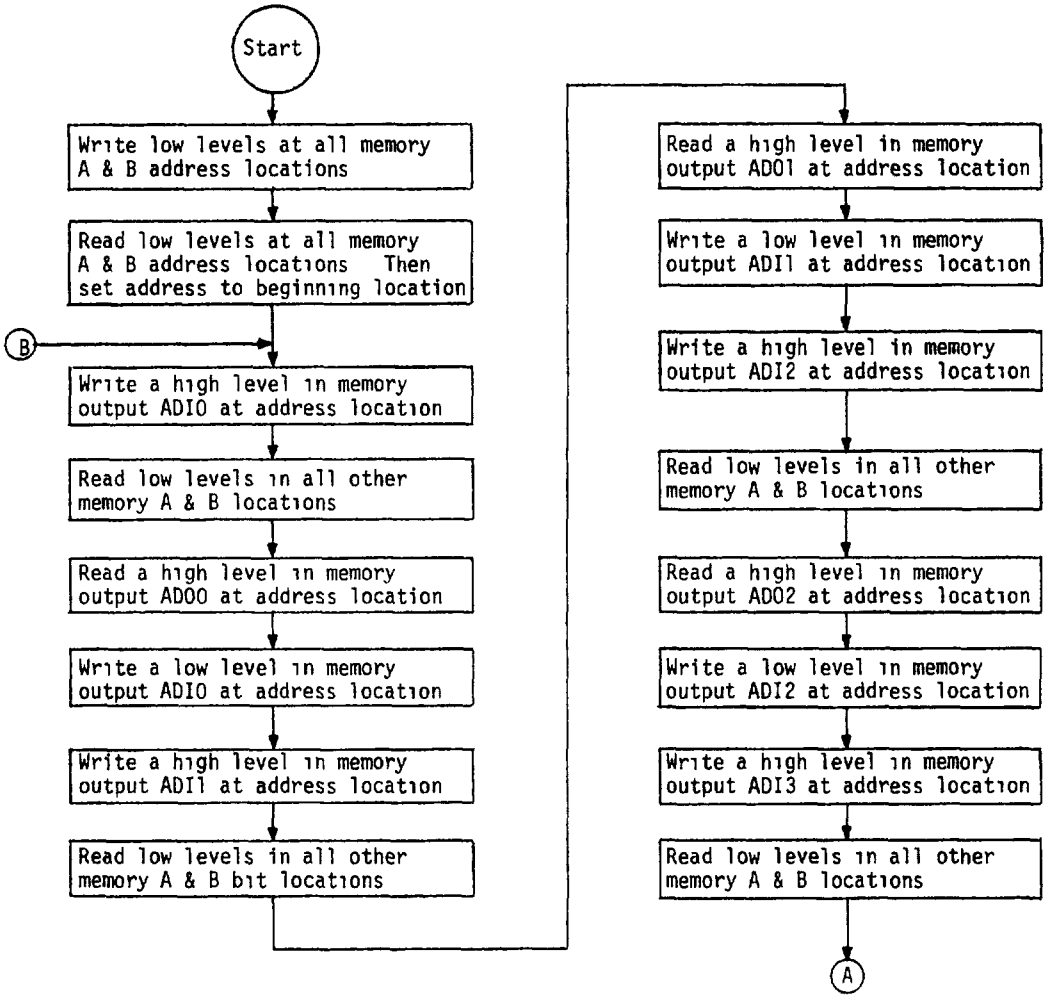


FIGURE 26. Walking-one test pattern flow chart

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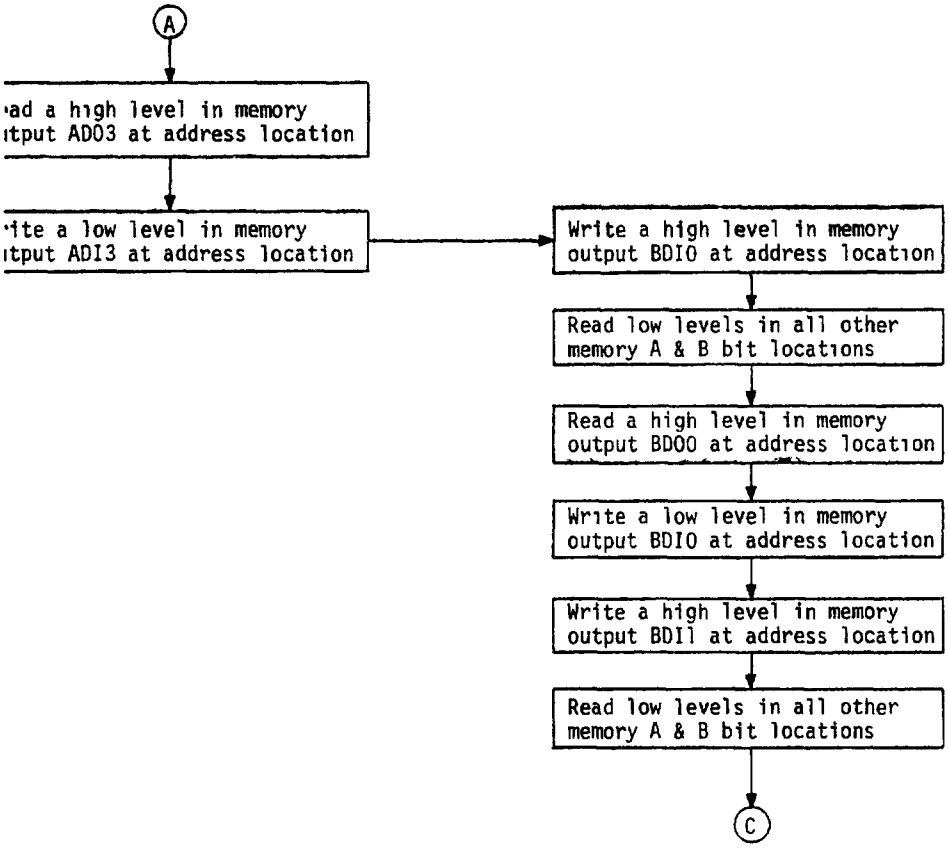


FIGURE 26. Walking-one test pattern flow chart - Continued.

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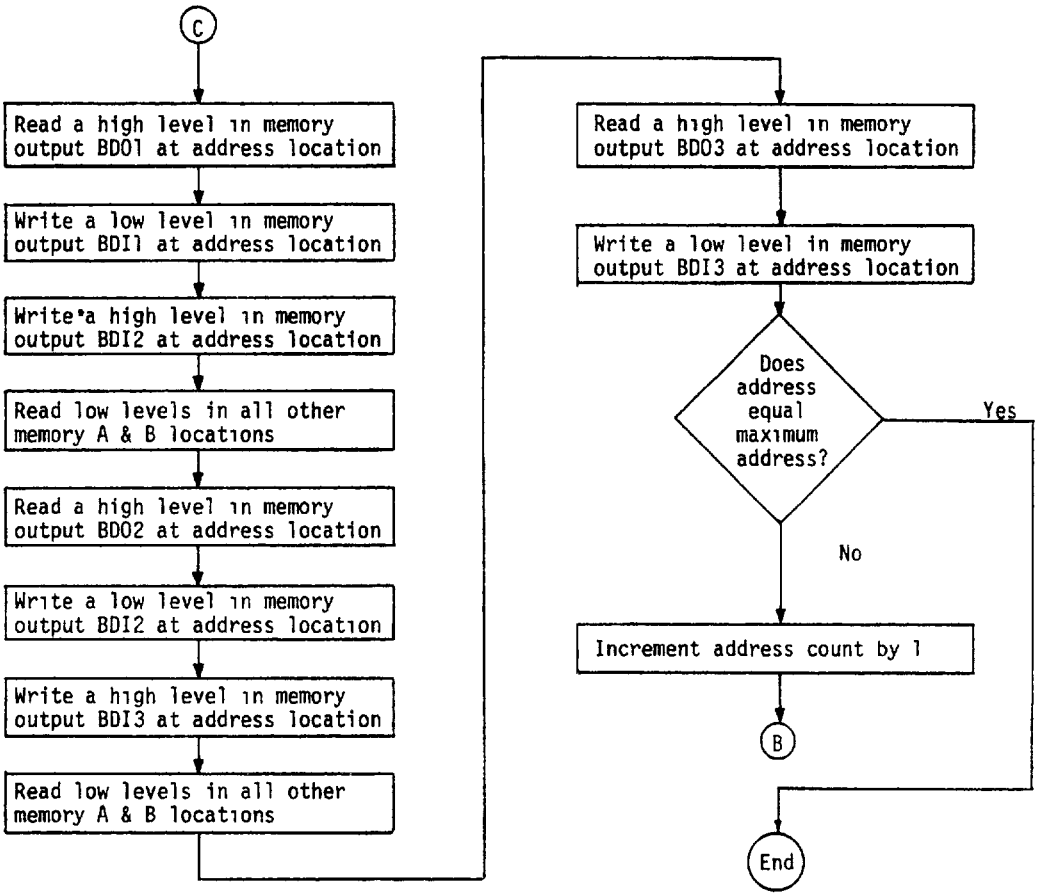


FIGURE 26 Walking-one test pattern flow chart - Continued

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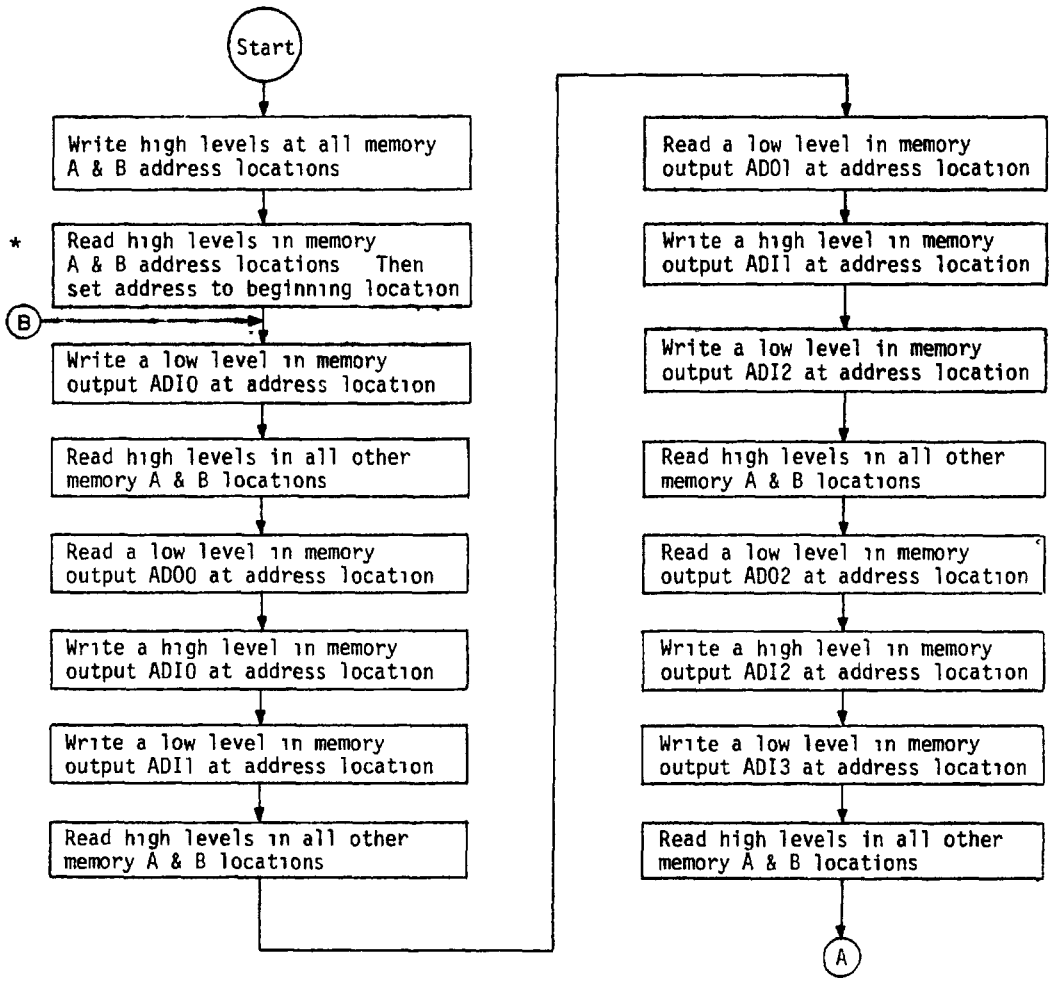


FIGURE 27 Walking-zero test pattern flow chart

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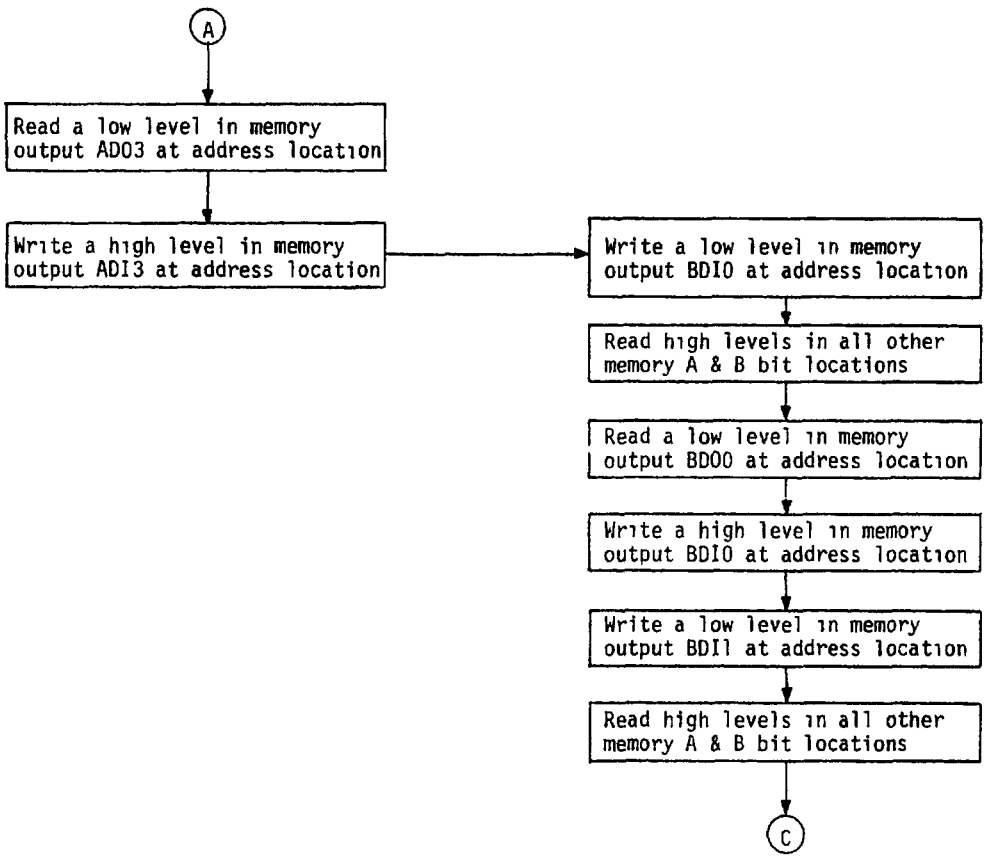


FIGURE 27 Walking-zero test pattern flow chart - Continued

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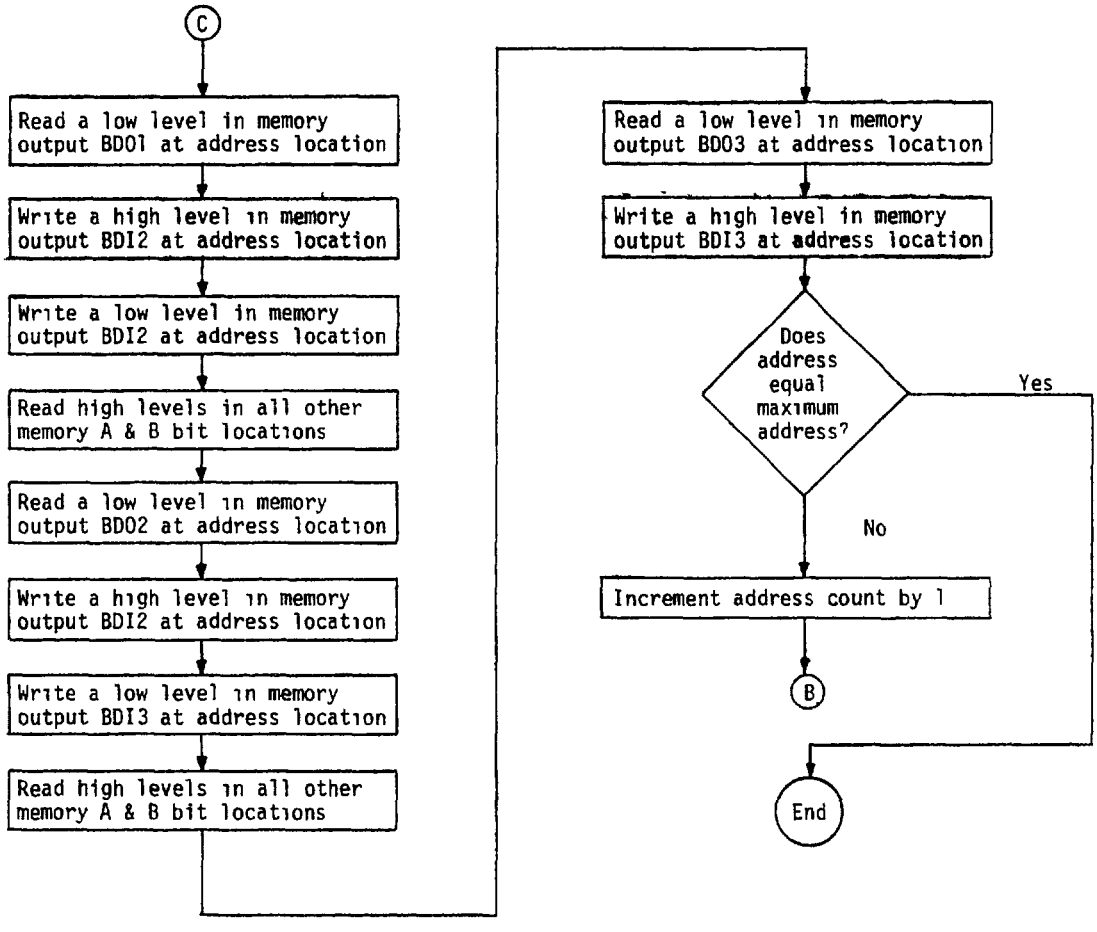


FIGURE 27 Walking-zero test pattern flow chart - Continued

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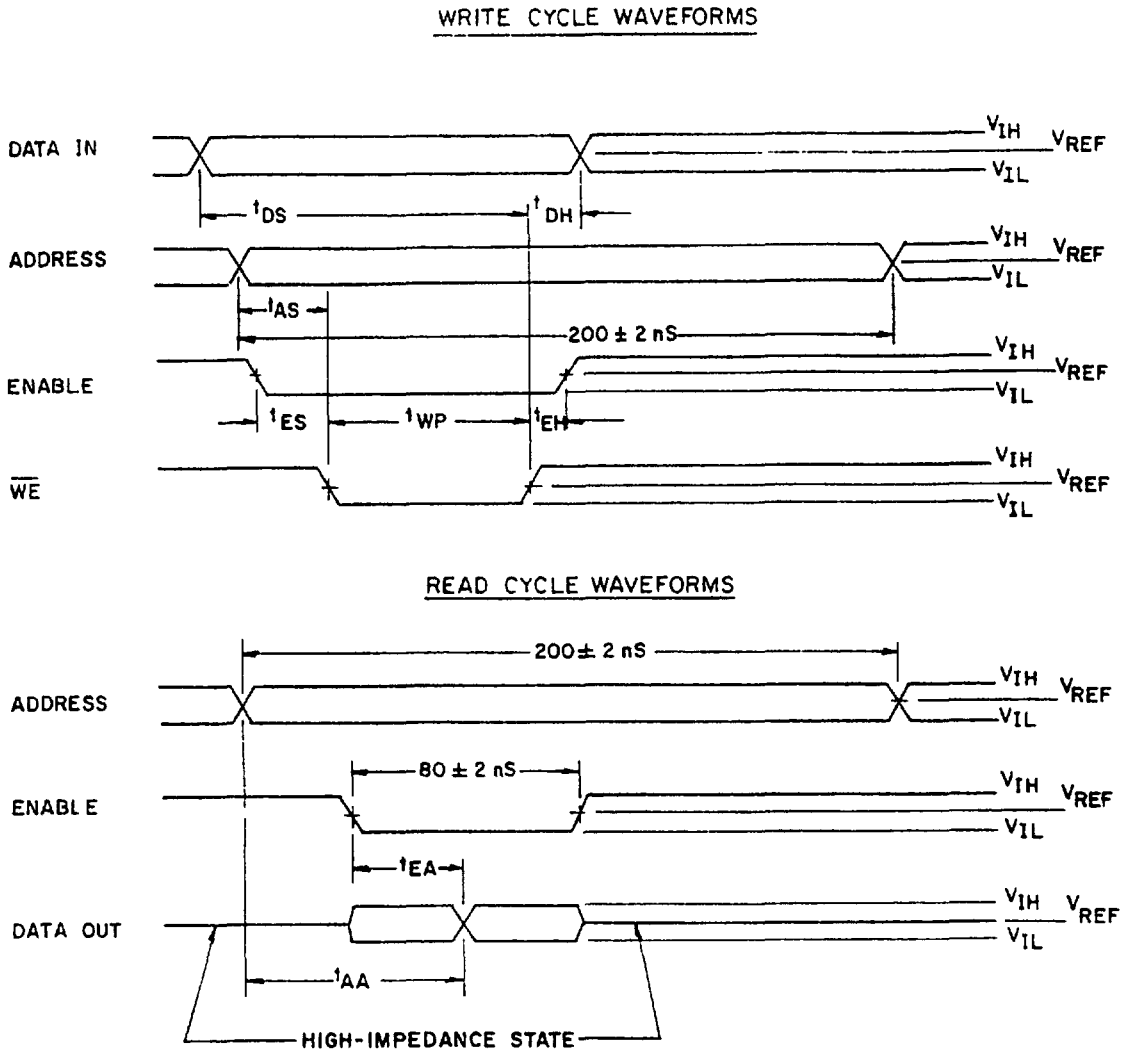


FIGURE 28. Walking-one-walking-zero test pattern waveforms for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:35Z
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NOTES:

1.	Address setup time	t_{AS} = 15 ± 1 ns.
2.	Data setup time	t_{DS} = 40 ± 2 ns.
3.	Enable setup time	t_{ES} = 10 ± 1 ns.
4.	Write pulse width	t_{WP} = 40 ± 2 ns.
5.	Data hold time	t_{DH} = 5 ± 1 ns.
6.	Enable hold time	t_{EH} = 0 ± 1 ns.
7.	Address access time (C_L = 50 pF)	t_{AA} = 60 ± 2 ns.
8	Enable access time (C_L = 50 pF)	t_{EA} = 30 ± 2 ns.
9	Address access time (C_L = 150 pF)	t_{AA} = 70 ± 2 ns
10	Enable access time (C_L = 150 pF)	t_{EA} = 40 ± 2 ns.
11.	Input high voltage	V_{IH} = 2.2 ± 0.2 V.
12	Input low voltage	V_{IL} = 0.6 ± 0.2 V.
13.	Reference voltage	V_{REF} = 1.5 ± 0.1 V.
14.	High-impedance state	≈1.6 V.

IGURE 28. Walking-one-walking-zero test pattern waveforms for table XVIII - Continued.

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TABLE XIX. Galloping-one-galloping-zero test pattern.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Test figure
		Min	Max	
S1	Galloping-one-galloping-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V) See notes 1, 2, 3, 4, and 6	See notes		29,30, 31,32
S2	Galloping-one-galloping-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V) See notes 1, 2, 3, 4, and 7			
S3	Galloping-one-galloping-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V) See notes 1, 2, 3, 5, and 6			
S4	Galloping-one-galloping-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V) See notes 1, 2, 3, 5, and 7	See notes		29,30, 31,32

NOTES

1. Galloping-one-galloping-zero pattern testing. The objective of galloping-one-galloping-zero pattern testing is to test access time. It is necessary to check all possible address transitions (addressing) from a read cycle to a read cycle. This pattern also verifies minimum write pulse width (t_{WP}), address setup time (t_{AS}), data setup time (t_{DS}), enable setup time (t_{ES}), data hold time (t_{DH}), enable hold time (t_{EH}), enable access time (t_{EA}), and address access time (t_{AA}).

2. A program that fulfills the galloping-one test pattern would be as follows:

   a. Write low levels in all memory A and memory B address locations.
   b. Read low levels at all memory A and memory B address locations.
   c. Write high levels at memory A and memory B address location 0.
   d. Read low levels at memory A and memory B address location 1.

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TABLE XIX. Galloping-one-galloping-zero test pattern - Continued.

NOTES -Continued

2. -Continued
   e. Read high levels at memory A and memory B address location 0.
   f. Read low levels at memory A and memory B address location 2.
   g. Read high levels at memory A and memory B address location 0.
   h. Read low levels at memory A and memory B address location 3.
   i. Read high levels at memory A and memory B address location 0.
   j. Continue the pattern until memory A and memory B address location 0 has been checked with respect to all other memory address locations.
* k. Write low levels at memory A and B address location 0.
   l. Write high levels at memory A and memory B address location 1.
   m. Read low levels at memory A and memory B address location 2.
   n. Read high levels at memory A and memory B address location 1.
   o. Read low levels at memory A and memory B address location 0.
   p. Read high levels at memory A and memory B address location 1.
   q. Write low levels at memory A and memory B address location 1.
   r. Write high levels at memory A and memory B address location 2.
   s. Continue the pattern until every memory A and memory B address location has been checked with respect to all other memory address locations.

3. A program that fulfills the galloping-zero test pattern would be as follows.

   a. Write high levels at all memory A and memory B address locations.
   b. Read high levels at all memory A and memory B address locations.
   c. Write low levels at memory A and memory B address location 0.
   d. Read high levels at memory A and memory B address location 1.
   e. Read low levels at memory A and memory B address location 0.
   f. Read high levels at memory A and memory B address location 2.
   g. Read low levels at memory A and memory B address location 0.
   h. Read high levels at memory A and memory B address location 3.
   i. Read low levels at memory A and memory B address location 0.
   j. Continue the pattern until memory A and memory B address location 0 has been checked with respect to all other memory address locations.
   k. Write high levels at memory A and B address location 0.
   l. Write low levels at memory A and memory B address location 1.
   m. Read high levels at memory A and memory B address location 2.
   n. Read low levels at memory A and memory B address location 1.
   o. Read high levels at memory A and memory B address location 0.
   p. Read low levels at memory A and memory B address location 1.
   q. Write high levels at memory A and memory B address location 1.
   r. Write low levels at memory A and memory B address location 2.
   s. Continue the pattern until every memory A and memory B address location has been checked with respect to all other memory address locations.

4. For test conditions S1 and S2, load output pins with the load specified on figure 14, note 4, test type S.

5. For test conditions S3 and S4, load output pins with the load specified on figure 14, note 4, test type Q.

6. For test conditions S1 and S3, set V_{CC} = 4.5 V $\pm$1%.

7. For test conditions S2 and S4, set V_{CC} = 5.5 V $\pm$1%.

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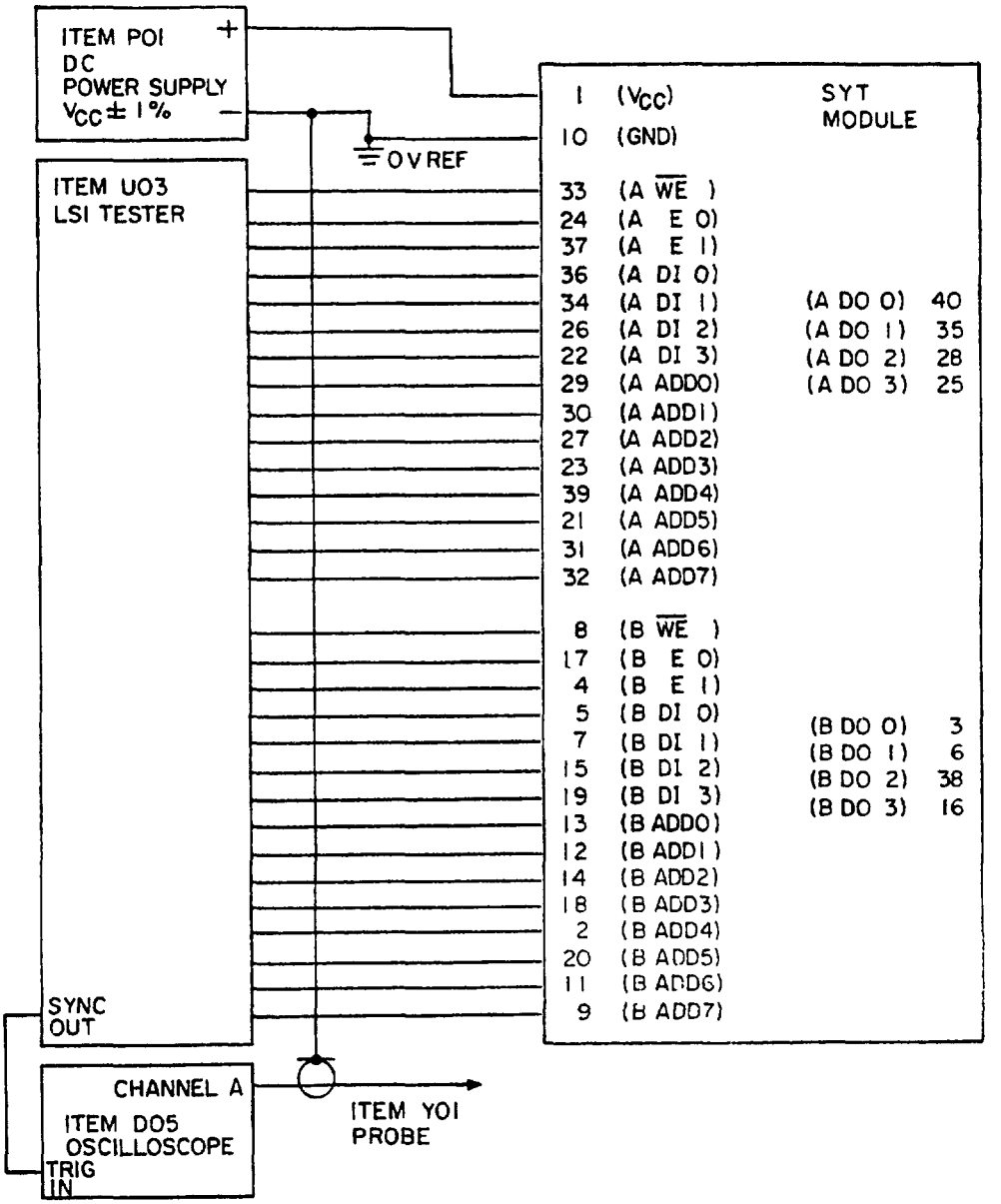


FIGURE 29. Galloping-one-galloping-zero test pattern circuit for table XIX.

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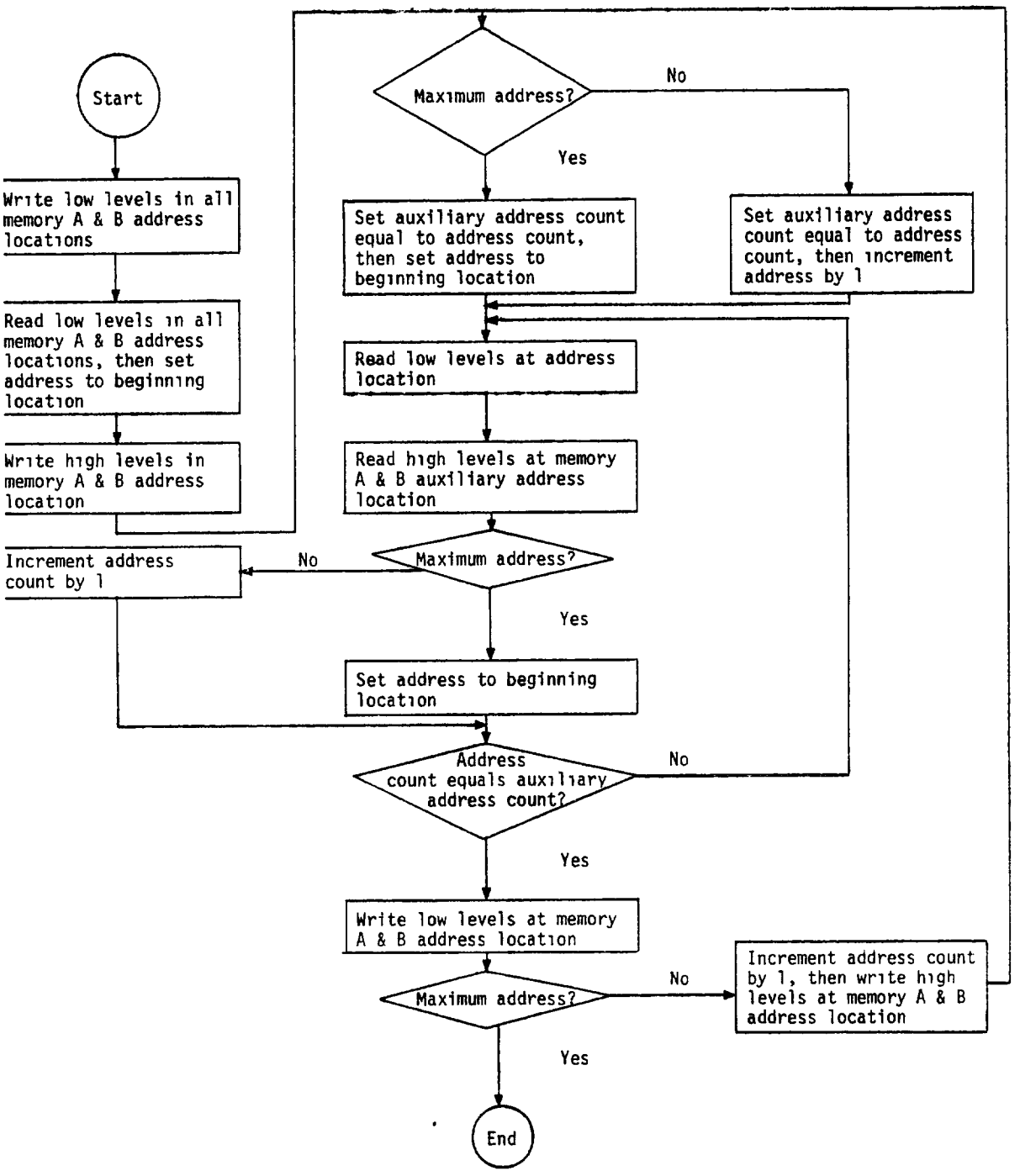


FIGURE 30. Galloping-one test pattern flow chart

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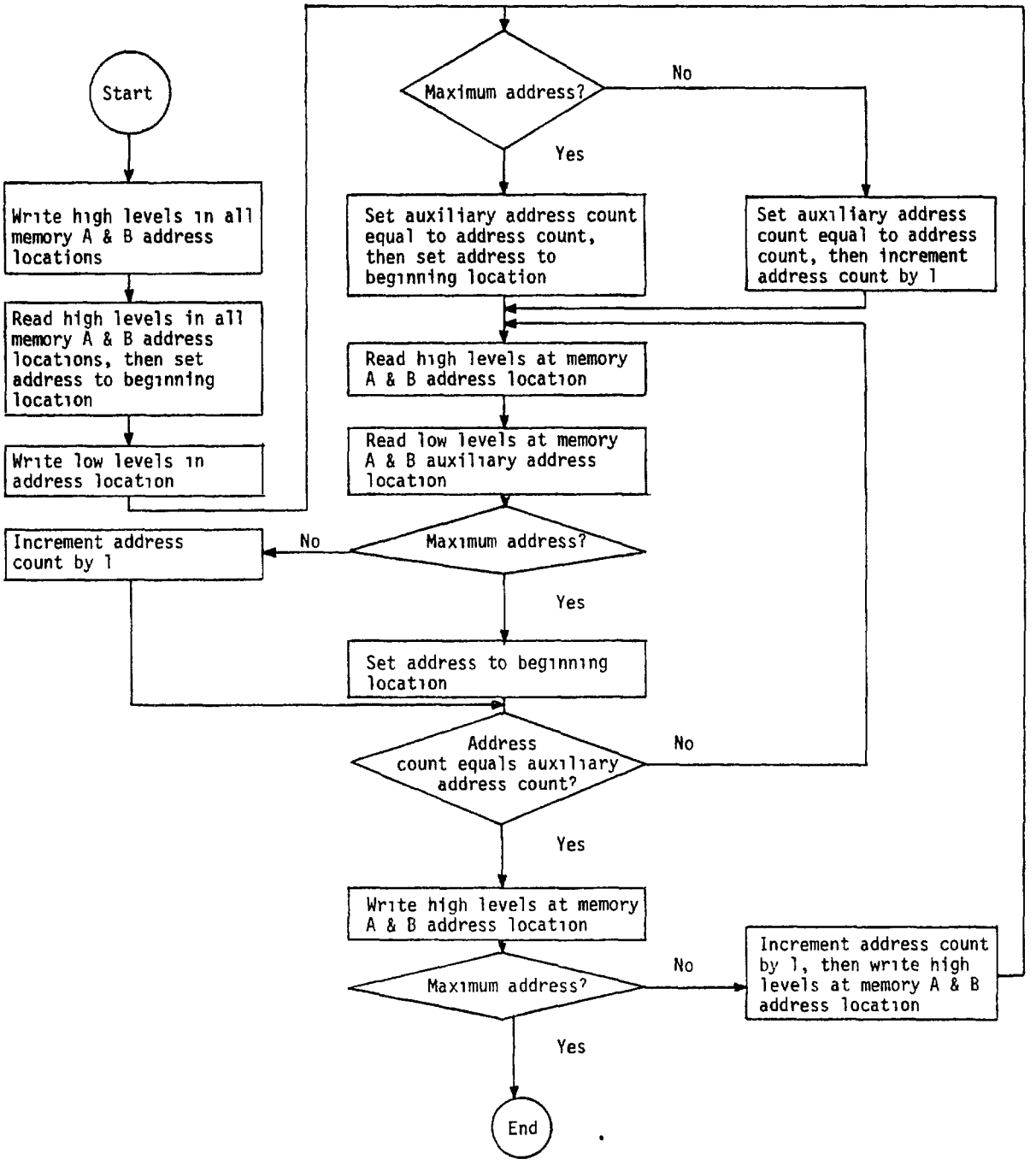


FIGURE 31. Galloping-zero test pattern flow chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:35Z
Check the source to verify that this is the current version before use.

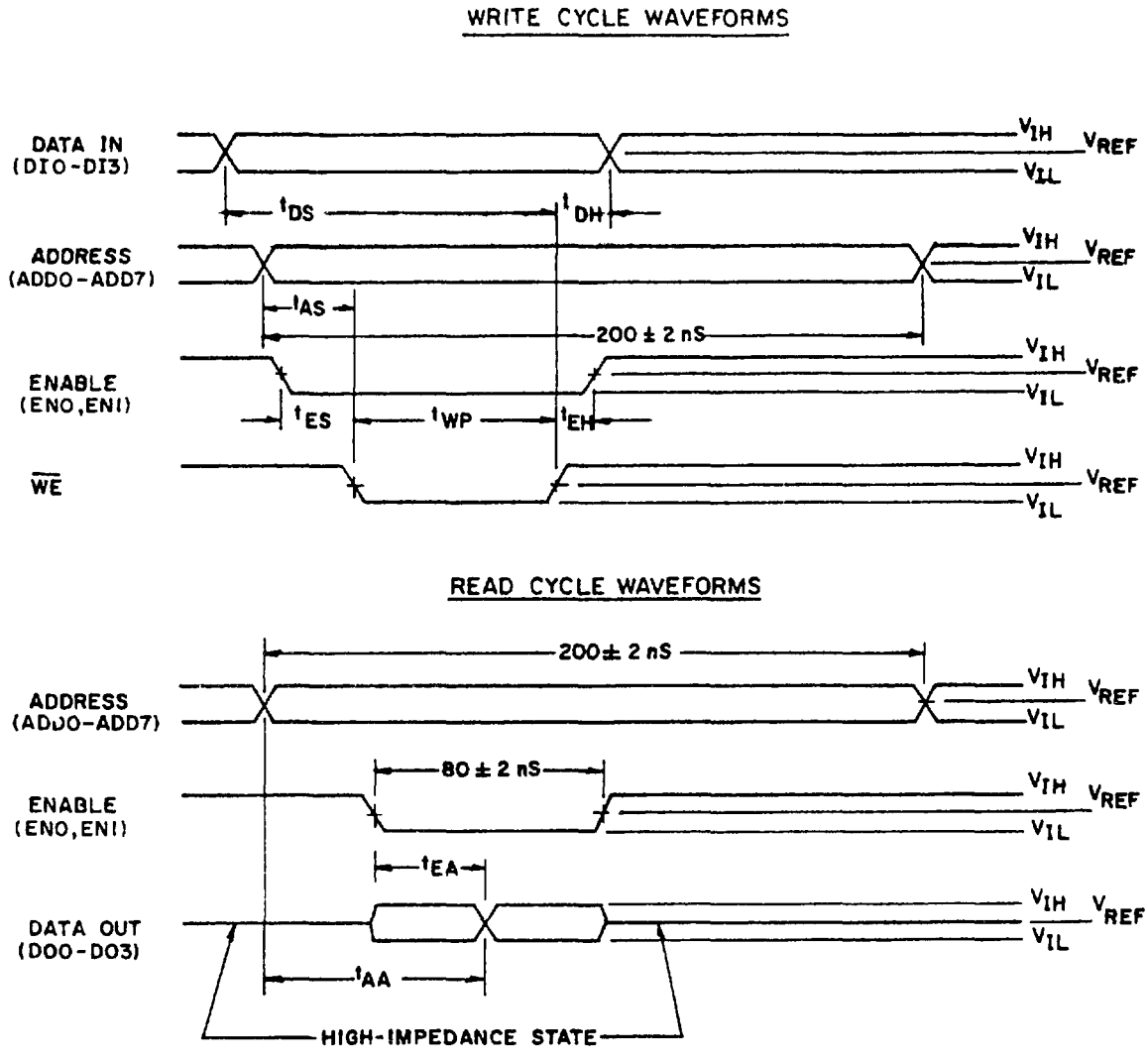


FIGURE 32. Galloping-one-galloping-zero test pattern waveforms for table XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:35Z
Check the source to verify that this is the current version before use.

NOTES:

1.	Address setup time	t_{AS} = 15 ± 1 ns.
2.	Data setup time	t_{DS} = 40 ± 2 ns.
3.	Enable setup time	t_{ES} = 10 ± 1 ns.
4.	Write pulse width	t_{WP} = 40 ± 2 ns.
5	Enable hold time	t_{EH} = 0 ± 1 ns.
6	Data hold time	t_{DH} = 5 ± 1 ns.
7	Address access time (C_L = 50 pF)	t_{AA} = 60 ± 2 ns.
8	Enable access time (C_L = 50 pF)	t_{EA} = 30 ± 2 ns.
9.	Address access time (C_L = 150 pF)	t_{AA} = 70 ± 2 ns.
10.	Enable access time (C_L = 150 pF)	t_{EA} = 40 ± 2 ns.
11.	Input high voltage	V_{IH} = 2.2 ± 0.2 V.
12.	Input low voltage	V_{IL} = 0.6 ± 0.2 V.
13.	Reference voltage	V_{REF} = 1 5 ± 0.1 V.
14.	High-impedance state	≈ 1.6 V.

FIGURE 32. Galloping-one-galloping-zero test pattern waveforms for table XIX - Continue

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TABLE XX. Inter-address write-recovery-one-write-recovery-zero test pattern.

Test condition	Parameter		Initial limits 25° ±5°C Ambient		Test figure
			Min	Max	
T1	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	See notes 1, 2, 3, 4, and 6	See notes		33,34, 35,36
T2	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	See notes 1, 2, 3, 4, and 7			
T3	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	See notes 1, 2, 3, 5, and 6			
T4	Inter-address write-recovery-one-write-recovery-zero test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	See notes 1, 2, 3, 5, and 7	See notes		33,34, 35,36

NOTES:

1. Inter-address write-recovery-one-write-recovery-zero pattern testing. The objective of inter-address write-recovery-one-write-recovery-zero pattern testing is to test recovery time from a write at one address location to a read at a different address location. The pattern checks recovery time for all possible address transitions from a write cycle to a read cycle. This pattern also verifies address to write enable hold time (t_{AH}), data to write enable setup time (t_{DS}), date to write enable hold time (t_{DH}), enable to write enable hold time (t_{EH}), minimum write pulse width (t_{WP}), address access time (t_{AA}), enable access time (t_{EA}), and enable to write enable setup time (t_{ES}).

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TABLE XX. Inter-address write-recovery-one-write-recovery-zero test pattern - Continued.

NOTES: -Continued

2. A program that fulfills the inter-address write-recovery-zero test pattern would be as follows:

a. Write high levels at all memory A and memory B address locations.
b. Read high levels at all memory A and memory B address locations.
c. Write low levels at memory A and memory B address location 1.
d. Read high levels at memory A and memory B address location 0.
e. Read low levels at memory A and memory B address location 1.
f. Write high levels at memory A and memory B address location 1.
g. Read high levels at memory A and memory B address location 0.
h. Read high levels at memory A and memory B address location 1.
i. Continue the pattern until memory address 0 has been checked with respect to all other memory address locations.
j. Write low levels at memory A and memory B address location 2.
k. Read high levels at memory A and memory B address location 1.
l. Read low levels at memory A and memory B at address location 2.
m. Write high levels at memory A and memory B address location 2.
n. Read high levels at memory A and memory B address location 1.
o. Read high levels at memory A and memory B address location 2.
* p. Continue the pattern until memory address 1 has been checked with respect to all other memory address locations.
q. Write low levels at memory A and memory B at address location 0.
r. Read high levels at memory A and memory B address location 1.
s. Read low levels at memory A and memory B address location 0.
t. Write high levels at memory A and memory B at address location 0.
u. Read high levels at memory A and memory B address location 1.
v. Read high levels at memory A and memory B address location 0.
w. Continue the pattern until all memory address locations have been checked with respect to all other memory address locations.

3. A program that fulfills the inter-address write-recovery-one test pattern would be as follows:

a. Write low levels at all memory A and memory B address locations.
b. Read low levels at all memory A and memory B address locations.
c. Write high levels of memory A and memory B at address location 1.
* d. Read low levels at memory A and memory B address location 0.
e. Read high levels at memory A and memory B address location 1.
f. Write low levels at memory A and memory B address location 1.
g. Read low levels at memory A and memory B address location 0.
h. Read low levels at memory A and memory B address location 1.
i. Continue the pattern until memory address location 0 has been checked with respect to all other memory address locations.
j. Write high levels at memory A and memory B address location 2.
k. Read low levels at memory A and memory B address location 1.
l. Read high levels at memory A and memory B address location 2.
m. Write low levels at memory A and memory B address location 2.
n. Read low levels at memory A and memory B address location 1.
o. Read low levels at memory A and memory B address location 2.
* p. Continue the pattern until memory address location 1 has been checked with respect to all other memory address locations.
q. Write high levels at memory A and memory B address location 0.
r. Read low levels at memory A and memory B address location 1.

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TABLE XX. Inter-address write-recovery-one-write-recovery-zero test pattern - Continued.

NOTES -Continued

s. Read high levels at memory A and memory B address location 0.
t. Write low levels at memory A and memory B address location 0.
u. Read low levels at memory A and memory B at address location 1.
v. Read low levels at memory A and memory B address location 0.
w. Continue the pattern until all memory address locations have been checked with respect to all other memory address locations.

4. For test conditions T1 and T2, load output pins with the load specified on figure 14, note 4, test type S.

5. For test conditions T3 and T4, load output pins with the load specified on figure 14, note 4, test type Q.

6. For test conditions T1 and T3, set V_{CC} = 4.5 V $\pm$1%.

7. For test conditions T2 and T4, set V_{CC} = 5.5 V $\pm$1%.

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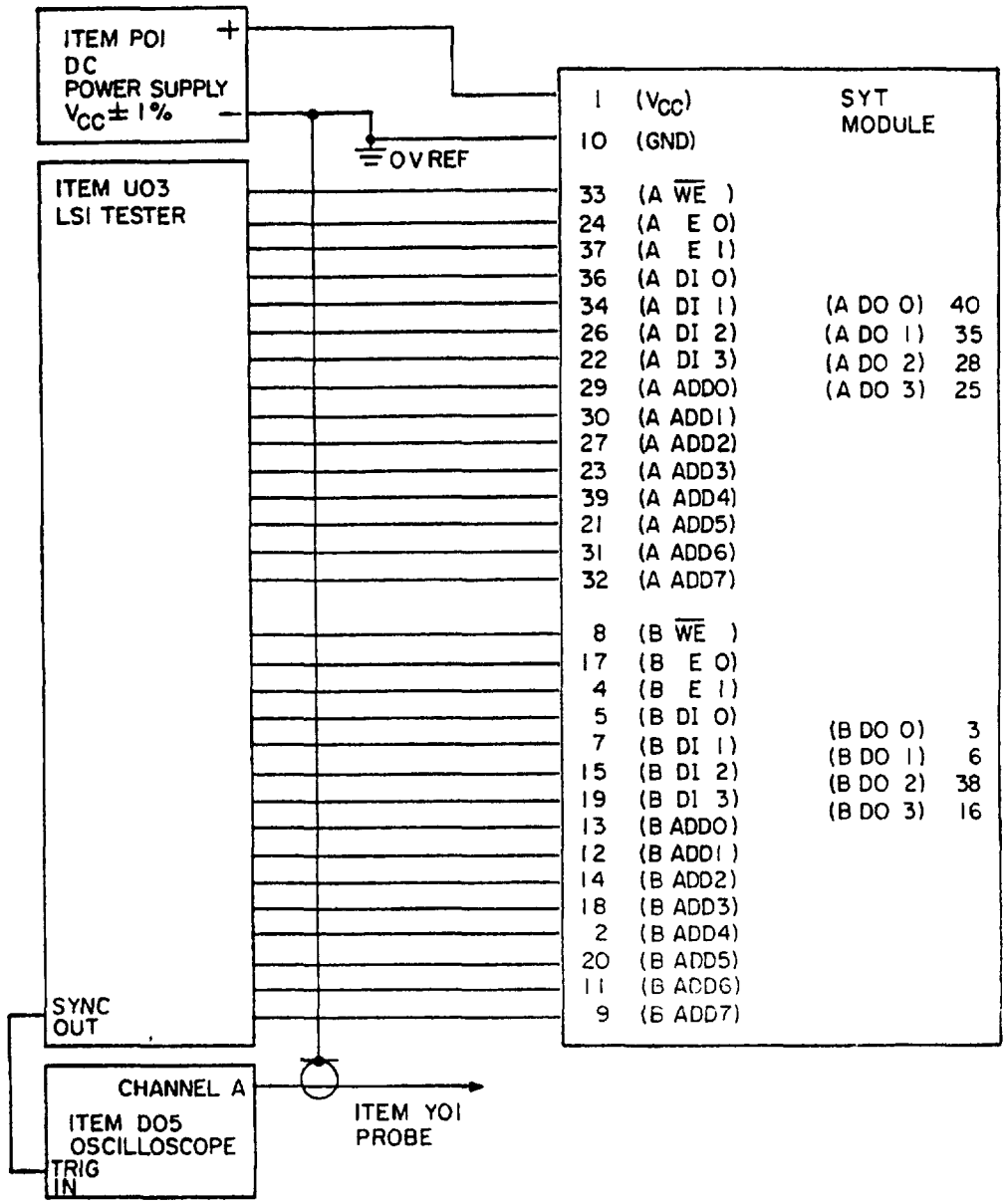


FIGURE 33. Inter-address write-recovery-one-write-recovery-zero test pattern circuit for table XX.

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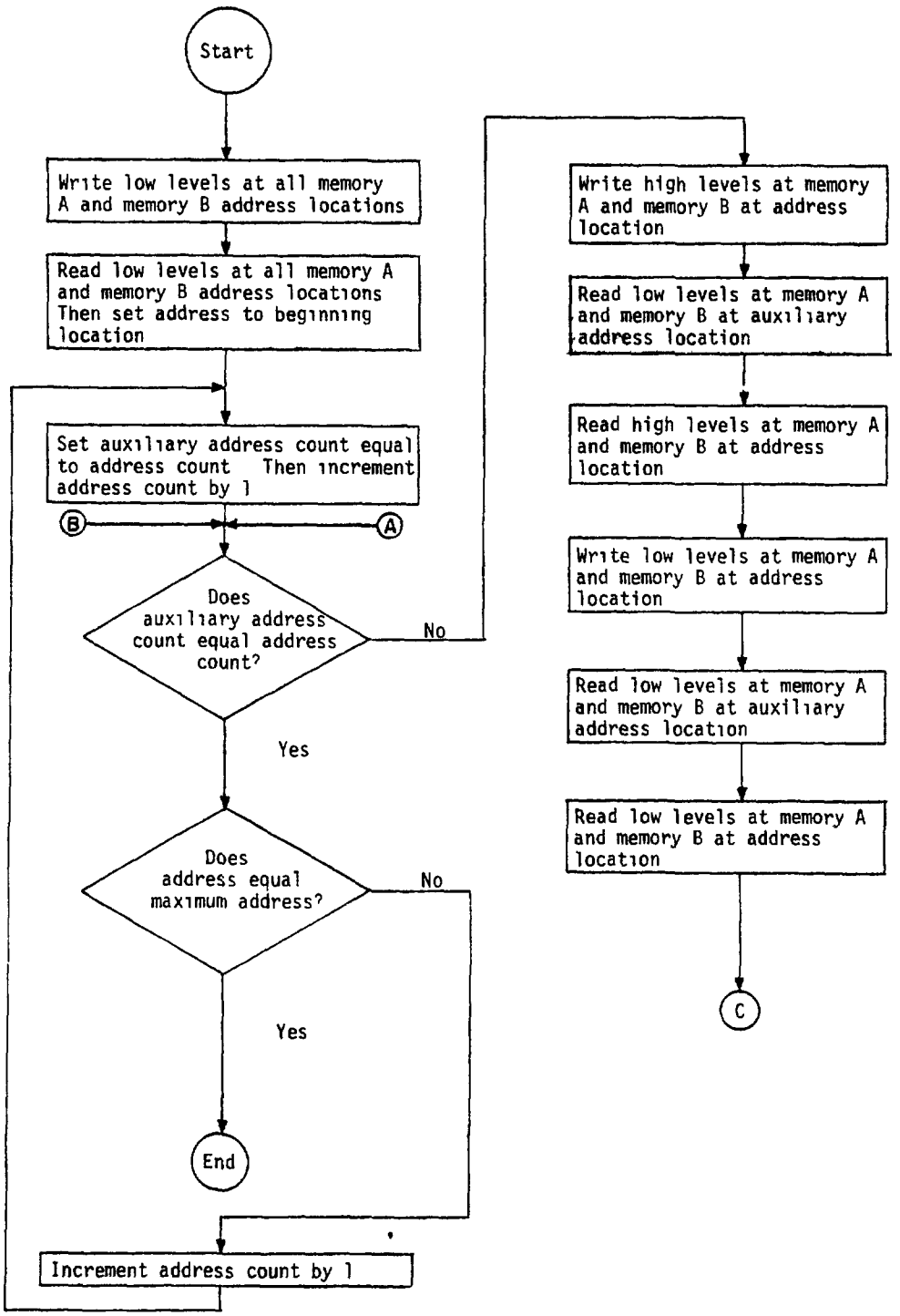


FIGURE 34 Inter-address write-recovery-one test pattern flow chart

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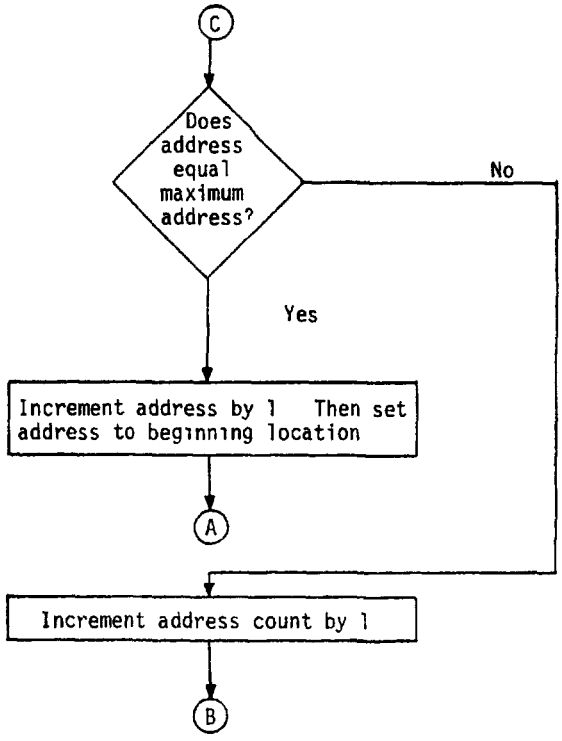


FIGURE 34 Inter-address write-recovery-one test pattern flow chart - Continued

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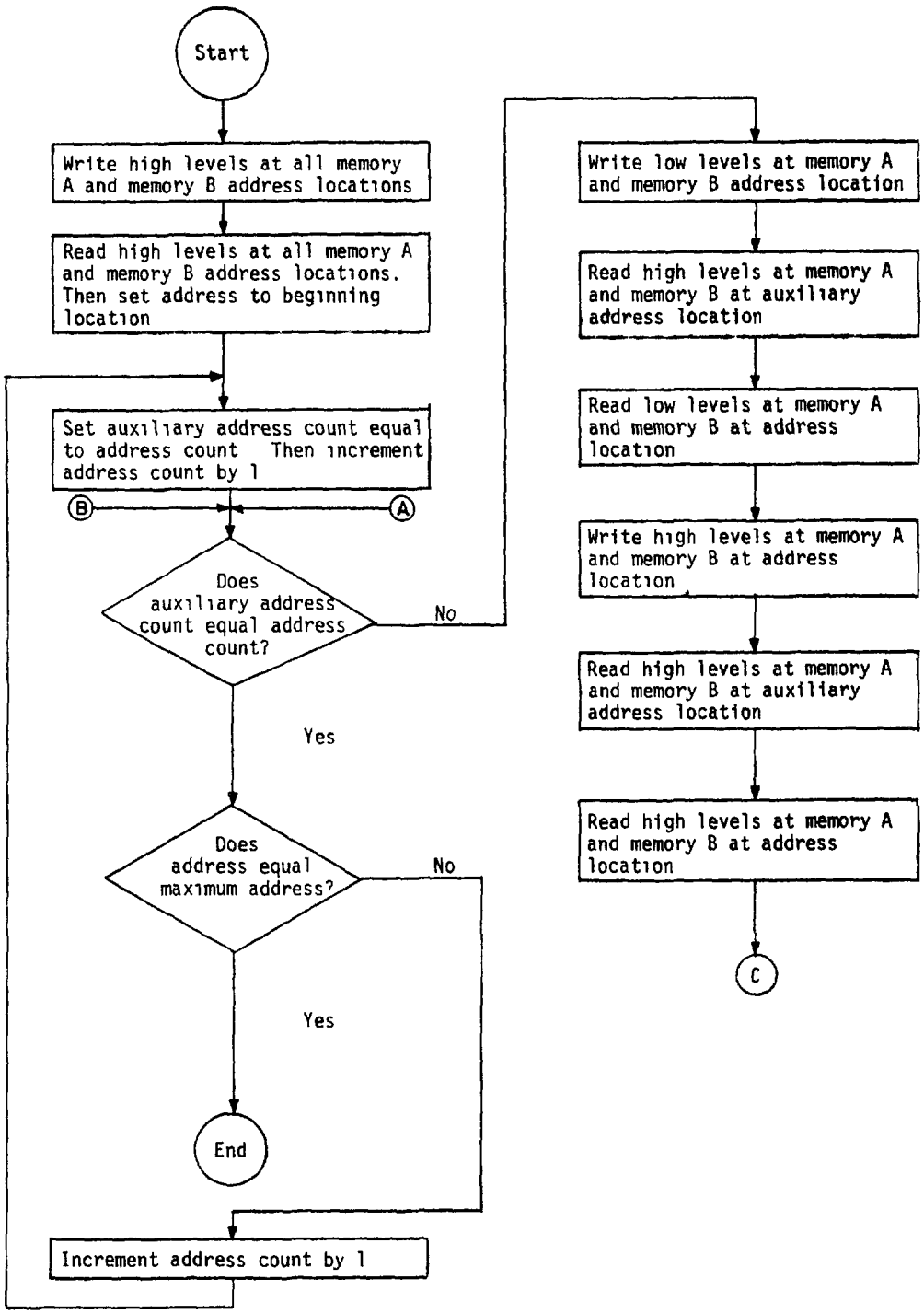


FIGURE 35. Inter-address write-recovery-zero test pattern flow chart.

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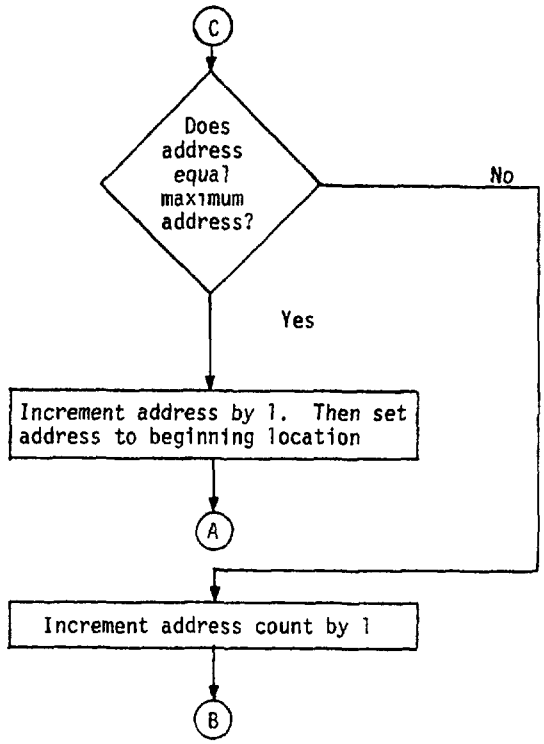


FIGURE 35. Inter-address write-recovery-zero test pattern flow chart - Continued

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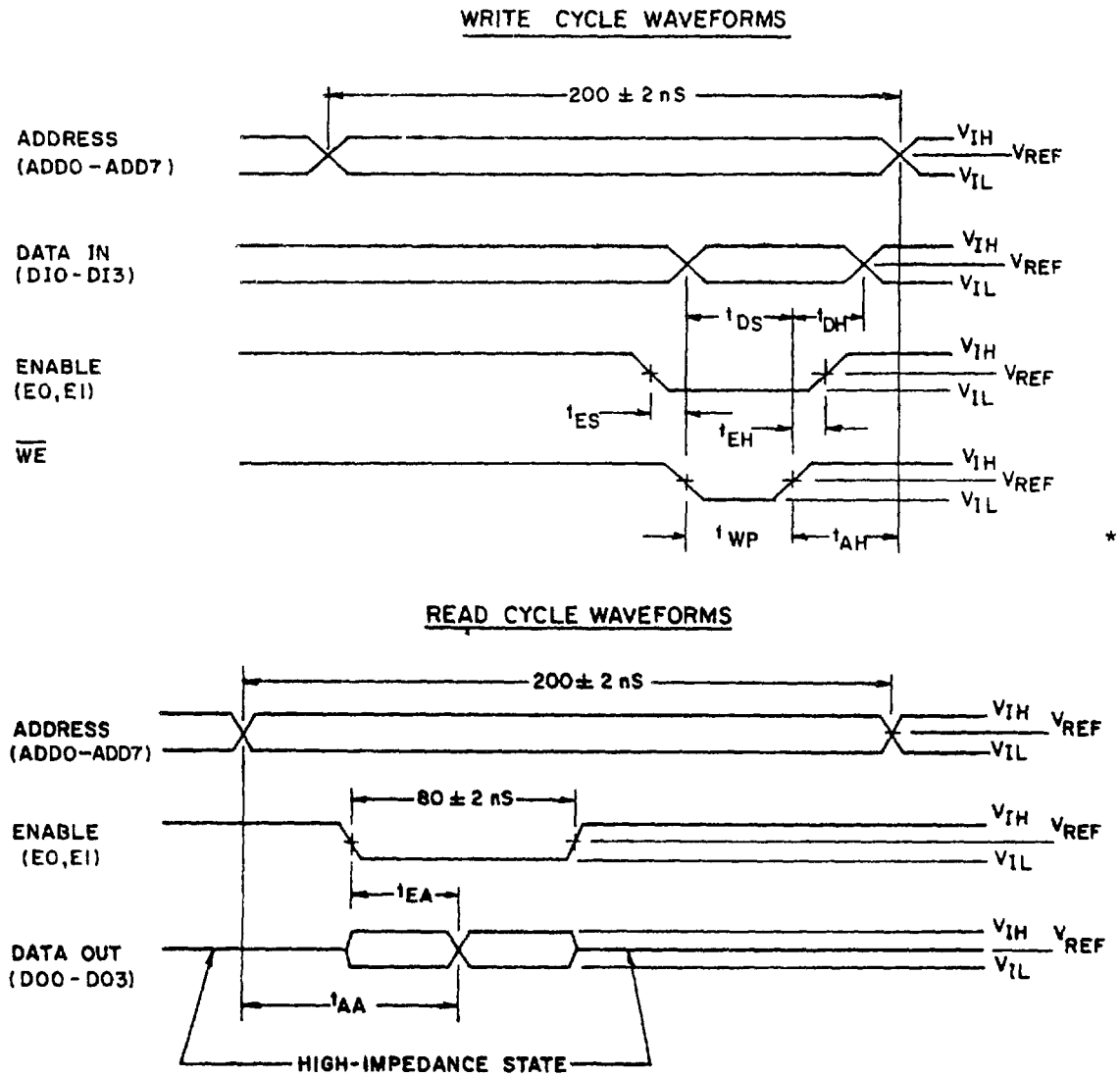


FIGURE 36. Inter-address write-recovery-one-write-recovery-zero test pattern waveforms for table XX.

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NOTES:

1.	Data setup time	t_{DS} = 40 ± 2 ns.
2.	Enable setup time	t_{ES} = 10 ± 1 ns.
3.	Write pulse width	t_{WP} = 40 ± 2 ns.
4.	Data hold time	t_{DH} = 5 ± 1 ns.
5.	Enable hold time	t_{EH} = 0 ± 1 ns.
6.	Address hold time	t_{AH} = 10 ± 1 ns.
7.	Address access time (C_L = 50 pF)	t_{AA} = 60 ± 2 ns.
8.	Enable access time (C_L = 50 pF)	t_{EA} = 30 ± 2 ns.
9.	Address access time (C_L = 150 pF)	t_{AA} = 70 ± 2 ns.
10.	Enable access time (C_L = 150 pF)	t_{EA} = 40 ± 2 ns.
11.	Input high voltage	V_{IH} = 2.2 ± 0.2 V.
12.	Input low voltage	V_{IL} = 0.6 ± 0.2 V.
13.	Reference voltage	V_{REF} = 1.5 ± 0.1 V.
14.	High-impedance state	≈1.6 V.

FIGURE 36. Inter-address write-recovery-one-write-recovery-zero test pattern waveforms for table XX - Continued.

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TABLE XXI. Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern.

Test condition	Parameter		Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Test figure
U1	Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 50 pF, V_{CC} = 4.5 V)	See notes 1, 2, 3, 4, and 6	See notes		37,38, 39,40
U2	Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 50 pF, V_{CC} = 5.5 V)	See notes 1, 2, 3, 4, and 7			
U3	Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 150 pF, V_{CC} = 4.5 V)	See notes 1, 2, 3, 5, and 6			
U4	Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern (C_L = 150 pF, V_{CC} = 5.5 V)	See notes 1, 2, 3, 5, and 7	See notes		37,38, 39,40

NOTES:

1. Marching intra-address write-recovery-one-write-recovery-zero and enable/disable testing. The objective of marching intra-address write-recovery-one-write-recovery-zero and enable/disable testing is to check for the proper functioning of the address and chip select circuitry in a dynamic environment. The pattern checks to assure the module 1024 unique bit locations can be independently addressed. The pattern also verifies address setup time (tAS), enable setup

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TABLE XXI. Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern - Continued.

NOTES: -Continued

time (t_{ES}), enable hold time (t_{EH}), write pulse width (t_{WP}), write recovery time (t_{WR}), and data setup time (t_{DS}).

2. A program that fulfills the marching intra-address-write-recovery-one and enable/disable test pattern would be as follows:

a. Write lows at all memory address locations.
b. Read lows at all memory address locations.
c. Read lows at address location 0.
d. Write and read highs at address location 0.
e. Apply disable pulse and attempt to write lows at address location 0.
f. Read highs at address location 0.
g. Read lows at address location 1023.
h. Write and read highes at address location 1023.
i. Apply disable pulse and attempt to write lows at address location 1023.
j. Read highs at address location 1023.
k. Read highs at address location 1023.
l Write and read lows at address location 1023.
m. Apply disable pulse and attempt to write highs at address location 1023.
n. Read lows at address location 1023.
o. Read highs at address location 0.
p. Write and read lows at address location 0.
q Apply disable pulse and attempt to write highs at address location 0.
r. Read lows at address location 0.

3. A program that fulfills the marching intra-address write-recovery-zero and enable/disable test pattern would be as follows:

a Write highs at all memory address locations.
b. Read highs at all memory address locations.
c. Read highs at address location 0.
d. Write and read lows at address location 0
e. Apply disable pulse and attempt to write highs at address location 0.
f Read lows at address location 0.
g. Read highs at address location 1023.
h. Write and read lows at address location 1023
i. Apply disable pulse and attempt to write highs at address location 1023.
j. Read lows at address location 1023.
k. Read lows at address location 1023.
l. Write and read highs at address location 1023.
m. Apply disable pulse and attempt to write lows at address location 1023.
n. Read highs at address location 1023.
o. Read lows at address location 0.
p. Write and read highs at address location 0.
q. Apply disable pulse and attempt to write lows at address location 0.
r. Read highs at address location 0.

4. For test conditions U1 and U2, load output pins with the load specified on figure 14, note 4, test type S.

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TABLE XXI. Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern - Continued.

NOTES: -Continued

5. For test conditions U3 and U4, load output pins with the load specified on figure 14, note 4, test type Q.

6. For test conditions U1 and U3, set V_{CC} = 4.5 V ±1%.

7. For test conditions U2 and U4, set V_{CC} = 5.5 V ±1%.

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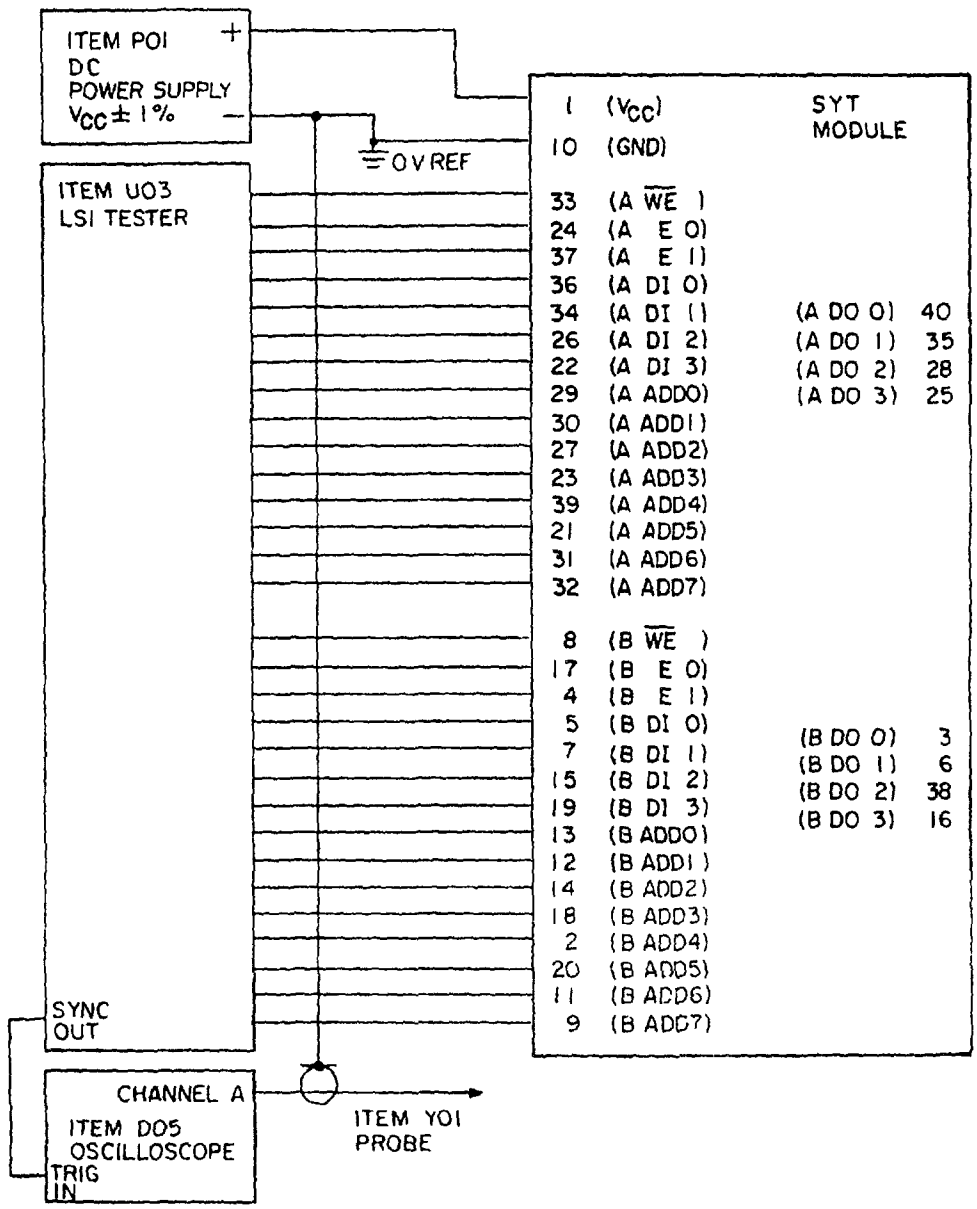


FIGURE 37. Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern circuit for table XXI.

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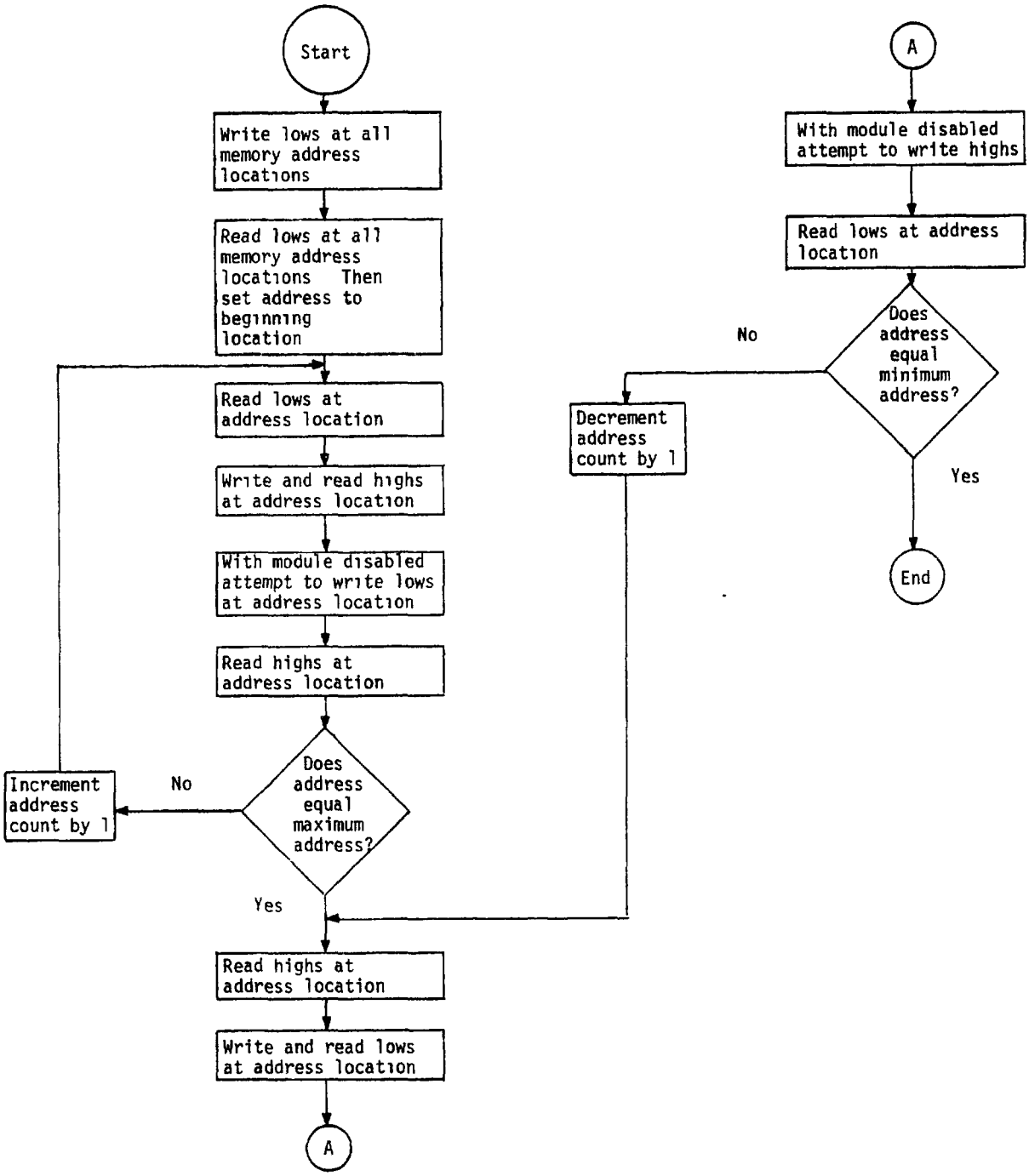


FIGURE 38 Marching intra-address write-recovery-one and enable/disable test pattern flow chart

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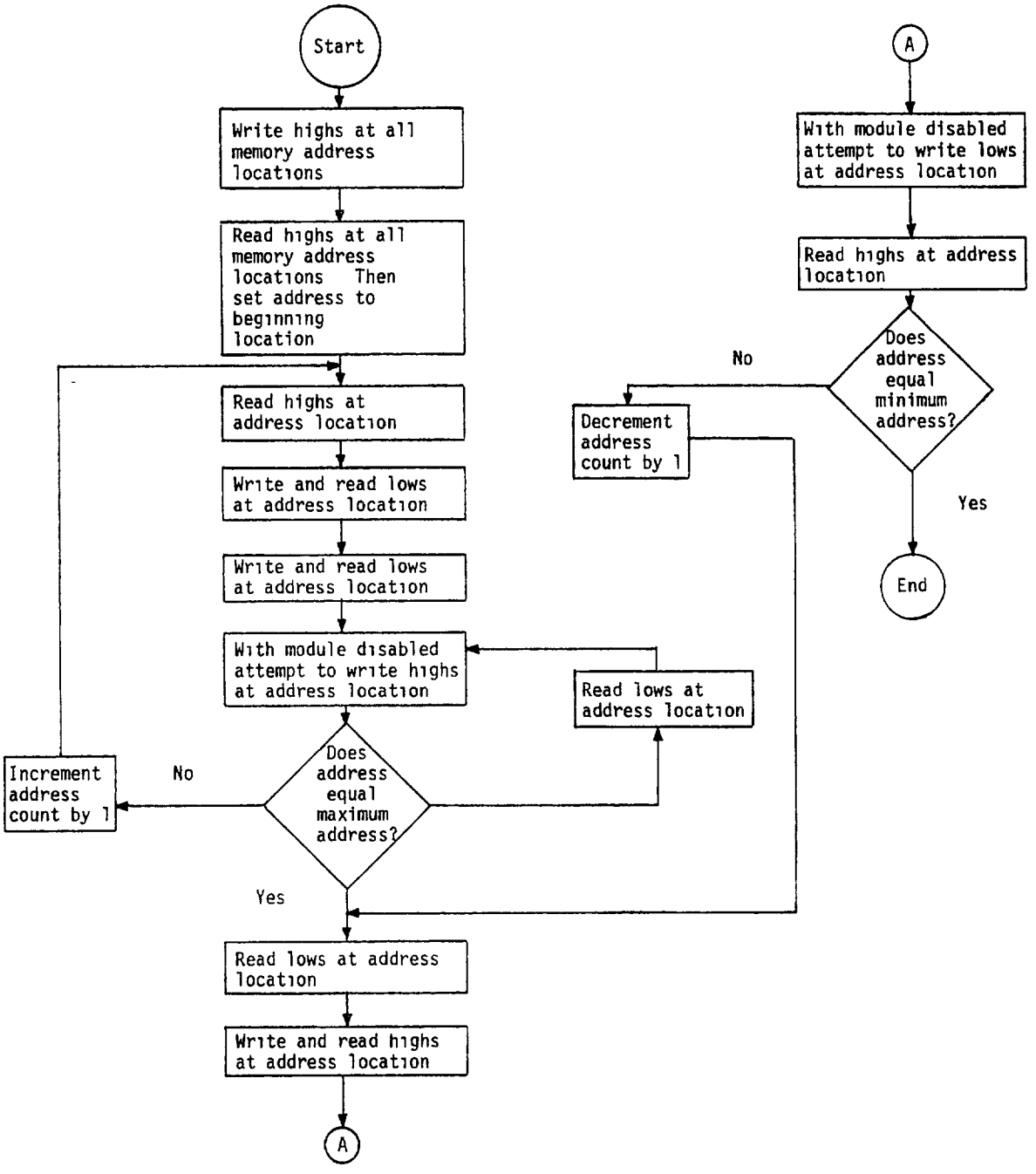


FIGURE 39 Marching intra-address write-recovery-zero and enable/disable test pattern flow chart

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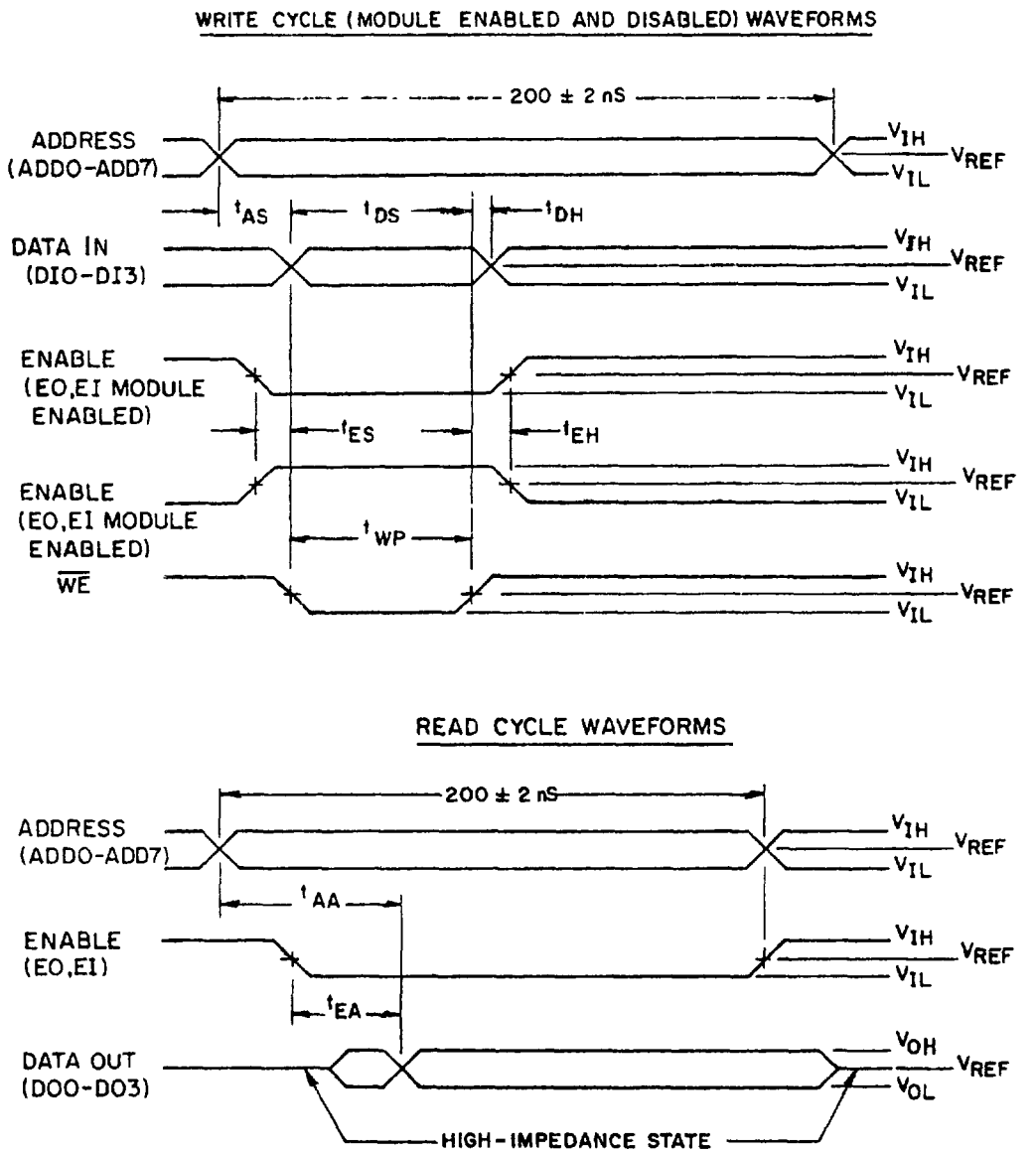


FIGURE 40. Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern waveforms for table XXI.

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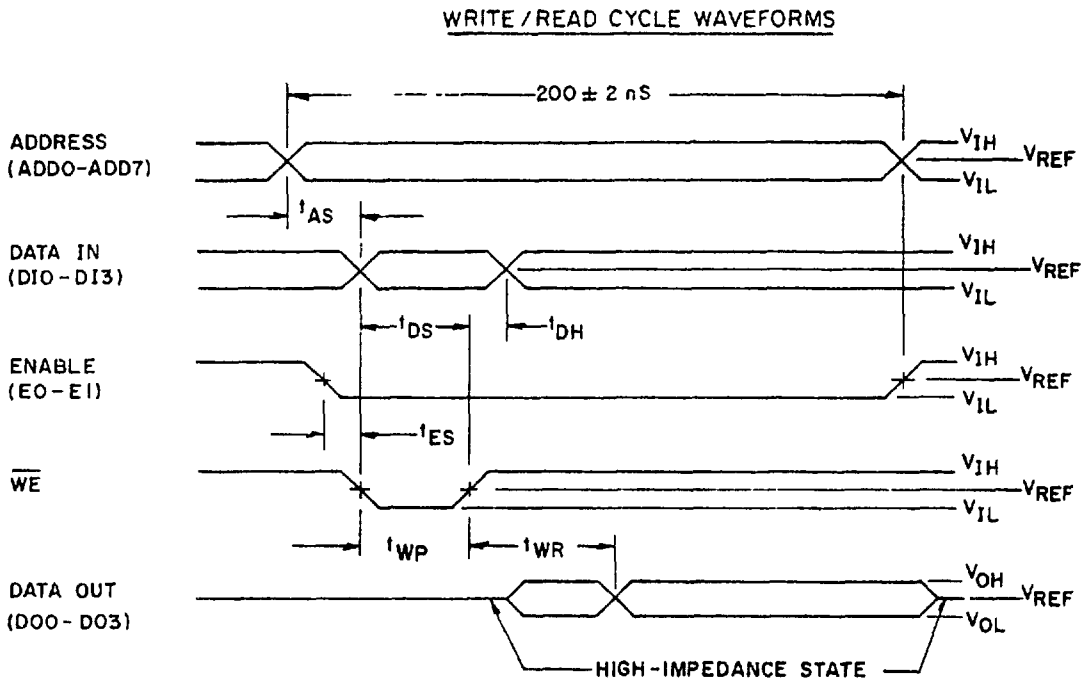


FIGURE 40. Marching intra-address write-recovery-one-write-recovery-zero and enable/ disable test pattern waveforms for table XXI - Continued.

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NOTES:

1.	Address setup time	t_{AS} = 15 ±1 ns.
2.	Data setup time	t_{DS} = 40 ±2 ns.
3.	Enable setup time	t_{ES} = 10 ±1 ns
4.	Write pulse width	t_{WP} = 40 ±2 ns.
5.	Data hold time	t_{DH} = 5 ±1 ns.
6.	Enable hold time	t_{EH} = 0 ±1 ns.
7.	Write recovery time (C_L = 50 pF)	t_{WR} = 50 ±2 ns.
8.	Address access time (C_L = 50 pF)	t_{AA} = 60 ±2 ns.
9.	Enable access time (C_L = 50 pF)	t_{EA} = 30 ±2 ns.
10.	Write recovery time (C_L = 150 pF)	t_{WR} = 60 ±2 ns.
11.	Address access time (C_L = 150 pF)	t_{AA} = 70 ±2 ns.
12.	Enable access time (C_L = 150 pF)	t_{EA} = 40 ±2 ns.
13.	Input high voltage	V_{IH} = 2.2 ±0.2 V.
14.	Input low voltage	V_{IL} = 0.6 ±0.2 V.
15.	Reference voltage	V_{REF} = 1.5 ±0.1 V.
16.	High-impedance state	≈ 1.6 V.

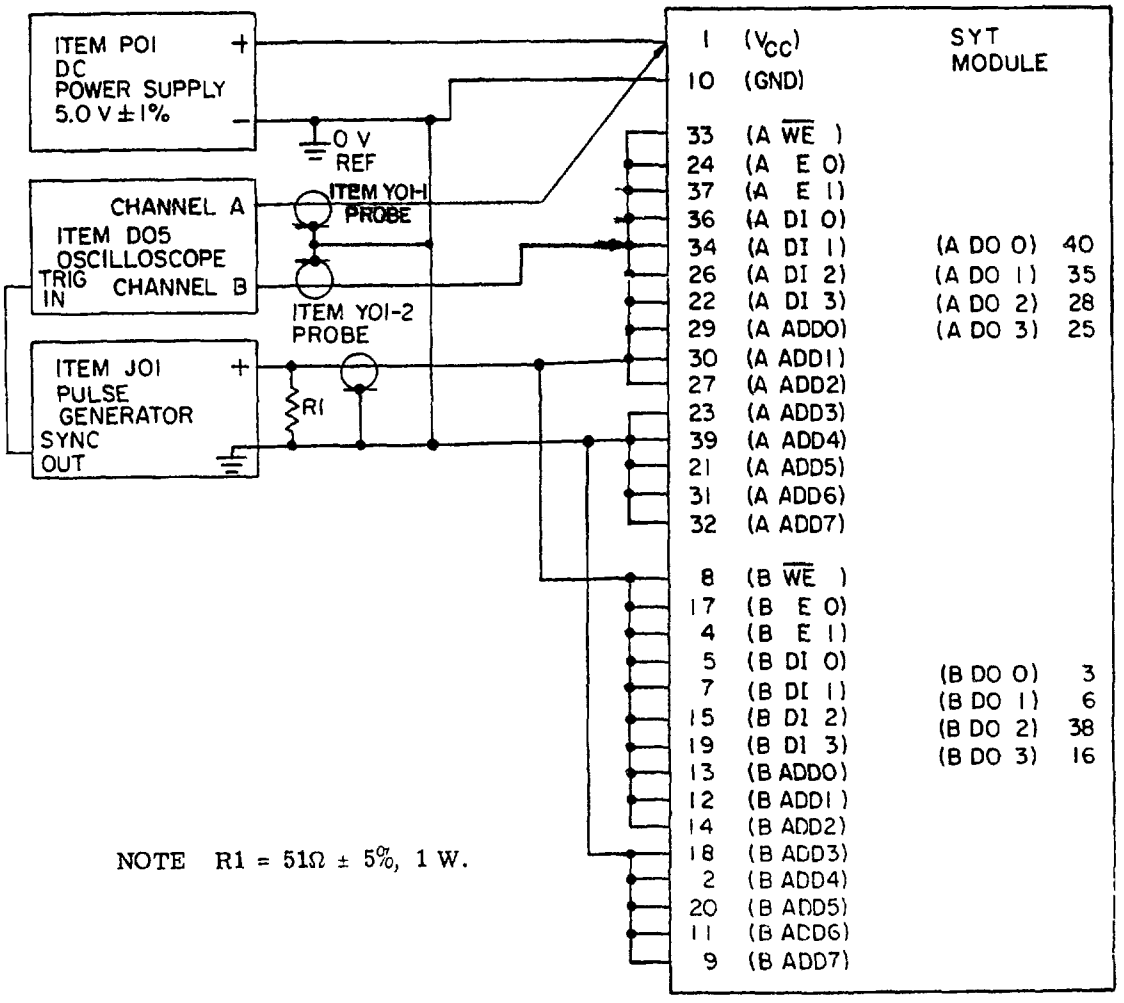
FIGURE 40 Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern waveforms for table XXI - Continued.

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TABLE XXII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	400	mV	41,42

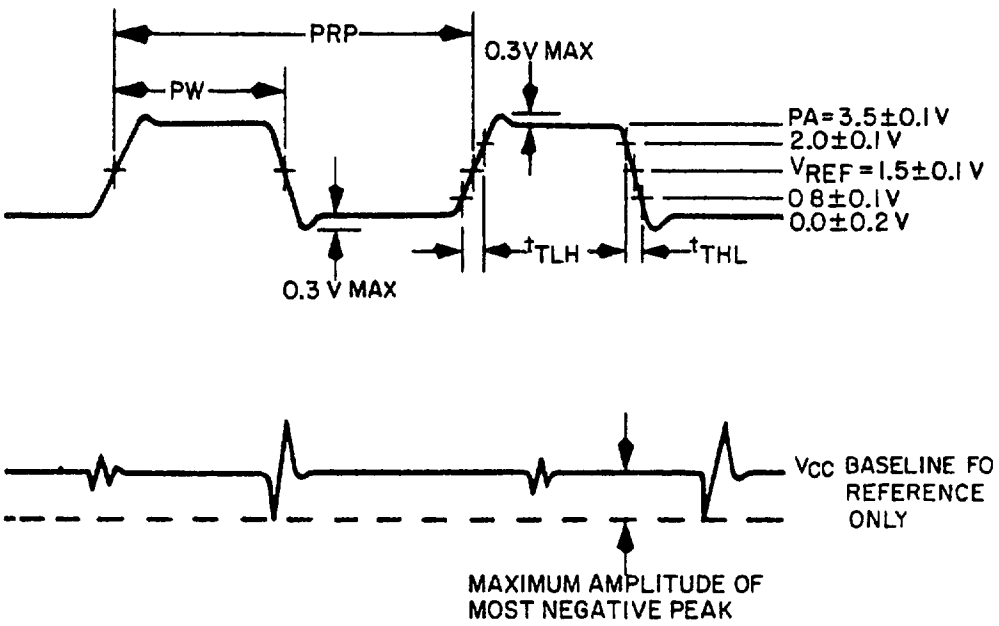
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NOTE R1 = 51Ω ± 5%, 1 W.

FIGURE 41. Power supply transient amplitude test circuit for table XXII.

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NOTES
1. Input pulse parameters
   $t_{TLH} = t_{THL}$ = 8.0 ±1.5 ns.
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

FIGURE 42. Power supply transient amplitude test circuit waveforms for table XXII.

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Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	$\overline{AWE}$	Input capacitance	33	Open	3	60	pF	43
W2	AE0		24			120		
W3	AE1		37			60		
W4	ADI0		36			15		
W5	ADI1		34					
W6	ADI2		26					
W7	ADI3		22			15		
W8	AADD0		29			60		
W9	AADD1		30					
W10	AADD2		27					
W11	AADD3		23					
W12	AADD4		39					
W13	AADD5		21					
W14	AADD6		31					
W15	AADD7		32					
W16	$\overline{BWE}$		8			60		
W17	BE0		17			120		
W18	BE1		4			60		
W19	BDI0	Input capacitance	5	Open	3	15	pF	43

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TABLE XXIII. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W20	BDI1	Input capacitance	7	Open	3	15	pF	43
W21	BDI2		15			15		
W22	BDI3		19			15		
W23	BADD0		13			60		
W24	BADD1		12					
W25	BADD2		14					
W26	BADD3		18					
W27	BADD4		2					
W28	BADD5		20					
W29	BADD6		11					
W30	BADD7	Input capacitance	9	Open	3	60	pF	43

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TABLE XXIV. Output capacitance test.

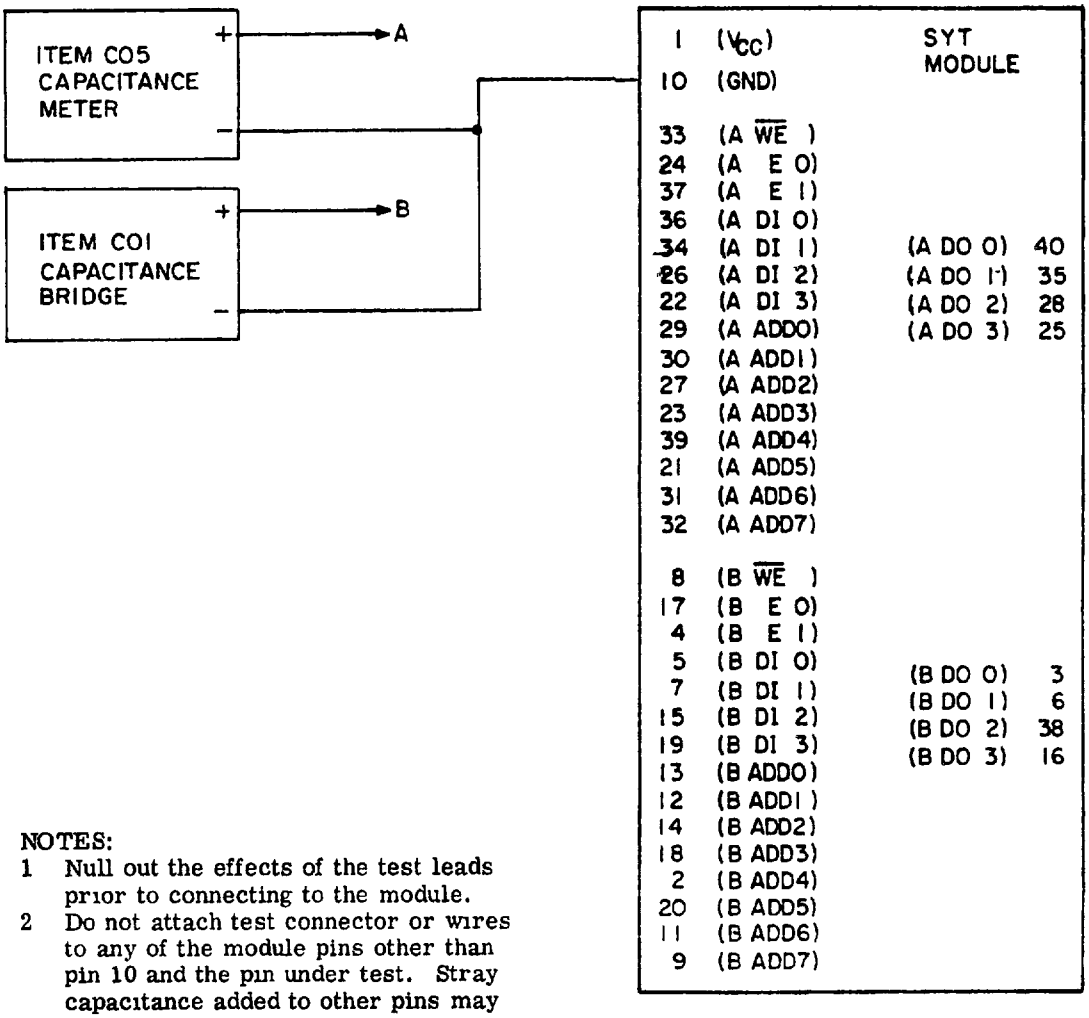
	Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
*	X1	ADO0 Output capacitance	40	Open	3	20	pF	43
	X2	ADO1	35					
	X3	ADO2	28					
	X4	ADO3	25					
	X5	BDO0	3					
	X6	BDO1	6					
	X7	BDO2	38					
*	X8	BDO3 Output capacitance	16	Open	3	20	pF	43

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TABLE XXV. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Power supply filter capacitance	Open	1	1.8	6.0	µF	43

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NOTES:
1. Null out the effects of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 43. Input, output, and filter capacitance test circuit for tables XXIII, XXIV, and XXV.

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